QUESTIONS ON HARDY SPACES AND HANKEL OPERATORS

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Answer. Open means no informed opinion. Expected yes or Expected no gives a belief with reasons, without a proof. Answered gives the proposed answer; Partial names what remains open.

Evidence. Heuristic means reasons or numerics; Sketch means an argument by Søvik, with or without AI assistance; arXiv means a posted preprint or a public institutional report; Published means a peer-reviewed result. Sketch arguments have not been through arXiv or peer review and should be checked by the reader. Conditional on names an unrefereed result or standing hypothesis used by the argument.

The labels describe the evidence collected here as of 13 October 2026; an Open label is not a claim that no solution exists.

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Part 1. Replacing factorization: point evaluations

1. Embeddings and boundary values

1.1. Removing a spectrum restriction in the infinite polydisc

Is it possible to remove the restriction on the Fourier spectrum in Theorem 1?

Aleman, Olsen, and Saksman, *Fatou and brothers Riesz theorems in the infinite-dimensional polydisc*, arXiv:1512.01509, Question 1, p. 16. Here Theorem 1 concerns the radial path $(r^j)_{j\geq 1}$ and measures with Fourier support in $\mathbb{N}_0^\infty \cup (-\mathbb{N}_0^\infty)$, meaning finitely supported multi-indices with all coordinates of one sign; its endpoint conclusions are a weak $(1,1)$ maximal bound for finite measures and almost-everywhere recovery of L^1 boundary values.
Partial. Evidence: Sketch.

The source's unrestricted part (i) is not the issue; the open endpoint content is the finite-measure weak $(1,1)$ bound and full almost-everywhere convergence in parts (ii) and (iii). Mixed-sign frequencies remain outside the established restricted-spectrum theorem.

1.2. Radial failure along decreasing coordinate radii

Does there exist a bounded analytic function $f \in H^\infty(\mathbb{T}^\infty)$ such that almost surely the radial convergence fails even if the approach is limited by assuming decreasing radii in n, i.e. $r_{n,k} \geq r_{n+1,k}$ for all n, k?

Aleman, Olsen, and Saksman, *Fatou and brothers Riesz theorems in the infinite-dimensional polydisc*, arXiv:1512.01509, Question 2, p. 16.
Answered: yes. Evidence: arXiv.

Sun, Zu and Lu's *Radial convergence along decreasing coordinate radii in $H^\infty(\mathbb{T}^\infty)$*, Theorem 2.1, arXiv:2608.03164v2, constructs the requested bounded analytic function and verifies failure of boundary recovery along an ordered approach. They take a product of analytic exponentials with summable masses at the origin, while selecting radii where successive block Poisson sums diverge. The product tends to zero although its boundary values are nonzero almost everywhere; interleaving good finite-section radii gives the stronger failure of a numerical limit.
Conditional on: Sun, Lu, and Zu, arXiv:2608.03164, latest unrefereed version.

1.3. Boundary-independent radial failure

What is the answer to the above question under the added condition that the radial approach does not depend on the point on the boundary?

Aleman, Olsen, and Saksman, *Fatou and brothers Riesz theorems in the infinite-dimensional polydisc*, arXiv:1512.01509, Question 3, p. 16. The "above question" asks for a bounded analytic function whose boundary values fail to be recovered almost everywhere along radii $r_{n,k} \geq r_{n+1,k}$, with $r_{n,k} \to 1$ for each fixed n; here the whole sequence of radius vectors must be independent of the boundary point.
Answered: yes. Evidence: arXiv.

Sun, Zu and Lu, *Radial convergence along decreasing coordinate radii in $H^\infty(\mathbb{T}^\infty)$*, Theorem 3.2 and Remark 3.3, arXiv:2608.03164v2, give a boundary-independent construction. Finite grids retain a fixed fraction of each block's maximal Poisson sum; enumerating all

grids gives a zero subsequence at almost every boundary point. Interleaving the good path from Aleman, Olsen and Saksman's Theorem 1(iii), arXiv:1512.01509v2, supplies a second subsequence converging to the nonzero boundary value.
Conditional on: Sun, Lu, and Zu, arXiv:2608.03164, latest unrefereed version.

1.4. **Nontangential Fatou theory in infinitely many variables**

What can one say about non-tangential approach for functions in $H^\infty(\mathbb{T}^\infty)$, or more generally, in $H^p(\mathbb{T}^\infty)$?

Aleman, Olsen, and Saksman, *Fatou and brothers Riesz theorems in the infinite-dimensional polydisc*, arXiv:1512.01509, Question 4, p. 16, corrected arXiv version.
Partial. Evidence: Sketch.

A written argument, based on Aleman, Olsen and Saksman's Theorem 1, doi:10.1007/s11854-019-0006-x, establishes Fatou limits in correlated one-parameter tube regions and sharpness for broader unrestricted approaches. A classification for general coordinatewise nontangential regions remains open.

1.5. **Uniqueness for radial approaches**

For which radial approaches is the brothers Riesz theorem on the uniqueness of boundary values true? That is, when is it true for all $f \in H^\infty(\mathbb{T}^\infty)$ that having vanishing radial boundary values on a set of positive Lebesgue measure implies $f \equiv 0$?

Aleman, Olsen, and Saksman, *Fatou and brothers Riesz theorems in the infinite-dimensional polydisc*, arXiv:1512.01509, Question 5, p. 16.
Partial. Evidence: Sketch.

A written argument proves that a predictable-tail class of approaches is sufficient and classifies a factorized model, using Aleman, Olsen and Saksman's Corollary 2 and Theorem 4(i), doi:10.1007/s11854-019-0006-x. Arbitrary boundary-dependent approaches are not classified.

1.6. **Carleson interpolation beyond reciprocal-even exponents**

We would like to know if this result extends to other values of p.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, §2, paragraph after Problem 2.4. The inferred target is a necessary-and-sufficient criterion for interpolation with the standard normalized data norm.
Partial. Evidence: Sketch.

Saksman and Seip prove the reciprocal-even cases; later necessary product conditions cover additional ranges. No full criterion for arbitrary finite p and bounded distinct sequences is known.

1.7. **Local vertical embeddings for $\mathscr{H}^p$**

Is the L^p integral of a Dirichlet polynomial over any segment of fixed length on the vertical line $\operatorname{Re} s = 1/2$ bounded by a universal constant times its $\mathscr{H}^p$ norm?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.1, pp. 3–4.
Answered: exactly for $p \geq 2$. Evidence: arXiv.

Chen, Fang, Guo, Hou, Shao, and Zhou prove the inequality for every finite $p \geq 2$ and rule it out below 2 in *The local embedding problem for Hardy spaces of Dirichlet series*, Theorem 1.1 and Corollary 1.2, arXiv:2609.11560v2; the negative range uses Harper's *Moments of random multiplicative functions, I: Low moments, better than squareroot cancellation, and critical multiplicative chaos*, Theorem 1, doi:10.1017/fmp.2019.7. For the positive range, duality isolates resonant prime-frequency blocks, whose fractional moments are bounded through a

covariance-preserving Gaussian replacement and a critical branching-random-walk estimate; a square-function bound then assembles the blocks.
Conditional on: Chen and collaborators, arXiv:2609.11560, latest unrefereed version.

1.8. Carleson measures for $\mathscr{H}^p$ below two

Characterize Carleson measures for $\mathscr{H}^p$ on $\{\operatorname{Re} s > 1/2\}$ for $p < 2$.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.2.
Partial. Evidence: Sketch.

Necessary Carleson-square estimates and failures for bounded-support examples are known. They do not give a necessary-and-sufficient condition for arbitrary finite measures.

1.9. The Blaschke condition below two

Do the zero sets of functions in $\mathscr{H}^p$ for $p < 2$ satisfy the Blaschke condition in the half-plane $\operatorname{Re} s > 1/2$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.3.
Answered: no. Evidence: Sketch.

A lacunary sum of long Dirichlet blocks belongs to every $\mathscr{H}^p$ with $p < 2$ and has zeros violating the global half-plane Blaschke sum. The full construction is in the answers entry.
Answers, "The Blaschke condition below two", p. 2.
Standalone proof note.

1.10. Local Nevanlinna membership below two

Are elements of $\mathscr{H}^p$ for $p < 2$ locally in the Nevanlinna class?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.4.
Answered: no. Evidence: Sketch.

The same block construction has unbounded local logarithmic means at the critical line, so it fails the classical boundary-local Nevanlinna condition. The full argument is in the answers entry.
Answers, "Local Nevanlinna membership below two", p. 2.
Standalone proof note.

1.11. Weighted critical-line trace inequalities

An interesting open problem is to describe the non-negative weights ω for which there is a finite constant $C(\omega)$ such that $\int_{\mathbb{R}} |f(1/2+it)|^2 \omega(t)\, \mathrm{d}t \le C(\omega)\|f\|^2_{\mathscr{H}^2}$.

Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, arXiv:2005.05094, §7, remark following Lemma 7.2 and equation (7.4).
Partial. Evidence: Sketch.

Necessarily $\omega \in L^1 \cap L^\infty$, and the source proves sufficiency for even weights decreasing on $[0, \infty)$. A characterization without monotonicity remains open.

1.12. Extension of Smirnov functions to the square-summable polydisc

Does there exist $f \in N_*(\mathbb{T}^\infty)$ that cannot be extended to a holomorphic function on $\mathbb{D}^\infty \cap \ell^2$?

Guo, Ni, and Zhou, *Nevanlinna class, Dirichlet series and Szegő's problem*, arXiv:2201.01993, Question 2.28, §2, p. 20.
Answered: no. Evidence: Sketch.

Every such function extends holomorphically to $\mathbb{D}^\infty \cap \ell^2$. A full proof is in the public note.
The proof

1.13. **Holomorphic Carleson measures on finite polydiscs**

> Nonetheless, a complete characterization of Carleson measures for the holomorphic Hardy spaces in higher dimensions has yet to emerge. In particular it is not clear whether the sufficient condition of Chang–Stein is also necessary, or if the class of $H^p(\mathbb{D}^d)$ Carleson measures is the same for every $p \geq 1$ for that matter.

Chalmoukis and Dayan, *Simply interpolating and Carleson sequences for Hardy spaces in the polydisc*, arXiv:2412.09099, §1, p. 3. We isolate those two questions for finite-dimensional polydiscs. The Chang–Stein condition is $\mu(S(U)) \leq Cm(U)$ for every open $U \subset \mathbb{T}^d$, where $S(U)$ is the union of product Carleson boxes over all product arcs contained in U.
Partial. Evidence: Sketch.

The written argument identifies the same Carleson class for $p = 1, 2$ using analytic weak factorization (Lacey and Terwilleger, *Hankel operators in several complex variables and product BMO*, Theorem 1.1 and (1.3), arXiv:math/0310348v3). Under a structured sector hypothesis it obtains Chang–Stein necessity by reflection and Gram comparison, using Chalmoukis and Dayan, *Simply interpolating and Carleson sequences for Hardy spaces in the polydisc*, Theorem 3.1 and Lemma 3.2, doi:10.1007/s12220-025-02155-5. The arbitrary-measure characterization and all $p \geq 1$ remain open.

1.14. **Window lengths for mean values of Dirichlet polynomials**

> How does $T_p(N, C)$ depend on p? In particular, is $T_p(N, C) = O(N)$ for $1 \leq p < 2$?

Seip, *Two Problems on Dirichlet Polynomials*, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, in *Dirichlet Series and Function Theory in Polydiscs*, Oberwolfach Report 06/2014, Problem 4(b), p. 387. Here $T_p(N, C)$ is the infimum of $T > 0$ for which $T^{-1} \int_0^T |f(it)|^p \, \mathrm{d}t \leq C\|f\|^p_{\mathscr{H}^p}$ for every $f(s) = \sum_{n \leq N} a_n n^{-s}$, with $C > 1$ fixed; $\|f\|^p_{\mathscr{H}^p}$ is its infinite-time mean on $\operatorname{Re} s = 0$.
Partial. Evidence: Published.

For $p = 2$, the Montgomery–Vaughan mean-value theorem gives the linear window bound: Montgomery and Vaughan, *Hilbert's inequality*, Corollary 3, doi:10.1112/jlms/s2-8.1.73. Expanding the square separates the diagonal coefficient sum from off-diagonal oscillatory terms; Hilbert's inequality bounds the latter by $O(N/T)\|f\|^2_{\mathscr{H}^2}$. Seip leaves the range $1 \leq p < 2$ open.

2. Nevanlinna classes

2.1. **Interchanging mean-counting limits**

> For $f \in \mathcal{N}_u$, can the $\sigma \to 0$ and $T \to \infty$ limits in $M_f(w)$ be interchanged for every $w \neq f(+\infty)$?

Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, arXiv:2005.05094, §8.1, Problem 1, p. 33. Here $\mathcal{N}_u$ consists of Dirichlet series uniformly convergent on every $\{\operatorname{Re} s > \varepsilon\}$ with bounded logarithmic means, $f(+\infty)$ is the constant coefficient, and $M_f(w) = \lim_{\sigma \to 0^+} \lim_{T \to \infty} (\pi/T) \sum_{f(s)=w,\, |\operatorname{Im} s| < T,\, \operatorname{Re} s > \sigma} \operatorname{Re} s$, counting multiplicities.
Answered: no. Evidence: Published.

Brevig, Kouroupis, and Perfekt give a counterexample to the interchange in *Almost periodicity and boundary values of Dirichlet series*, Theorems 1.6–1.7, doi:10.1090/tran/9420. They adapt Saksman–Seip's construction to separate scalar boundary values from vertical-limit boundary values, then use a Jensen-type counting formula to turn the difference into incompatible zero-counting limits.

2.2. **A logarithmic local embedding for $\mathcal{N}_u$**

Does there exist a constant $C > 0$ such that, for every $f \in \mathcal{N}_u$ with $\sigma_u(f) \le 0$, $\sup_{\sigma>0,\,\tau\in\mathbb{R}} \int_\tau^{\tau+1} \log(1 + |f(\sigma + it)|)\, \mathrm{d}t \le C\|f\|_0$?

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, Problem 4.13, p. 25. Here $\sigma_u(f)$ is the abscissa of uniform convergence, $\|f\|_0 = \lim_{\sigma\to 0^+} \lim_{T\to\infty} (2T)^{-1} \int_{-T}^{T} \log(1 + |f(\sigma + it)|)\, \mathrm{d}t$, and $\mathcal{N}_u$ consists of Dirichlet series with $\sigma_u(f) \le 0$ and finite logarithmic means.

Answered: no. Evidence: Sketch.

A prime-clustered construction violates the proposed estimate. The proof is in the public local-embedding note; it concerns the stated local logarithmic bound. The proof

2.3. **Characterizing composition operators on $\mathcal{N}_u$**

It is natural to investigate which analytic symbols Φ generate bounded composition operators C_Φ on the Dirichlet–Nevanlinna class $\mathcal{N}_u$.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, §1. The curated question asks for correct criteria for continuity or Lipschitz continuity in d_0; Theorems 1.3–1.4 motivate but do not settle that inferred target.

Here $C_\Phi f = f \circ \Phi$ and $d_0(f, g) = \|f - g\|_0$, with $\|h\|_0$ the limiting mean of $\log(1+|h(\sigma+it)|)$ as $T \to \infty$ and then $\sigma \to 0^+$; $\mathcal{N}_u$ consists of Dirichlet series uniformly convergent on every strict right half-plane with finite such means.

Partial. Evidence: Sketch.

Explicit symbols show that the displayed half-plane inclusion and a compact-image condition do not suffice for continuity. The exact correct conditions for boundedness or continuity remain undetermined.

2.4. **Removing a compact-image assumption in boundary composition**

It would be desirable to eliminate the assumption that $\Phi(\mathbb{C}_0)$ is a compact subset of $\mathbb{C}_0$ in our Lemma 4.4, following the approach of Brevig and Perfekt [12].

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, Remark 4.5, p. 27. The curated question infers whether the lemma remains true without that compact-image hypothesis.

Here $\mathbb{C}_0 = \{\mathrm{Re}\, s > 0\}$, and Lemma 4.4 is the boundary push-forward identity $\|f \circ \Phi\|_0 = \int_{\mathbb{T}^\infty} \log(1 + |f(\Phi^*(\chi))|)\, \mathrm{d}m(\chi)$ for $f \in \mathcal{N}_u$, where Φ^* is the almost-everywhere boundary value of its Bohr lift and $\|\cdot\|_0$ is the limiting logarithmic mean. The reference [12] is Brevig and Perfekt, *Norms of composition operators on the H^2 space of Dirichlet series*, doi:10.1016/j.jfa.2019.108320.

Answered: no. Evidence: Sketch.

A boundary-approaching symbol makes the proposed unrestricted push-forward identity fail. The question is inferred from the compact-image hypothesis in the quoted source passage.

2.5. **A counting formula for the completed Nevanlinna class**

However, for the Nevanlinna–Dirichlet class $\mathcal{N}$, defined as the completion of $\mathcal{N}_u$, the existence of a corresponding counting function yielding a Stanton-type formula remains an open question.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, Remark 4.2, p. 22.

Partial. Evidence: Sketch.

A conditional potential identity is established for a restricted completion-level setup. A general counting function and Stanton formula for the completed class remain open.

2.6. **Extending boundary composition to the completion**

It remains an open question whether Lemma 4.4 can be extended to the full Nevanlinna class $\mathcal{N}$.

Allu and Mondal, *Composition operators and Carleson embeddings for the Nevanlinna class of Dirichlet series*, arXiv:2609.17133, Remark 4.5, pp. 27–28.

Here $\mathbb{C}_0 = \{\operatorname{Re} s > 0\}$, and Lemma 4.4 is the boundary push-forward identity $\|f \circ \Phi\|_0 = \int_{\mathbb{T}^\infty} \log(1 + |f(\Phi^*(\chi))|)\, \mathrm{d}m(\chi)$ for $f \in \mathcal{N}_u$, where Φ^* is the almost-everywhere boundary value of its Bohr lift and $\|\cdot\|_0$ is the limiting logarithmic mean. The completed class $\mathcal{N}$ is the metric completion for $d_0(f, g) = \|f - g\|_0$.

Answered: no. Evidence: Sketch.

A lacunary H^1 function composed with a suitable characteristic-zero symbol has divergent local logarithmic means, contradicting any finite completion-valued extension. The full proof is in the answers entry.

Answers, "Extending boundary composition to the completed Nevanlinna class", p. 3.

Standalone proof note.

3. Paley–Wiener spaces

3.1. **A quantitative lower bound below exponent two**

There is a positive constant A such that $\mathcal{C}_p \geq p/2(1 - A(p-2))$ for $1 \leq p \leq 2$.

Conjecture 9.2, printed p. 56, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, arXiv:2210.13922. The source's stated interval and sign convention are part of the question.

Here $\mathcal{C}_p = \sup_{f \neq 0} |f(0)|^p / \|f\|_p^p$ on PW^p, the entire functions of type at most π whose restrictions belong to $L^p(\mathbb{R})$.

Answered: yes. Evidence: Sketch.

A computer-assisted proof gives the bound with $A = 7/100$ throughout $1 \leq p \leq 2$. The certificate and proof are in the point-evaluation note.

3.2. **Uniform boundedness of Fourier interpolation basis functions**

There exists $C > 0$ such that $|f_n(x)| < C$ holds for all $n \in \mathbb{Z}_{\geq 0}$, $x \geq 0$.

Conjecture 8.1, §8.2, printed p. 38, of David Berghaus, Andriy Bondarenko, Danylo Radchenko, Kristian Seip and Qihang Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. Here $f_n(x) = b_n^+(\sqrt{x})$ is the Fourier-invariant Radchenko–Viazovska basis function in the source's notation.

Open. Evidence: Heuristic.

The source reports numerical values mostly below two, but its discussion supplies no uniform analytic bound in both n and x.

3.3. **Statistical distribution of the basis-function values**

Does the sequence $\{n^{1/4} f_n(x)\}_{n \geq 0}$ have a statistical distribution with finite moments? If not, is this true for $\{n^{1/4} \log^\alpha(n) f_n(x)\}_{n \geq 2}$ for some $\alpha \in \mathbb{R}$?

The unnumbered question in §8.2, printed p. 38, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. The source fixes x and asks about the sequence of values as n varies.

Here $f_n(x) = b_n^+(\sqrt{x})$, where b_n^+ is the Fourier-invariant Radchenko–Viazovska interpolation basis function.

Open. Evidence: Heuristic.

The plotted values suggest a limiting distribution, while the authors' moment calculations do not support normality. No finite-moment distribution is established.

3.4. **Moving-point decay of the basis functions**

For any $\varepsilon, \delta > 0$ we have $f_n(x) = O_{\varepsilon,\delta}(n^{-1/2+\delta})$ for $\varepsilon n \le x \le n - \sqrt{n}$.

Conjecture 8.2, §8.2, printed p. 39, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. In this range the conjecture is equivalent to subpower growth of the source's cosine sum Ψ.

Here $f_n(x) = b_n^+(\sqrt{x})$, where b_n^+ is the Fourier-invariant Radchenko–Viazovska interpolation basis function.

Open. Evidence: Heuristic.

The source proves an $O(n^{-0.392})$ estimate on the smaller range $n/8 < x < n - \sqrt{n}$ and notes that an exponent-pair conjecture would imply the stated decay. Neither gives the requested bound unconditionally.

3.5. **Polylogarithmic growth of the cosine sum**

There exists $k > 0$ such that

$$\Psi(x) = \sum_{n \ge 1} \frac{2}{\sqrt{n}} \cos(\pi(\frac{3n-1}{4} - \frac{x}{n}))$$

satisfies

$$\Psi(x) = O(\log^k |x|), \qquad x \to \pm\infty.$$

Conjecture 8.3, §8.2, printed p. 39, of Berghaus, Bondarenko, Radchenko, Seip and Sun, *The basis functions of Fourier interpolation*, arXiv:2512.18677v1. The sum is denoted by Ψ in their formula (4.6).

Open. Evidence: Heuristic.

Their exponent-pair reduction gives subpower growth under an additional conjecture, but the checked source proves no polylogarithmic estimate.

3.6. **Fourier-invariant functions vanishing on near-square-root samples**

> "this result fails when $\beta > 1/2$"; "explore further the finer details of the transition".

The first passage is in the abstract and the second in the introduction, printed p. 2, of Andriy Bondarenko and Kristian Seip, *Fourier-invariant functions with dense zero sets*, arXiv:2608.13468v1. Together they motivate two targets: whether a nonzero Fourier-invariant continuous $L^1 \cap L^2$ function can vanish at $\lambda_n = \sqrt{n}/(\log n)^\beta$ for $\beta > 1/2$, and the regularity threshold for Fourier uniqueness at $\lambda_n = \sqrt{n}$.

Partial. Evidence: Sketch.

A written argument proves uniqueness in Fourier-invariant H^s for every $s > 1/2$ and every $\beta > 0$. The $L^1 \cap L^2$ existence question above $\beta = 1/2$, and the endpoint regularity threshold at $\beta = 0$, remain open.

Conditional on: The endpoint examples for $0 \le \beta \le 1/2$ use Bondarenko–Seip's unrefereed Theorem 2.3.

3.7. **Complex interpolation of ball Paley–Wiener spaces**

> "possible to interpolate between"

The discussion before Theorem 1.3, printed p. 4, of Konstantinos Bampouras and Karl-Mikael Perfekt, *Boundary-weighted Fourier inequalities for convex domains*, arXiv:2608.19806v1, says the authors do not know whether the ball spaces interpolate. The reconstructed question is whether $[PW^{p_1}(B), PW^{p_2}(B)]_\theta = PW^{p_\theta}(B)$ with equivalent norms for $1 < p_1 < p_2 < \infty$ and $1/p_\theta = (1-\theta)/p_1 + \theta/p_2$; here $PW^p(B) = \{f \in L^p(\mathbb{R}^n) : \operatorname{supp} \widehat{f} \subset \overline{B}\}$ for a Euclidean ball B.

Partial. Evidence: Sketch.

A written argument proves interpolation for radial subspaces when both exponents lie in a common power-weight strip. It does not establish the full nonradial ball identity.

3.8. **Subcritical weighted Fourier inequalities on the ball**

"We do not know whether the hypothesis of Theorem 1.5 is ever satisfied."

The passage after Theorem 1.5, printed p. 4, of Bampouras and Perfekt, *Boundary-weighted Fourier inequalities for convex domains*, arXiv:2608.19806v1. The inferred target is whether for every $n \geq 3$ some $1 \leq q < 2$ satisfies

$$\int_B |\widehat{f}(x)|^q \omega_B(x)^{-d}\, \mathrm{d}x \leq C_{q,d}\|f\|_q^q, \qquad d < 2 - q, \quad f \in PW^q(B),$$

where B is the unit ball in $\mathbb{R}^n$ and $\omega_B(x) = |B \cap (2x - B)|$. For each fixed n, the same q must work for every strict subcritical d.

Partial. Evidence: Sketch.

Tomas's *A restriction theorem for the Fourier transform*, doi:10.1090/S0002-9904-1975-13790-6, gives a smaller range. The public institutional report *Diagonal Fourier Extension for Positively Curved Surfaces in Three Dimensions* (Theorem 1.1), by OpenAI, https://github.com/openai/math/blob/main/preprints/Diagonal-Fourier-extension-for-positively-curved-surfaces-in-three-dimensions-September-24-2026/Diagonal-Fourier-extension-for-positively-curved-surfaces-in-three-dimensions-September-24-2026.pdf, conditionally supplies the full target for $n = 3$ and $3/2 \leq q < 2$; the all-function problem in dimensions $n \geq 4$ remains open.

Conditional on: The three-dimensional claim depends on that unreviewed report's named inputs, *Elliptic Capacity Propagation and Fourier Restriction to the Sphere* (Theorem 6.4), https://github.com/openai/math/blob/main/preprints/Elliptic-capacity-propagation-and-Fourier-restriction-to-the-sphere-September-24-2026/Elliptic-capacity-propagation-and-Fourier-restriction-to-the-sphere-September-24-2026.pdf, and *The Kakeya Maximal Conjecture in Three Dimensions* (Theorem 1.1), https://github.com/openai/math/blob/main/preprints/The-Kakeya-maximal-conjecture-in-three-dimensions-September-23-2026/paper.pdf.

3.9. **Monotonicity of the normalized evaluation constant**

The function $p \mapsto \mathcal{C}_p/p$ is strictly decreasing on $(0, \infty)$.

Conjecture 9.1, printed p. 56, of Ole Fredrik Brevig, Andrés Chirre, Joaquim Ortega-Cerdà and Kristian Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z, J. Anal. Math. 153 (2024), 595–670. Here $\mathcal{C}_p$ is the sharp constant in $|f(0)|^p \leq \mathcal{C}_p\|f\|_p^p$ for entire functions of type at most π whose real restrictions belong to $L^p(\mathbb{R})$.

Partial. Evidence: Sketch.

A full written proof gives differentiability of $\mathcal{C}_p$ on $(1, \infty)$, strict monotonicity near $p = 2$, and eventual monotonicity. The conjecture for all pairs of positive exponents remains open.

3.10. **The imaginary-axis evaluation envelope**

Conjecture 9.3 asserts $K_p = \mathcal{C}_p$ for $0 < p < 2$ and $K_p = p/2$ for $2 \leq p < \infty$.

Conjecture 9.3, printed p. 57, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. Here K_p is the supremum over $y \geq 0$ of the best normalized point-evaluation constant at iy, with the source's factor $\operatorname{sinc}(i\pi py)$.

More explicitly, $K_p(y) = \sup_{f \neq 0} |f(iy)|^p / (\operatorname{sinc}(i\pi py)\|f\|_p^p)$ on the type-π space PW^p, $\operatorname{sinc} z = \sin z/z$, and $K_p = \sup_{y \geq 0} K_p(y)$.

Partial. Evidence: Sketch.

The source's asymptotic as $y \to \infty$ and the power trick settle integer exponents. A written argument also gives a strict gap at bounded heights for even exponents; the full envelope identity remains open.

3.11. **Uniqueness of the low-exponent extremal**

The extremal problem (1.6) has a unique solution for $0 < p < 1$.

Conjecture 9.4, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. Problem (1.6) minimizes $\int_{\mathbb{R}} |\varphi(x)|^p \, \mathrm{d}x$ over PW^p, the entire functions of type at most π in $L^p(\mathbb{R})$, subject to $\varphi(0) = 1$.
Open. Evidence: Heuristic.

Existence, reality and structural zero identities are known, and uniqueness holds for $p \geq 1$ by their Lemma 3.5, doi:10.1007/s11854-024-0338-z. At low exponents the known arguments do not synchronize the signs of two candidate extremizers.

3.12. **Evenness of low-exponent extremizers**

Any solution of (1.6) for $0 < p < 1$ is even.

Conjecture 9.5, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The assertion concerns every minimizer, not merely the existence of an even one.

Here (1.6) minimizes $\|\varphi\|_p^p$ over PW^p, the entire functions of exponential type at most π in $L^p(\mathbb{R})$, subject to $\varphi(0) = 1$.
Open. Evidence: Heuristic.

Reflection preserves the minimizing problem, but averaging the reflected functions need not preserve minimality when $p < 1$. The written variation calculations give necessary conditions without proving symmetry of the zero divisor.

3.13. **Uniform separation of extremal zero sets**

The zero set of any solution of (1.6) is uniformly discrete for all $0 < p < \infty$.

Conjecture 9.6, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. Uniform discreteness here is for the distinct zero locations of each fixed extremizer.

Here (1.6) minimizes $\|\varphi\|_p^p$ over PW^p, the entire functions of exponential type at most π in $L^p(\mathbb{R})$, subject to $\varphi(0) = 1$.
Answered: yes. Evidence: Sketch.

The remaining range below $p = 1/2$ follows from a cluster argument using the canonical-product identity and a uniform polynomial estimate with real exponential weight. The full written proof is in the proof note.
Answers, "Uniform separation of extremal zero sets", p. 4.

3.14. **The full-range distributional reproducing identity**

Theorem 8.1 extends to the full range $0 < p < \infty$.

Conjecture 9.7, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. For every extremizer φ, the source defines $\Phi = |\varphi|^{p-2}\varphi/\|\varphi\|_p^p$ and asks that Φ be tempered and $f(0) = \int f\Phi$ for every $f \in \mathcal{S} \cap PW^2$.

Here PW^2 is the space of entire type-π functions in $L^2(\mathbb{R})$.
Answered: yes. Evidence: Sketch.

For $p < 1/2$, the written proof combines uniform zero separation with simplicity of the zeros to obtain polynomial lower bounds, then differentiates the extremal problem to establish the identity. The remaining exponents are covered in the source. A full proof appears in the proof note.
Answers, "The distributional reproducing identity", p. 5.

3.15. **Monotonicity of positive zeros**

The function $p \mapsto t_n(p)$ is strictly decreasing for all $n \geq 1$.

Conjecture 9.8, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The source assumes uniqueness of the extremizer for $p < 1$ when defining $t_n(p)$.

Here $t_n(p)$ is the nth distinct positive zero of the normalized extremizer $\varphi_p(0) = 1$ of $\|f\|_p^p$ in the type-π space PW^p.

Partial. Evidence: Sketch.

A horizontal-strip expansion near $p = 2$ proves the comparison for any fixed finite set of zero indices on a two-sided neighborhood of 2. It does not give a neighborhood uniform over all indices or monotonicity at arbitrary exponents.

Conditional on: Uniqueness of the extremizer for $0 < p < 1$, as assumed by the source.

3.16. **Asymptotic unit spacing of extremal zeros**

We have $\lim_{n\to\infty}(t_{n+1} - t_n) = 1$ for all $p > 0$.

Conjecture 9.9, printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. Below $p = 1$ the source continues to use its standing uniqueness assumption.

Here $t_n(p)$ is the nth distinct positive zero of the normalized extremizer $\varphi_p(0) = 1$ of $\|f\|_p^p$ in the type-π space PW^p.

Partial. Evidence: Heuristic.

The unit-spacing limit is known for the sinc extremizer at $p = 2$ and follows from Bondarenko, Ortega-Cerdà, Radchenko and Seip's *The Hörmander–Bernhardsson extremal function: A preliminary study*, Theorem 1, doi:10.1007/978-3-031-95551-8_4, at $p = 1$. A constructed family of nonextremizers shows why the coarse zero bounds and moment information alone do not force convergence.

Conditional on: Uniqueness of the extremizer for $0 < p < 1$, as assumed by the source.

3.17. **Endpoint limits of the first positive zero**

We have $\lim_{p\to\infty} t_1(p) = 1/2$ and $\lim_{p\to 0^+} p\, t_1(p) > 0$.

Conjecture 9.10, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The source assumes uniqueness for small p when it uses $t_1(p)$.

Here $t_n(p)$ is the nth distinct positive zero of the normalized extremizer $\varphi_p(0) = 1$ of $\|f\|_p^p$ in the type-π space PW^p.

Answered: yes. Evidence: Sketch.

The large-exponent limit follows from local convergence of extremizers to $\cos(\pi z)$. Under the stated low-exponent uniqueness assumption, a rescaling and zero-potential argument proves that the second limit exists and is finite and positive. Both proofs are written out in the proof note and the point-evaluation note.

Conditional on: Uniqueness of the extremizer for all sufficiently small $p > 0$.

Answers, "Limits of the first positive extremal zero", p. 5.

3.18. **Extremal zeros as a uniqueness set**

$Z(\varphi_p) \cup \{0\}$ is a uniqueness set for PW^p if and only if $p \geq 1$.

Conjecture 9.11, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The source assumes a unique extremizer below $p = 1$ when using the notation φ_p.

Here $\varphi_p(0) = 1$ minimizes $\|f\|_p^p$ in the type-π space PW^p, and $Z(\varphi_p)$ is its zero set; a uniqueness set is one on which only the zero function in PW^p vanishes.

Answered: yes. Evidence: Sketch.

For $p \geq 1$, Theorem 3 of Søvik's point-evaluation note gives the assertion. For $0 < p < 1$, a dilation-times-sinc competitor gives a tail estimate implying $x\varphi(x) \in PW^p$; this nonzero function vanishes on $Z(\varphi) \cup \{0\}$. The argument works for any extremizer; the source extquotesingle s uniqueness convention is only used to name φ_p.

Answers, "Extremal zeros as a uniqueness set", p. 6.

3.19. **An A_p weight from the extremal zeros**

Suppose that $1 < p < \infty$. The function $\left(\frac{|x\varphi_p(x)|}{\operatorname{dist}(x, Z(\varphi_p)\cup\{0\})}\right)^p$ is a Muckenhoupt A_p weight.

Conjecture 9.12, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The A_p condition is uniform over all finite intervals, as in the standard Muckenhoupt definition.

Here $\varphi_p(0) = 1$ minimizes $\|f\|_p^p$ in PW^p, the space of entire type-π functions in $L^p(\mathbb{R})$, and $Z(\varphi_p)$ denotes its zero set.

Partial. Evidence: Sketch.

The $p = 2$ weight is bounded above and below. A written criterion reduces the general question to bounded midpoint logarithmic slopes and a discrete weighted A_p condition; near-minimizers show that several tempting geometric estimates alone do not imply this criterion.

3.20. **Fourier regularity of the extremal**

If $1 < p < \infty$, then the Fourier transform $\widehat{\varphi_p}$ is a continuous integrable function on $(-\pi, \pi)$.

Conjecture 9.13, printed p. 59, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. Continuity at the endpoints is not part of the assertion.

Here $\varphi_p(0) = 1$ minimizes $\|f\|_p^p$ in PW^p, the space of entire type-π functions in $L^p(\mathbb{R})$, and $Z(\varphi_p)$ denotes its zero set.

Partial. Evidence: Sketch.

The source settles $p \le 1$. A written argument establishes the Fourier conclusion for $1 < p \le 2$ and Fourier integrability for $2 < p < 4$; the remaining range is open. A workshop announcement claims interior continuity at $p = 4$, without a public proof manuscript.

3.21. **Positivity and shape of the Fourier transform**

Let φ be a solution of (1.6). Then $\widehat{\varphi}$ is nonnegative. Moreover, (a) if $0 < p < 2$, it is decreasing on $(0, \pi)$ and tends to 0 as $\xi \to \pi^-$; (b) if $2 < p < \infty$, it is increasing on $(0, \pi)$ and tends to ∞ as $\xi \to \pi^-$.

Conjecture 9.14, printed p. 60, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The assertions concern every solution of (1.6).

Here (1.6) minimizes $\|\varphi\|_p^p$ over PW^p, the entire functions of exponential type at most π in $L^p(\mathbb{R})$, subject to $\varphi(0) = 1$.

Partial. Evidence: Sketch.

A written certificate proves the full Fourier-shape conclusion at $p = 1$. The general ranges, including the endpoint behavior for other exponents, remain open; a workshop announcement about $p = 4$ has no public proof manuscript.

3.22. **Real-valued solutions of the convolution equation**

(a) There is exactly one real-valued admissible solution of (9.5) when n is an even integer, namely $\widehat{\psi} = \widehat{\varphi}_{n+2}/\|\varphi_{n+2}\|_{n+2}^{1+(n+1)^{-1}}$. (b) There are no real-valued admissible solutions of (9.5) when n is an odd integer.

Conjecture 9.15, printed p. 62, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. For $n \ge 0$, admissible $\widehat{\psi}$ is integrable, supported on $[-\pi, \pi]$ and continuous in the open interval; (9.5) is $C_n\widehat{\psi}(\xi) = 1$ for $|\xi| \le \pi$, where $C_n\widehat{\psi}$ is the convolution of $n+1$ copies and $(u * v)(\eta) = \int_{\mathbb{R}} u(\xi)v(\eta - \xi)\, d\xi/(2\pi)$.

Here $\varphi_{n+2}(0) = 1$ is the type-π point-evaluation extremizer. The real-valued restriction is part of the source conjecture.

Partial. Evidence: Sketch.

For odd n, a written argument proves that every solution must violate a certain global sector bound on an open interval, extending the source's even-solution obstruction. It does not exclude all real-valued solutions or prove the even-n uniqueness assertion.

3.23. **Monotonicity in the imaginary-axis height**

> For fixed p, the best constant $K_p(y)$ "may increase with y when $p > 2$ and decrease with y when $0 < p < 2$."

The paragraph before Conjecture 9.3, printed p. 57, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The inferred question is whether $K_p(y)$ is monotone in y in the indicated direction for each fixed p.

More explicitly, $K_p(y) = \sup_{f \neq 0} |f(iy)|^p / (\operatorname{sinc}(i\pi p y) \|f\|_p^p)$ on the type-π space PW^p, $\operatorname{sinc} z = \sin z / z$, and $K_p = \sup_{y \geq 0} K_p(y)$.

Partial. Evidence: Sketch.

A written argument proves the proposed order for each fixed pair of heights when p is sufficiently close to 2. Full monotonicity for fixed $p \neq 2$, including collapsing height pairs, remains open.

3.24. **Sharp bounds for positive zeros**

> "$n - 1/2 + 1/p \leq t_n \leq n$" for $2 \leq p < \infty$, and "$n \leq t_n \leq n - 1/2 + 1/p$" for $0 < p \leq 2$.

The paragraph following (9.4), printed p. 58, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The source presents these inequalities as a reasonable expectation for the positive zeros of the normalized extremizer.

Here $t_n(p)$ is the nth distinct positive zero of the normalized extremizer $\varphi_p(0) = 1$ of $\|f\|_p^p$ in the type-π space PW^p.

Partial. Evidence: Sketch.

The bounds are known at $p = 1$ and $p = 2$. A written perturbation argument proves both inequalities strictly for each fixed finite set of indices in a two-sided neighborhood of $p = 2$; no all-index neighborhood or global range is known.

Conditional on: Uniqueness of the extremizer for $p < 1$, as assumed by the source.

3.25. **The small-exponent evaluation constant**

> "It remains an interesting problem to determine the constant c_0" of Theorem 1.3.

The paragraph before Conjecture 9.1, printed p. 56, of Brevig, Chirre, Ortega-Cerdà and Seip, *Point evaluation in Paley–Wiener spaces*, doi:10.1007/s11854-024-0338-z. The source defines $c_0 = \lim_{p \to 0^+} 2\mathcal{C}_p/p$, where $\mathcal{C}_p = \sup_{f \in PW^p \setminus \{0\}} |f(0)|^p / \|f\|_p^p$ and PW^p has type at most π, and places c_0 in $[1.1393830, 1.1481785]$.

Partial. Evidence: Sketch.

A written, computer-assisted construction improves the lower bound to $c_0 > 57/50$. The exact value and a matching upper bound remain open.

3.26. **Schwartz-convergent Fourier interpolation formulas**

> "we do not know whether there exists a similar interpolation formula convergent in the topology of the Schwartz space"

The final paragraph of §2.2, arXiv p. 13, of Aleksei Kulikov, Fedor Nazarov and Mikhail Sodin, *Fourier uniqueness and non-uniqueness pairs*, doi:10.15407/mag21.01.04, arXiv:2306.14013v1; the paragraph follows formula (2.1) in the published version and (2.2.1) in the preprint. The source asks whether its interpolation formula for supercritical pairs

can converge in the Schwartz topology. Here $1 < p, q < \infty$, $1/p + 1/q = 1$, and supercriticality is the condition $\limsup_{j\to\pm\infty} |\lambda_j|^{p-1}(\lambda_{j+1} - \lambda_j) < 1/2$ and its counterpart $\limsup_{j\to\pm\infty} |\mu_j|^{q-1}(\mu_{j+1} - \mu_j) < 1/2$ for the two increasing sample sequences. The source's formula reconstructs f as $\sum_j f(\lambda_j)a_j + \sum_j \widehat{f}(\mu_j)b_j$, with basis functions depending on the chosen smoothness index.

Partial. Evidence: Sketch.

Published formulas converge in weighted Hilbert spaces, and later work treats a smaller dense class of supercritical pairs. Cassese and Ramos, *On the regularity of Fourier interpolation formulas*, Theorem 1.3, arXiv:2503.15733v1, give Schwartz-convergent formulas for a proper, sufficiently dense subclass. A written criterion reduces Schwartz convergence to all-order off-diagonal decay of a range projection; that decay has not been proved in the general supercritical range.

3.27. **Higher-multiplicity Fourier interpolation uniqueness**

> Let d and k be positive integers. If $f \in \mathcal{S}_{\mathrm{rad}}(\mathbb{R}^d)$ satisfies $f^{(j)}(\sqrt{kn}) = \widehat{f}^{(j)}(\sqrt{kn}) = 0$ for all integers $n \geq 0$ and $0 \leq j < k$, then must f vanish identically?

Open Problem 7.1, printed p. 93, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, doi:10.4007/annals.2022.196.3.3, Ann. Math. 196 (2022), 983–1082.

Partial. Evidence: Heuristic.

The $k = 1$ case is proved in all dimensions, and $k = 2$ is known for $d = 8, 24$. The general multiplicity and dimension problem remains open; Rolle-type arguments do not provide the required spacing of the extra zeros.

3.28. **Interpolation at Gaussian-optimal radii**

> Let $r_1, r_2, \ldots$ be the radii from Conjecture 7.2 for some d. For which d is every $f \in \mathcal{S}_{\mathrm{rad}}(\mathbb{R}^d)$ uniquely determined by the values $f(r_n), f'(r_n), \widehat{f}(r_n), \widehat{f}'(r_n)$ for $n \geq 1$ through an interpolation formula?

Open Problem 7.3, printed p. 94, of Cohn, Kumar, Miller, Radchenko and Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, doi:10.4007/annals.2022.196.3.3. The question remains conditional on the radii provided by Conjecture 7.2: these are the common positive contact radii where an optimal radial Schwartz auxiliary function for the density-one Gaussian energy bound satisfies $f(x) = e^{-\alpha|x|^2}$ and $\widehat{f}(x) = 0$, forming a discrete infinite set independent of $\alpha > 0$.

Partial. Evidence: Sketch.

A written criterion reduces the question to polynomial separation of the given squared radii and a supported order-one inverse-Laplace kernel. The criterion is conditional: neither hypothesis has been proved for the conjectural radii outside the known dimensions.

Conditional on: Existence of the radii in Conjecture 7.2 for the dimension in question.

3.29. **Hexagonal radii and nonuniqueness**

> Let $r_1, r_2, \ldots$ be the positive real numbers of the form $(4/3)^{1/4}\sqrt{j^2 + jk + k^2}$, where j and k are integers. Then radial Schwartz functions $f : \mathbb{R}^2 \to \mathbb{R}$ are not uniquely determined by the values $f(r_n), f'(r_n), \widehat{f}(r_n), \widehat{f}'(r_n)$ for integers $n \geq 1$.

Conjecture 7.5, printed p. 95, of Cohn, Kumar, Miller, Radchenko and Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, doi:10.4007/annals.2022.196.3.3.

Answered: yes. Evidence: arXiv.

Naser Talebizadeh Sardari, *Higher Fourier interpolation on the plane*, Theorem 1.7, arXiv:2102.08753v2, constructs infinitely many linearly independent radial Schwartz functions with the stated vanishing data. The proof covers the integers represented by j^2+jk+k^2 by a periodic set of arbitrarily small density, then uses Hecke-triangle-group interpolation identities to construct functions vanishing to order two on that set. The paper is listed as in press in the *Journal d'Analyse Mathématique*.

3.30. **Finite perturbations of Fourier interpolation nodes**

For example, does moving (but not removing) finitely many interpolation points preserve the interpolation property?

The paragraph after Conjecture 7.5, printed p. 96, of Henry Cohn, Abhinav Kumar, Stephen D. Miller, Danylo Radchenko and Maryna Viazovska, *Universal optimality of the E_8 and Leech lattices and interpolation formulas*, doi:10.4007/annals.2022.196.3.3, Ann. Math. 196 (2022), 983–1082. The canonical data are $f, f', \widehat{f}, \widehat{f}'$ at radii $\sqrt{2n}$ in dimensions 8 and 24.
Answered: no. Evidence: Sketch.

A full written proof constructs, in each of these two systems, a collision-free finite move for which the data fail to determine the radial Schwartz function.
Answers, "Finite perturbations can destroy Fourier interpolation", p. 6.
Standalone proof note.

3.31. **Orthogonality at smaller exponential type in a zeta-weighted space**

"orthogonal to all functions in H of strictly smaller exponential type"

Problem 5, p. 32, proposed by Jean-François Burnol in the open-problems session of *Mini-Workshop: Interplay between Number Theory and Analysis for Dirichlet Series*, https://publications.mfo.de/bitstream/handle/mfo/3615/OWR_2017_51.pdf, Oberwolfach Report 51/2017. The question asks whether there are $\tau > 0$ and a nonzero entire function f of exponential type τ in $H = L^2(\mathbb{R}, |\zeta(1/2+ix)|^2\,\mathrm{d}x)$ orthogonal in H to every function of strictly smaller type.
Open. Evidence: Heuristic.

The report gives an entire function in the weighted space but does not establish the required orthogonality.

4. Bernstein–Nikolskii inequalities

4.1. **A finite-q Levin–Lubinsky limit formula**

Prove a Levin–Lubinsky type result for $q \neq \infty$.

Problem 1, Section 7, printed p. 36, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. Levin and Lubinsky, *Asymptotic behavior of Nikolskii constants for polynomials on the unit circle*, Theorem 1, https://lubinsky.math.gatech.edu/Research%20papers/LevLubSharpNikolskiiJanuary2015.pdf, prove the asymptotic formula for $q = \infty$; for finite q, Gorbachev reports only the required lower estimate, with no matching asymptotic equality.

Here the requested limit is $n^{1/q-1/p}C_{pq}(T_n) \to C_{pq}(E_1)$ for $0 < p < q < \infty$, where C_{pq} is the sharp L^p-to-L^q norm ratio, T_n consists of trigonometric polynomials of degree at most n, and E_1 consists of entire functions of exponential type at most one.
Open. Evidence: Heuristic.

The periodization argument for the supremum norm does not give the finite-q upper bound. The checked sources identify no method that closes this gap.

4.2. **Exponential improvement of the multidimensional bounds**

Can the upper and lower bounds in (32) for the multidimensional constant L_1^d be improved exponentially in d?

Problem 6, Section 7, printed p. 36, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. The target is to improve both bounds in (32) by exponential factors as the dimension grows.

Here $L_1^d = \sup_{f\neq 0} |f(0)|/\|f\|_{L^1(\mathbb{R}^d)}$ over entire functions of spherical type at most one, and (32) states

$$2^{-d} \le L_1^d/A_d \le {}_1F_2(d/2; d/2+1, d/2+1; -\beta_d^2/4), \qquad A_d = |B^d|/(2\pi)^d,$$

where B^d is the unit ball, β_d is the first positive zero of $J_{d/2}$, and ${}_1F_2$ is the generalized hypergeometric function.

Open. Evidence: Heuristic.

Dai, Gorbachev and Tikhonov, *Estimates of the asymptotic Nikolskii constants for spherical polynomials*, Theorem 1.2, doi:10.1016/j.jco.2021.101553, give these estimates, but the checked results do not establish exponential improvements in dimension.

4.3. **Alternation of extremal Taylor coefficients**

Prove that the signs of the nonzero Taylor coefficients of extremal functions in the problems $c_{n,p}^{(r)}$ and $L_p^{(r)}$ alternate.

Problem 8, Section 7, printed p. 37, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. At the endpoint $p=\infty$, $r=0$, the source does not specify a unique choice of extremizer.

Here $c_{n,p}^{(r)} = \sup_{T\neq 0} |T^{(r)}(0)|/\|T\|_p$ over trigonometric polynomials of degree at most n, and $L_p^{(r)}$ is the analogous supremum over entire functions of exponential type at most one on $\mathbb{R}$.

Partial. Evidence: Sketch.

At $p=\infty$, $r=0$, an explicit normalized extremizer has two consecutive nonzero Taylor coefficients with the same sign, so a universal assertion fails. The existence of a selected alternating extremizer at that endpoint and the conjecture for finite p remain open.

4.4. **The Bessel extremal problem below the known range**

Solve problem (55) for $-1/2 < \alpha < -0.272$.

Problem 9, Section 7, printed p. 37, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. Problem (55) asks for

$$I_\alpha = \inf_{a_k\in\mathbb{R}} \Big\| j_{\alpha+1}(t) - \sum_{k\ge 1} a_k j_\alpha(q_{\alpha+1,k}t/q_{\alpha+1,1}) \Big\|_{L^\infty(\mathbb{R}_+)}.$$

Here $j_\alpha(t) = 2^\alpha\Gamma(\alpha+1)J_\alpha(t)/t^\alpha$, $q_{\alpha,k}$ is its kth positive zero, and the series is required to converge absolutely almost everywhere.

Partial. Evidence: Sketch.

For every $\alpha > -1/2$, integration against $t^{2\alpha+1}$ up to the first positive zero gives the lower bound $I_\alpha \ge a_0^*$, where $a_0^* = \int_0^{q_{\alpha+1,1}} j_{\alpha+1}(t)t^{2\alpha+1}\,\mathrm{d}t / \int_0^{q_{\alpha+1,1}} t^{2\alpha+1}\,\mathrm{d}t$. Equality is equivalent to a uniform tail bound for the Bessel extremizer, which remains open on the stated interval.

4.5. **Uniqueness of the radial extremal among all functions**

Prove that the radial extremal function in the problem L_1^d is unique not only among radial functions.

Problem 7, printed p. 37, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072. Reconstructed question: is the normalized radial minimizer for L_1^d unique in the full class of integrable entire functions of spherical exponential type at most one on $\mathbb{R}^d$?

Answered: yes. Evidence: Sketch.

The full written proof uses raywise evenization to identify the even part with the radial minimizer, then a bounded-type division lemma and the simple real zeros of its profile to eliminate every odd perturbation.

Answers, "Uniqueness of the radial L_1^d extremal", p. 7.

Standalone proof note.

4.6. Degree monotonicity of sharp Bernstein–Nikolskii constants

The sequences $A^{(s)}_{p,q,n}$ and $A^{(s)}_{p,q,n,\mathbb{R}}$ are decreasing in $n \in \mathbb{N}$.

Conjecture 1.3, printed p. 452, of Michael Ganzburg and Sergey Tikhonov, *On sharp constants in Bernstein–Nikolskii-type inequalities*, doi:10.1007/s00365-016-9363-1, Constructive Approximation 45 (2017), 449–466. Here $s \in \mathbb{Z}_{\geq 0}$, $0 < p < q \leq \infty$, and $A^{(s)}_{p,q,n}$ is the normalized best constant for trigonometric polynomials of degree at most n on $[-\pi, \pi]$; the $\mathbb{R}$ subscript restricts to real-valued polynomials.

Partial. Evidence: Sketch.

The known $p = 2, q = \infty$ endpoint decreases by the cited Bennett–Jameson result (Bennett and Jameson, *Monotonic averages of convex functions*, Proposition 3, doi:10.1006/jmaa.2000.7087). Our written estimates prove adjacent decrease in an explicit large-q range for $p = 2$, but leave general p and fixed finite q open.

4.7. Equality of real and complex polynomial constants

The questions as to whether $A^{(s)}_{p,q,n} = A^{(s)}_{p,q,n,\mathbb{R}}$ and $D^{(s)}_{p,q} = D^{(s)}_{p,q,\mathbb{R}}$ still remain open. However, these equalities are valid for $q = \infty$.

The paragraph "Complex and Real Constants," printed p. 451, of Ganzburg and Tikhonov's *On sharp constants in Bernstein–Nikolskii-type inequalities*, doi:10.1007/s00365-016-9363-1 raises these equalities. The question here is the polynomial equality for $0 < p < q < \infty$, all derivative orders $s \geq 0$ and degrees $n \geq 1$.

Here $A^{(s)}_{p,q,n} = n^{-s-1/p+1/q} \sup_{T \neq 0} \|T^{(s)}\|_q / \|T\|_p$ for trigonometric polynomials of degree at most n on $[-\pi, \pi]$; $D^{(s)}_{p,q}$ is the same ratio without the factor in n, over entire functions of type at most one on $\mathbb{R}$, and the real subscript restricts to real-valued competitors.

Answered: yes. Evidence: Sketch.

The phase-averaging argument in the public note proves equality for the polynomial constants throughout this range. The proof.

4.8. Equality of real and complex entire-function constants

The questions as to whether $A^{(s)}_{p,q,n} = A^{(s)}_{p,q,n,\mathbb{R}}$ and $D^{(s)}_{p,q} = D^{(s)}_{p,q,\mathbb{R}}$ still remain open. However, these equalities are valid for $q = \infty$.

The same paragraph of Ganzburg and Tikhonov, *On sharp constants in Bernstein–Nikolskii-type inequalities*, doi:10.1007/s00365-016-9363-1, printed p. 451, raises the entire-function equality. For $0 < p < q < \infty$ and integer $s \geq 0$, $D^{(s)}_{p,q}$ is the sharp derivative ratio over entire functions of exponential type at most one with the indicated L^p and L^q norms; the real subscript restricts to real-valued functions.

Answered: yes. Evidence: Sketch.

The public note proves equality of the real and complex entire-function constants for every stated parameter. The proof.

4.9. Finite-q limit of the polynomial constants

The limit $\lim_{n\to\infty} A^{(s)}_{p,q,n}$ exists and $D^{(s)}_{p,q} = \lim_{n\to\infty} A^{(s)}_{p,q,n}$ for $s \in \mathbb{Z}_+$ and $0 < p < q < \infty$. The real version of Conjecture 1.10 is believed to be true as well.

Conjecture 1.10 and the following paragraph, printed p. 454, of Ganzburg and Tikhonov's *On sharp constants in Bernstein–Nikolskii-type inequalities*, doi:10.1007/s00365-016-9363-1.

Here $A^{(s)}_{p,q,n} = n^{-s-1/p+1/q} \sup_{T \neq 0} \|T^{(s)}\|_q / \|T\|_p$ for trigonometric polynomials of degree at most n on $[-\pi, \pi]$; $D^{(s)}_{p,q}$ is the same ratio without the factor in n, over entire functions of type at most one on $\mathbb{R}$, and the real subscript restricts to real-valued competitors.

Partial. Evidence: Sketch.

Ganzburg and Tikhonov's Theorems 1.4 and 1.5, doi:10.1007/s00365-016-9363-1, prove a lower-semicontinuity bound and the $q = \infty$ limit. A public note states the finite-q identity for real and complex constants, but that argument remains unchecked. The finite-q identity is therefore not treated here as an established answer. The finite-q note.

4.10. Four-digit accuracy of the monotone-delta constant

The upper bound is sharp "at least up to the first four significant digits."

The paragraph after Theorem 1, Section 1.2, p. 2, of Andrés Chirre, Dimitar K. Dimitrov, Emily Quesada-Herrera and Mateus Sousa, *An extremal problem and inequalities for entire functions of exponential type*, doi:10.1090/proc/16764, Proc. Amer. Math. Soc. 152 (2024). In the paper's notation, the question is whether $A_1 = \inf_f \|f\|_1/f(0)$, over nonnegative radially decreasing real entire functions of type at most 2π, agrees with the exhibited upper bound 1.27714 through the four significant digits requested by the authors.

Answered: yes. Evidence: Sketch.

A finite-section Gram estimate improves the lower bound to 1.2771345, while the published certified upper bound is 1.277135042105; together they determine the requested digits, without determining the exact constant.

Answers, "Four-digit accuracy of the monotone-delta constant", p. 7.

Standalone proof note.

4.11. Tangential Bernstein inequality on algebraic manifolds

"Prove that there is a constant C such that for any polynomial p"

Problem-session item 7, p. 389, in *Dirichlet Series and Function Theory in Polydiscs*, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014; proposed by Joaquim Ortega-Cerdà. The reconstructed question is whether, for every smooth compact algebraic manifold $M \subset \mathbb{R}^n$, there is C such that $\int_M |\nabla_t p|\, \mathrm{d}m \le C \deg(p) \int_M |p|\, \mathrm{d}m$ for all polynomials p, where ∇_t denotes the tangential gradient.

Answered: yes. Evidence: Published.

Berman and Ortega-Cerdà prove the L^q tangential-gradient Bernstein inequality for smooth compact real algebraic varieties; the $q = 1$ case is Theorem 9 of *Sampling of real multivariate polynomials and pluripotential theory*, doi:10.1353/ajm.2018.0019. Cauchy estimates reduce the gradient integral to the polynomial's mass in a complex tube of width $1/\deg p$. A plurisubharmonic three-circles estimate bounds that mass by $\deg(p)^{-\dim M}\|p\|_{L^1(M)}$.

5. Nehari theorems and Hankel forms

5.1. A bounded symbol for the multiplicative Hilbert functional

Is the symbol $\varphi_0(s) = 1 + \sum_{n=2}^{\infty} (\log n)^{-1} n^{-1/2-s}$ the Riesz projection of a function in $L^\infty(\mathbb{T}^\infty)$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 3.2, p. 4.

Answered: yes. Evidence: arXiv.

Perfekt, *A Fejér–Riesz inequality for Dirichlet series*, arXiv:2609.03855, Theorem 1.1 and Corollary 1.2 prove boundedness of the coefficient functional and hence existence of a bounded symbol. The proof localizes reproducing densities on rotation-invariant sets at geometric prime-curve scales; concentration estimates make distant scales disjoint, giving the absolute H^1 embedding. Hahn–Banach then supplies the symbol. This relies on his latest unrefereed preprint.

Conditional on: Karl-Mikael Perfekt, arXiv:2609.03855, latest version.

5.2. **Local L^q embeddings of the H^1 dual**

It is natural to ask if the situation is even worse, namely that $(H^1)^*$ fails to be contained in $H^q_i(\mathbb{C}_{1/2})$ for any $q > 2$.

Konyagin, Queffélec, Saksman, and Seip, *Riesz projection and bounded mean oscillation for Dirichlet series*, arXiv:2005.11951, §3.1, p. 9. Here $H^q_i(\mathbb{C}_{1/2})$ consists of analytic f for which $\sup_{\sigma>1/2} \int_{\mathbb{R}} |f(\sigma+it)|^q \,\mathrm{d}t/(1+t^2) < \infty$, and H^1 is the Dirichlet Hardy space.
Partial. Evidence: Sketch.

Konyagin, Queffélec, Saksman, and Seip, doi:10.4064/sm200601-22-5, §3.1, prove failure for $q > 4$; our argument extends this to $q > 7/2$. Whether failure holds throughout $2 < q \le 7/2$ remains open.

6. Composition operators

6.1. **The endpoint for a boundary circle restriction**

Source passage., arXiv:1602.03446 For the symbol $\varphi(s) = 3/2 - 2^{-s}$, Bayart and Brevig discuss the integration functional (34), then state: "Theorem 25. The functional L defined by (34) is bounded on H^p for any $p > 1$." The inferred endpoint question is whether C_φ is bounded on H^1; the quoted result concerns L, rather than C_φ. See Theorem 25 and the discussion preceding it in *Composition operators and embedding theorems for some function spaces of Dirichlet series*.

Bayart and Brevig, *Composition operators and embedding theorems for some function spaces of Dirichlet series*, Theorem 25, arXiv:1602.03446v2: their weighted Bergman embedding reduces boundedness of L to the coefficient summability of the representing function for integration, which holds when $p > 1$. This does not establish boundedness of the circle-restriction operator.

Here (34) is $L(f) = \int_{1/2}^{\infty} (f(s) - a_1)\,\mathrm{d}s$, with $a_1 = f(+\infty)$; the preceding discussion proves that boundedness of C_φ on H^1 implies boundedness of L on H^1.
Partial. Evidence: Published.

Bayart and Brevig prove boundedness of L for every $p > 1$ in Theorem 25. No boundedness theorem for C_φ at $p = 1$ is given there.

Part 2. Where Nehari's theorem survives: structured Helson matrices

7. Embeddings and boundary values

7.1. **Carleson classes on the infinite polydisc**

Theorem 22 invites the following question. Question. If μ defines an H^p-Carleson measure on $\mathbb{D}^\infty$ for some $0 < p < \infty$, does it also define an H^q-Carleson measure for every $p < q < \infty$?

Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, arXiv:1701.06897, §5, question after Theorem 22, p. 26. Here H^p-Carleson means $\int |F|^p \,\mathrm{d}\mu \le C\|F\|^p_{H^p(\mathbb{T}^\infty)}$ for every analytic polynomial F. Theorem 22 constructs μ by mapping product normalized area measure under $(z_1, z_2, \ldots) \mapsto (z_1, z_1, z_3, z_3, \ldots)$, and proves this estimate for even integers p, but not for $p < 2$.
Open. Evidence: Heuristic.

The cited Theorem 22 and the integer-power argument give specific even-integer and $p \to kp$ implications, but no general passage from an arbitrary H^p-Carleson measure to all larger exponents.

8. Nehari theorems and Hankel forms

8.1. **Intrinsic duals of Dirichlet Hardy spaces**

Describe $(\mathscr{H}^p)^*$ for p other than 2.

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.8; the preceding paragraph includes $0 < p < 1$.
Partial. Evidence: Published.

The $p = 2$ dual is explicit, and for $p \geq 1$ a quotient description follows from general duality. An intrinsic description for other exponents, including the quasi-Banach range, remains open. Konyagin, Queffélec, Saksman, and Seip, *Riesz projection and bounded mean oscillation for Dirichlet series*, doi:10.4064/sm200601-22-5, §3.1 and Corollary 3.1, give a concrete BMOA subclass of the $p = 1$ dual. They polarize the Littlewood–Paley identity and apply the BMOA Carleson measure estimate to bound the coefficient pairing.

8.2. **A Nehari counterpart on the infinite torus**

> What is the counterpart to Nehari's theorem on $\mathbb{T}^\infty$? In particular, what can be said about the Riesz projection of $L^\infty(\mathbb{T}^\infty)$ and other BMO-type spaces on $\mathbb{T}^\infty$?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 3.1.
Partial. Evidence: Published.

Konyagin, Queffélec, Saksman, and Seip in *Riesz projection and bounded mean oscillation for Dirichlet series*, doi:10.4064/sm200601-22-5, §3.1 and Corollary 3.1, embed half-plane BMOA in the dual and show that the dual is not contained in any H^q with $q > 2$. The inclusion follows from the polarized Littlewood–Paley identity and a Carleson estimate; the obstruction uses Fourier polytopes and circular Dirichlet kernels to show that the infinite-dimensional Riesz projection is unbounded from L^∞ to L^q for $q > 2$. A useful full symbol-space characterization is still missing.

8.3. **Positive Nehari theorem for Helson matrices**

> In general there are Carleson measures for $H^2(\mathbb{T}^\infty)$ which are not Carleson measures for $H^1(\mathbb{T}^\infty)$, but we do not know if these can be chosen to be concentrated on $(-1, 1)^\infty$.

Perfekt and Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, arXiv:1611.03772, question after Remark 4.11, p. 15. Their equivalent bounded-symbol formulation is also answered by the construction.
Answered: yes. Evidence: Sketch.

A finite real-cube measure has bounded H^2 embedding and unbounded absolute H^1 embedding. A positive polynomial test detects the strict two-atom gain, and disjoint tensor blocks amplify it.
Answers, "Positive Nehari theorem for Helson matrices", p. 8.

The proof

8.4. **Carleson moment representations of bounded Helson matrices**

> We do not know the answer to the following. Question 4.6. Let $M(\alpha)$ be bounded on $\ell^2(\mathbb{N})$. Does there exist a Carleson measure μ for $H^2(\mathbb{T}^\infty)$ such that α is represented as the moment sequence of μ, as in (4.4)?

Perfekt and Pushnitski, *On Helson matrices: moment problems, non-negativity, boundedness, and finite rank*, arXiv:1611.03772, Question 4.6, p. 12. Here $M(\alpha) = (\alpha(nm))_{n,m\geq 1}$, and (4.4) means $\alpha(p^\kappa) = \int_{\mathbb{D}^\infty} z^\kappa \, \mathrm{d}\mu(z)$, where $p^\kappa = \prod_j p_j^{\kappa_j}$ for the primes p_j; the Carleson condition is $\int |f|^2 \, \mathrm{d}|\mu| \leq C\|f\|_{H^2}^2$.
Partial. Evidence: Sketch.

Finite-rank cases and several weighted ℓ^2 coefficient classes admit such representations. The general question allows complex measures on the open product disc and remains unresolved.

8.5. **Necessity of the apicality criteria**

It is natural to ask which, if any, of these two conditions may also be necessary for φ not to be saturated.

Brevig and Seip, *Maximal norm Hankel operators*, arXiv:2301.07937, §3, after Problem 1. The conditions are apicality of $|\varphi|$ at some level and nonempty interior of the low-modulus set. Here saturation means $\|H_\varphi\| = \|\varphi\|_\infty$ for $H_\varphi f = P_+(\varphi\overline{f})$; the conditions require $\log|\varphi| = u + v$ with real $u \in L^\infty$, $v \in L^1$ and $\operatorname{ess\,sup}_{\{|\varphi|\geq\delta\|\varphi\|_\infty\}} |\widetilde{v}| < \pi/2$, and nonempty interior of $\{|\varphi| < \delta\|\varphi\|_\infty\}$ for some $0 < \delta < 1$.

Answered: no. Evidence: Sketch.

Neither condition is necessary: separate constructions show that neither is required for nonsaturation. A full written proof appears in the answers entry.

Answers, "Are the apicality criteria necessary for nonsaturation?", p. 9.

8.6. **Classifying saturated outer functions**

Describe the outer functions in H^∞ that are saturated.

Brevig and Seip, *Maximal norm Hankel operators*, arXiv:2301.07937, Problem 1. Saturation means equality in their equation (4), $\|H_\varphi\| = \|\varphi\|_\infty$.

Partial. Evidence: Published.

Their Theorems 4–6 give criteria and substantial classes, including apical weights, but not a complete description (doi:10.1016/j.jmaa.2023.127221). Their nonsaturation proof constructs an H^∞_0 perturbation decreasing the distance of $\overline{\varphi}$ to H^∞_0, using conjugate control on the high-modulus set. The necessity question above is separate from the original classification problem.

8.7. **Power invariance of apical weights**

Is it true that w is an apical Helson–Szegő weight if and only if w^κ is an apical Helson–Szegő weight for all positive numbers κ? In other words, we ask if (7) could be replaced simply by (27) $\operatorname{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \infty$.

Brevig and Seip, *Maximal norm Hankel operators*, arXiv:2301.07937, Problem 2 and condition (27). Here $L^+_w(\delta) = \{w \geq \delta\|w\|_\infty\}$ for $0 < \delta < 1$, $\widetilde{v}$ is the harmonic conjugate of v, and $w = \exp(u + v)$ with real $u \in L^\infty$, $v \in L^1$; (7) requires $\operatorname{ess\,sup}_{L^+_w(\delta)} |\widetilde{v}| < \pi/2$, whereas (27) requires only finiteness.

Answered: yes. Evidence: Sketch.

The finite localized conjugate bound can be reduced below any prescribed threshold by an analytic truncation. Brevig and Seip's criterion in *Maximal norm Hankel operators*, doi:10.1016/j.jmaa.2023.127221, Theorem 5 and §3, then applies to every positive power.

Answers, "Power invariance of apical weights", p. 9.

The proof

8.8. **Norm-attaining vectors in complete Pick spaces**

It would be interesting to know what can be said of the norm attaining vectors and Hilbert points in this context.

Bampouras and Brevig, *Norm attaining vectors and Hilbert points*, arXiv:2310.20346, paragraph after Theorem 4, p. 4. The scoped question is whether every $\mathcal{H}$-extremal vector is norm attaining. Here $\mathcal{H}$ is a reproducing-kernel Hilbert space with complete Pick kernel normalized by $k_{x_0} = 1$; a nonzero f is norm attaining when $\|f\|_{\mathcal{H}\odot\mathcal{H}} = \|f\|_{\mathcal{H}}$, and extremal when $\langle qf, f\rangle = q(x_0)\|f\|^2_{\mathcal{H}}$ for every multiplier q.

Answered: yes in separable normalized complete Pick spaces. Evidence: Sketch.

For normalized separable complete Pick spaces, norm attainment is equivalent to the $\mathcal{H}$-extremal identity. In H^2, this says precisely that the boundary modulus is constant; the proof is in the answers entry.

Answers, "Norm-attaining vectors in complete Pick spaces", p. 10.

Full proof

8.9. **A product Hardy-space Nehari theorem**

Does every bounded small Hankel operator on $H^2(\mathbb{C}_+^2)$ have a symbol in product BMO?

Bampouras and Perfekt, *Recovering Product BMO from Schatten Hankel Operators*, arXiv:2604.26722, Introduction, equation (1).

Open. Evidence: Heuristic.

Bampouras and Perfekt prove a product-BMO conclusion for Schatten-class operators with finite exponent in Theorem 1.1 of arXiv:2604.26722; their result does not cover every bounded operator. The endpoint question remains unresolved.

Conditional on: Bampouras and Perfekt's unrefereed public preprint.

8.10. **An intrinsic description of the analytic H^1 dual**

Give a concrete intrinsic description of all analytic symbols in $(\mathscr{H}^1)^*$.

Søvik, the book's open question in the H^1 discussion.

Partial. Evidence: Sketch.

BMOA symbols and conditional-square subclasses provide descriptions in special cases, while Hahn–Banach only gives a nonintrinsic quotient representation. A general concrete characterization remains open. Book discussion

8.11. **Products of Hankel operators in Schatten classes**

An interesting and important problem is to describe functions φ and ψ in L^2 for which $H_\varphi^* H_\psi \in S_p$; this problem is still open.

Vladimir V. Peller, *Hankel Operators and Their Applications*, doi:10.1007/978-0-387-21681-2, Chapter 6, Concluding Remarks, printed p. 301. Our scoped question concerns bounded extension of the cross form on analytic polynomials.

Partial. Evidence: Sketch.

The sheet gives a necessary and sufficient criterion in terms of a bounded cross form for a substantial class, but does not settle all pairs or all possible composition domains. A full joint symbol characterization remains open.

8.12. **Recovering positive Fourier coefficient powers**

The very interesting problem of whether (12.1) holds for other values of $s \geq 1/2$ is still open.

Vladimir V. Peller, *Hankel Operators and Their Applications*, doi:10.1007/978-0-387-21681-2, Chapter 7, §12, after Corollary 12.3, printed p. 373. Formula (12.1) asks whether for every complex finite measure μ on $\mathbb{T}$ whether $\sum_{n\geq 1} |\widehat{\mu}(-n)|^{2s}/n < \infty$ implies $\sum_{n\geq 1} |\widehat{\mu}(n)|^{2s}/n < \infty$.

Partial. Evidence: Published.

The source proves the implication for positive integer s in Corollary 12.3: apply the square-energy symmetry of Corollary 12.2 to μ^{*s}, whose Fourier coefficients are $\widehat{\mu}(n)^s$. Other exponents, including the endpoint $s = 1/2$, remain undetermined.

8.13. **Compact and noncompact inverse Hankel choices**

It is unknown if it is possible to choose $q \notin \operatorname{Range} R$ so that $I - Q$ is compact.

It is also unclear whether it is possible to find q so that $I - Q$ is noncompact.

Megretskii, Peller, and Treil, *The Inverse Spectral Problem for Self-Adjoint Hankel Operators*, doi:10.1007/BF02392468, §III.7, p. 295; repeated by Peller, *Hankel Operators and Their Applications*, doi:10.1007/978-0-387-21681-2, Chapter 12, §13, p. 537. The question is for each admissible operator in the stated construction. Here $W = |\Gamma|$ in the source spectral model, J is its commuting unitary involution, $R = JW$, and Q is the contraction satisfying $QW^{1/2} = |A|^{1/2}$, where $A^2 = W^2 - (\cdot, Rq)Rq$ for an admissible unit vector q; I denotes the identity.

Answered: both choices exist. Evidence: Sketch.

Rescaling the spectral bands of an admissible vector yields a compact choice and, separately, a noncompact one while preserving the construction's hypotheses. The full proof is in the answers entry.

Answers, "Compact and noncompact inverse Hankel choices", p. 11.

Full proof

8.14. Simon's Schatten majorization property

Simon [2] conjectured that B_p has the majorization property for $p \in [2, \infty)$.

Peller, *Hankel Operators and Their Applications*, doi:10.1007/978-0-387-21681-2, Chapter 6, Concluding Remarks, printed p. 300. Here [2] is Barry Simon, *Trace Ideals and Their Applications* (Cambridge, 1979), and $B_p = S_p$ is the class of compact operators with p-summable singular values. The property is the implication from pointwise domination $|Tf| \le R|f|$ to $T \in S_p$ whenever $R \in S_p$.

Answered: no for non-even finite p. Evidence: Published.

Peller's counterexamples disprove the conjecture for every finite $p \ge 2$ that is not an even integer; the even-integer cases hold. Peller, *Hankel operators of class S_p and their applications*, *Math. USSR-Sb.* 41 (1982), doi:10.1070/SM1982v041n04ABEH002242, constructs analytic symbols with coefficientwise domination but opposite S_p membership; his Hankel–Besov criterion detects the failure. The monograph records this construction in Chapter 6, Concluding Remarks.

8.15. Generation of Douglas algebras

Is every closed subalgebra of L^∞ containing H^∞ generated by H^∞ and the complex conjugates of inner functions?

Douglas's conjecture, stated by Donald Sarason in *Functions of vanishing mean oscillation*, doi:10.1090/S0002-9947-1975-0377518-3, §4, after Corollary 2, printed p. 401.

Answered: yes. Evidence: Published.

Marshall, *Subalgebras of L^∞ containing H^∞*, doi:10.1007/BF02392414, Theorem 2, gives the generation theorem, using Chang, *A characterization of Douglas subalgebras*, doi:10.1007/BF02392413, Theorem 7. Marshall constructs interpolating Blaschke products to match the maximal ideal space of the generated Douglas algebra to the original algebra; Chang proves that this match forces equality.

8.16. Continuity of the modulus of a Poisson extension

Let f be a function in L^∞ with the property that $|f|$ is continuous in $\overline{D}$. Then f is in QC.

Donald Sarason, *Functions of vanishing mean oscillation*, doi:10.1090/S0002-9947-1975-0377518-3, Theorem 3, §4, printed p. 398. This gives the affirmative answer to the implicit question whether the modulus of the Poisson extension of f extending continuously to the closed disc forces $f \in QC = VMO \cap L^\infty$.

Answered: yes. Evidence: Published.

Sarason's Theorem 3 proves that this continuity forces f into QC. After a continuous normalization of the boundary modulus, Poisson averages near a maximum force the boundary values to concentrate near a single unimodular value. This bounds the oscillation on shrinking arcs and yields $f \in VMO \cap L^\infty$.

8.17. A converse Jensen-measure criterion

An open problem is to obtain a converse to Lemma 5.1, characterizing $\sup_{\tau \in J_\phi} \|\tau_a\|$ in terms of A-subharmonic functions.

Cole and Gamelin, *Representing Measures and Hardy Spaces for the Infinite Polydisk Algebra*, doi:10.1112/plms/s3-53.1.112, §5, after Lemma 5.1, printed p. 119. Here J_ϕ is the set of Jensen measures and τ_a is the part absolutely continuous with respect to the fixed

probability measure. Lemma 5.1 says that if A is a uniform algebra on X, $\phi \in M_A$ is a character, μ is a probability measure, and A-subharmonic functions u_n satisfy $u_n \le 0$ on M_A, $\limsup_n u_n \le -1$ μ-almost everywhere and $u_n(\phi) \to -c$ with $c > 0$, then $\|\tau_a\| \le c$ for every Jensen measure τ for ϕ. Jensen measures satisfy $u(\phi) \le \int u \, d\tau$ for all A-subharmonic u, and τ_a is their absolutely continuous part relative to μ.
Answered: yes. Evidence: Sketch.

A real Hahn–Banach separation identifies the envelope with the supremum over Jensen measures; truncating their densities and applying minimax gives the sharp mass criterion. The full proof is in the answers entry.
Answers, "A converse Jensen-measure criterion", p. 13.

Full proof

8.18. **Weak factorization on the unit ball**

Is every $f \in H^1(\mathbb{B}_n)$, $n > 1$, a product gh with $g, h \in H^2(\mathbb{B}_n)$?

Hayman and Lingham, *Research Problems in Function Theory*, arXiv:1809.07200, 50th Anniversary Edition, Update 8.16, p. 193; the update attributes the ball problem to Rudin.
Answered: no. Evidence: Published.

Gowda, *Nonfactorization theorems in weighted Bergman and Hardy spaces on the unit ball of* $\mathbb{C}^n$ $(n > 1)$, doi:10.1090/S0002-9947-1983-0690048-9, Theorem 2, shows that for $n > 1$ the products $H^2(\mathbb{B}_n)H^2(\mathbb{B}_n)$ form a first-category subset of $H^1(\mathbb{B}_n)$. The proof shows that the product map cannot be open at the origin, using polynomial norm estimates and ball automorphisms. Normal families make bounded-factor product sets closed, and the same obstruction gives them empty interior; their countable union is therefore first category.

8.19. **A BMO route to Lindelöf for Dirichlet series**

Is there a BMO-type space of Dirichlet series between H^∞ and the symbols of bounded Hankel forms, such that every member has finite vertical p-means and $\zeta_1(\sigma + \cdot)$ belongs to it for every $\sigma > 1/2$?

McCarthy, *The Lindelöf Hypothesis and BMO*, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, Problem session, item 2, pp. 384–385. Here $\zeta_1(s) = \sum_{n\ge1}(-1)^n n^{-s}$, and finite vertical p-means means $\limsup_{T\to\infty}(2T)^{-1}\int_{-T}^{T}|f(\sigma+it)|^p\,dt < \infty$ for every $\sigma > 0$ and finite $p > 0$.
Partial. Evidence: Sketch.

McCarthy explains that neither endpoint candidate, $BMO_1 = (H^1)^*$ nor BMO_2, has the required all-moments property. Our tests rule out natural rotation-invariant BMO candidates; existence of a different space is unresolved.

8.20. **A Nehari theorem for multiplicative Hankel forms**

Does every bounded multiplicative Hankel form have a bounded symbol ψ on the polytorus $\mathbb{T}^\infty$?

Brevig and Perfekt, *Failure of Nehari's theorem for multiplicative Hankel forms in Schatten classes*, doi:10.4064/sm228-2-1, Introduction, p. 1, quoting Helson's question in *Dirichlet Series*, pp. 52–54; Helson also discusses the lifting question in *Hankel forms and sums of random variables*, doi:10.4064/sm176-1-6, §1.
Answered: no. Evidence: Published.

Ortega-Cerdà and Seip construct bounded forms for which the least finite-dimensional symbol norms grow without bound, ruling out a bounded symbol on the infinite torus. Their unnumbered theorem in *A lower bound in Nehari's theorem on the polydisc*, arXiv:1107.0175, doi:10.1007/s11854-012-0038-y, uses $\prod_j(z_{2j-1}+z_{2j})$: its Hankel norm is $2^{d/4}$ while testing the coefficient functional gives norm at least $(\pi/2)^{d/2}$ for even d. The growing ratio precludes an infinite-dimensional bounded-symbol theorem.

8.21. **Contractivity of diagonal restriction**

Numerical estimation of the norms, with up to fifth degree polynomials for $g(w)$ and $(D_d f)(w)$, suggests the following conjectures: $\|D_d\|_p = 1$, $p \geq 2$, and $\|E_d\|_p > 1$, $p \neq 2$.

Birk D. Ramberg, *Nehari's theorem for multiplicative Hankel matrices*, https://www.math.ntnu.no/studforsk/ramberg_nehari.pdf, numerical conjectures before Theorem 2, printed p. 6. Here $d \geq 2$ and $0 < p < \infty$, $D_d f(w) = f(w, \dots, w)$ maps $H^p(\mathbb{T}^d)$ to the weighted Bergman space A^p_d with probability density $(d-1)(1-|w|^2)^{d-2}\,\mathrm{d}A(w)/\pi$, and $E_d(\sum_n a_n w^n) = \sum_n a_n \binom{n+d-1}{n}^{-1} \sum_{|\alpha|=n} z^\alpha$ is its canonical right inverse; the subscripts p denote operator norms.

Partial. Evidence: arXiv.

The public NTNU institutional report proves contractivity for $p = 2$ and all positive even integers. It remains a public institutional report, not an arXiv posting; the general range $p \geq 2$ is open. Degree-group Cauchy–Schwarz proves the $p = 2$ estimate, and applying it to f^k gives $p = 2k$. This is also Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, doi:10.1090/tran/7290, Lemma 18.

Conditional on: Birk D. Ramberg's unrefereed NTNU institutional report.

8.22. **Strict noncontractivity of the diagonal right inverse**

Numerical estimation of the norms, with up to fifth degree polynomials for $g(w)$ and $(D_d f)(w)$, suggests the following conjectures: $\|D_d\|_p = 1$, $p \geq 2$, and $\|E_d\|_p > 1$, $p \neq 2$.

Birk D. Ramberg, *Nehari's theorem for multiplicative Hankel matrices*, https://www.math.ntnu.no/studforsk/ramberg_nehari.pdf, numerical conjectures before Theorem 2, printed p. 6. Here $d \geq 2$ and $0 < p < \infty$, $D_d f(w) = f(w, \dots, w)$ maps $H^p(\mathbb{T}^d)$ to the weighted Bergman space A^p_d with probability density $(d-1)(1-|w|^2)^{d-2}\,\mathrm{d}A(w)/\pi$, and $E_d(\sum_n a_n w^n) = \sum_n a_n \binom{n+d-1}{n}^{-1} \sum_{|\alpha|=n} z^\alpha$ is its canonical right inverse; the subscripts p denote operator norms.

Answered: yes. Evidence: arXiv.

The public NTNU institutional report proves strict noncontractivity for every $p \neq 2$. It is an institutional report, not an arXiv posting. In Theorem 2, the report tests $g(w) = 1 + \lambda w$ and $E_d g = 1 + (\lambda/d)\sum_j z_j$: their constant and quadratic norm terms agree, while the quartic difference is strictly positive for every finite $p \neq 2$.

Conditional on: Birk D. Ramberg's unrefereed NTNU institutional report.

8.23. **A bounded symbol for Hilbert–Schmidt forms**

Helson proved that any Hankel form on $H^2 \times H^2$ which is of Hilbert–Schmidt class S_2 is induced by a bounded symbol on the infinite polytorus $\mathbb{T}^\infty$.

Brevig and Perfekt, *Weak product spaces of Dirichlet series*, arXiv:1510.02019, §2. They identify the corresponding question as Helson's Question 3 in *Dirichlet Series*, p. 54; the exact wording of that question was not available for verification. The inferred target is whether every Hilbert–Schmidt multiplicative Hankel form has a bounded symbol.

Answered: yes. Evidence: Published.

Helson, *Hankel forms and sums of random variables*, doi:10.4064/sm176-1-6, the unnumbered theorem in §2, proves the bounded-symbol assertion for every Hilbert–Schmidt multiplicative Hankel form. The exact wording in the original book was unavailable locally; the source and location are verified through Helson's citation and answer. Iterating the one-variable coefficient inequality gives $\sum_n |a_n|^2/d(n) \leq \|f\|^2_{H^1}$, where $d(n)$ counts divisors. Cauchy–Schwarz pairs this with the Hilbert–Schmidt condition $\sum_n d(n)|\rho_n|^2 < \infty$, and Hahn–Banach produces a bounded symbol.

9. Hilbert points

9.1. Equal-modulus coefficients of Hilbert points

Suppose that $1 \le p < 2$. If $\varphi(z) = \sum_{j=1}^{d} c_j z_j$ is a Hilbert point in $H^p(\mathbb{T}^d)$, then the nonzero coefficients of φ all have the same modulus.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, arXiv:2106.07532, Conjecture 3.1. The source poses this question for $d \ge 3$. A Hilbert point is a nonzero φ such that $\|\varphi\|_{H^p} \le \|\varphi + f\|_{H^p}$ for every analytic f orthogonal to φ; below $p = 2$, orthogonality means the H^p closure of orthogonal analytic polynomials.

Answered: no. Evidence: Sketch.

A three-variable example violates the equal-modulus conclusion. The public remarks give a full proof of the counterexample. The counterexample

9.2. Unbounded Hilbert points below and above two

We do not know, for instance, whether there exists an unbounded Hilbert point in $H^p(\mathbb{T}^d)$ for some $p \neq 2$.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, arXiv:2106.07532, §4, paragraph following Question 4.1, p. 22. The curated target asks for a finite dimension and exponent with such an example. A Hilbert point is a nonzero φ such that $\|\varphi\|_{H^p} \le \|\varphi + f\|_{H^p}$ for every analytic f orthogonal to φ; below $p = 2$, orthogonality means the H^p closure of orthogonal analytic polynomials.

Partial. Evidence: Sketch.

Finite-dimensional tests and a bootstrap rule out several candidate ranges and structures. They do not decide whether an unbounded Hilbert point exists in general.

9.3. A cubic Hilbert-point exponent question

Numerical evidence suggests that this φ is a Hilbert point in $H^p(\mathbb{T}^3)$ only when $p = 2, 4$. Unfortunately, we are only able to verify analytically that there is possibly a finite number of p in $[1, 4) \setminus \{2\}$ for which φ is a Hilbert point in $H^p(\mathbb{T}^3)$.

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, arXiv:2106.07532, discussion following Question 4.1, p. 19. The curated target is whether the displayed cubic is a Hilbert point at any exponent in the source's remaining range. A Hilbert point is a nonzero φ such that $\|\varphi\|_{H^p} \le \|\varphi + f\|_{H^p}$ for every analytic f orthogonal to φ; below $p = 2$, orthogonality means the H^p closure of orthogonal analytic polynomials. The cubic in the quote is $\varphi(z) = z_1^3 + z_2^3 + z_1 z_2 z_3$.

Partial. Evidence: Sketch.

Our argument excludes the displayed cubic for exponents sufficiently close to 2, with the threshold depending on dimension. The source's broader remaining exponent range is unresolved.

9.4. Dynamics of the nonlinear Hilbert-point projection

Suppose that $\varphi_0(z) = \sum_{j=1}^{d} c_j z_j$ is in the unit sphere of $H^2_1(\mathbb{T}^d)$ and that $c_1 > c_j \ge 0$ for every $j = 2, \dots, d$. Is it true that $\lim_{n\to\infty}(P_p^n \varphi_0)(z) = z_1$ whenever $1 \le p < 2$?

Brevig, Ortega-Cerdà, and Seip, *Hilbert points in Hardy spaces*, arXiv:2106.07532, Question 4.1, p. 15. A Hilbert point is a nonzero φ such that $\|\varphi\|_{H^p} \le \|\varphi + f\|_{H^p}$ for every analytic f orthogonal to φ; below $p = 2$, orthogonality means the H^p closure of orthogonal analytic polynomials. Here H^2_1 is the space of 1-homogeneous analytic polynomials and $P_p\varphi = P(|\varphi|^{p-2}\varphi)/\|P(|\varphi|^{p-2}\varphi)\|_2$, where P is the analytic Riesz projection.

Partial. Evidence: Sketch.

Our argument proves convergence for p sufficiently near 2, with a threshold depending on d. The full range $1 \le p < 2$ in the source question remains unresolved.

10. Fock-space spectral optimization

10.1. **Nearly invariant subspaces of Fock-type spaces**

Is it true that any nearly invariant subspace without common zeros in a Fock-type space

Aleman, Baranov, Belov, and Hedenmalm, *Backward shift and nearly invariant subspaces of Fock-type spaces*, arXiv:2007.06107, final problem in the Introduction, p. 6. The complete question asks whether each such subspace of $\mathcal{F}^p_\alpha$ is one-generated, of the form $\mathcal{M}_G$; the paper defines these spaces with radial weights. Here $\mathcal{F}^p_\alpha$ consists of entire f with $\int_{\mathbb{C}} |f(z)|^p e^{-a|z|^\alpha}\, \mathrm{d}A(z) < \infty$, $a, \alpha > 0$. Nearly invariant means $f \in \mathcal{M}$ and $f(0) = 0$ imply $f/z \in \mathcal{M}$. For simple zeros of $G \in \mathcal{F}^p_\alpha$, $\mathcal{M}_G = \overline{\operatorname{span}}\{G(z)/(z-\lambda) : G(\lambda) = 0\}$; zeros of multiplicity m contribute the quotients with powers $1, \dots, m$. The source's Theorem 1.3 and Corollary 1.4 give ordering and the polynomial classification at zero exponential type. The proof uses annihilating functionals and Cauchy transforms to produce entire quotients; a Phragmén–Lindelöf argument forces comparability of the subspaces.

Partial. Evidence: arXiv.

The source proves ordering results in slow-growth regimes. The general radial Fock-type case remains without a verified answer in the checked literature.

10.2. **Uniqueness of discs for Ky Fan sums**

Which sets maximize the sum

Nicola, Riccardi, and Tilli, *The isoperimetric inequality for partial sums of Toeplitz eigenvalues in the Fock space*, arXiv:2503.07069, Introduction. They ask which sets of prescribed area maximize the Ky Fan sum $\sum_{n=1}^N \lambda_n(T_\Omega)$. Abreu's later Theorem 1 proves disc optimality, while Remark 1 says that uniqueness up to translations and null sets was left open. Here $T_\Omega F = P_{\mathcal{F}}(\mathbf{1}_\Omega F)$ on the entire-function space with norm squared $\int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\, \mathrm{d}A(z)$, and the λ_n are its eigenvalues in decreasing order. Abreu, *The isoperimetric inequality for the Ky Fan norm*, arXiv:2607.28376v1, Theorem 1 and Remark 1, gives the stated optimality without uniqueness. The argument maximizes over rank-N projections, applies Husimi-function majorization at a fixed threshold, and then uses the bathtub principle to compare with a disc.

Partial. Evidence: arXiv.

Conditional on: arXiv:2607.28376v1.

The disc is a maximizer for every Ky Fan sum, according to an unrefereed preprint. Whether every maximizer is a translate of a disc, up to null sets, remains open.

10.3. **Maximizing a single higher Fock Toeplitz eigenvalue**

which sets maximize the second eigenvalue

Nicola, Riccardi, and Tilli, *The isoperimetric inequality for partial sums of Toeplitz eigenvalues in the Fock space*, arXiv:2503.07069v2, Introduction. The source asks which sets $\Omega \subset \mathbb{C}$ of prescribed area maximize $\lambda_2(T_\Omega)$; it also raises the analogous question for a single λ_k, $k \geq 2$. Here $T_\Omega F = P_{\mathcal{F}}(\mathbf{1}_\Omega F)$ in the Gaussian Fock space with norm squared $\int_{\mathbb{C}} |F(z)|^2 e^{-\pi|z|^2}\, \mathrm{d}A(z)$, and λ_k is the kth eigenvalue in decreasing order. Theorem 1.3 and Proposition 1.4 of the linked paper give the radial optimum and symmetry breaking. Circular symmetry diagonalizes the operator in monomials, and the bathtub principle selects an annulus; for sufficiently small area, two widely separated discs give a larger second eigenvalue.

Partial. Evidence: arXiv.

Conditional on: arXiv:2503.07069.

For radial sets, an annulus is optimal for λ_2, but nonradial sets can improve on that radial optimizer. A global characterization for λ_2, and the analogous questions for higher single eigenvalues, remain open.

11. Hankel forms and Schatten classes

11.1. Explicit diagonalization of a multiplicative Hilbert matrix

Despite the similarity with Theorem A, we do not have an explicit diagonalisation of either $M(g_a)$ or $M_2(g_0)$; to find one is an interesting open problem.

Perfekt and Pushnitski, *On the spectrum of the multiplicative Hilbert matrix*, arXiv:1705.01959, Introduction, paragraph after Theorem 1.2. The known spectral description is abstract and does not give an explicit diagonalizing transform. Here $M(g_a) = (g_a(mn))_{m,n\geq 1}$ with $g_a(n) = 1/(\sqrt{n}(a + \log n))$, $a > 0$, and $M_2(g_0) = ((\sqrt{mn}\log(mn))^{-1})_{m,n\geq 2}$. Theorem A describes the classical additive Hilbert matrix, including its explicit spectral transform.

Open. Evidence: Heuristic.

Perfekt–Pushnitski Theorems 1.1–1.2 give $M_2(g_0)$ purely absolutely continuous spectrum $[0, \pi]$. The shifted $M(g_a)$ has the same absolutely continuous part and, for $0 < a < a_*$, one simple eigenvalue above π; for $a \geq a_*$ it has no eigenvalues, where $a_* > 0$ is the source's critical parameter. No explicit kernel for either requested transform was found in the checked sources. The shifted and unshifted operators are both part of the question.

11.2. Eigenvalues above the critical band for sparse prime sets

eigenvalues above Λ_α

Ma, Lu, and Zu, *On the Spectra of Multiplicative Hilbert Operator on Bergman Spaces of Dirichlet Series*, arXiv:2608.22882v3, Theorem 1.1(5), p. 3. The inferred extension asks whether a critical Hardy matrix on a regular sparse-prime semigroup can have eigenvalues above its essential band or embedded in it. This is our question, motivated by the cited Bergman theorem, not a question stated by the authors. In the source, $\Lambda_\alpha = \Gamma(\alpha/2)^2/\Gamma(\alpha)$ is the essential spectral edge of the weighted Bergman operator T_α.

Answered: depends on the prime set. Evidence: Sketch.

Finite deletions can force arbitrarily many eigenvalues above the band, whereas for primes congruent to 1 modulo 4 the norm equals the band edge and no above-band eigenvalue occurs. Embedded eigenvalues are excluded under an explicit C^2 regularity hypothesis. The complete argument is in the book.

Answers, "Sparse-prime eigenvalues above the critical band", p. 14.

Book draft, eigenvalues of critical prime-set matrices section

11.3. The unimodality of a multiplicative Hilbert norm

Does there exist a unique $\alpha_{\min} > 1$ such that $\|T_\alpha\|$ is strictly decreasing on $(1, \alpha_{\min})$ and strictly increasing on $(\alpha_{\min}, \infty)$?

Ma, Lu, and Zu, *On the Spectra of Multiplicative Hilbert Operator on Bergman Spaces of Dirichlet Series*, arXiv:2608.22882v3, §6.2, Question 1, p. 73 (version 3). Here T_α acts on Dirichlet series $f(s) = \sum_{n\geq 2} a_n n^{-s}$ with $\|f\|^2_{\mathcal{A}^2_{\alpha,0}} = \sum_{n\geq 2} |a_n|^2/d_\alpha(n)$, where $\zeta(s)^\alpha = \sum_{n\geq 1} d_\alpha(n)n^{-s}$, by

$$(T_\alpha f)(s) = \int_{1/2}^{\infty} f(t)(\zeta(s + t)^\alpha - 1)(t - 1/2)^{\alpha-1}\,\mathrm{d}t.$$

The band edge is $\Lambda_\alpha = \Gamma(\alpha/2)^2/\Gamma(\alpha)$, and $\alpha_* \in (1, 2)$ is the unique parameter above which eigenvalues occur.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2608.22882v3.

An interval-certified argument gives a unique minimum with $\alpha_{\min} \in (8/5, 3)$. The proof uses the source's kernel model, essential-spectrum description, and convexity theorem.

Answers, "Unimodality of a multiplicative Hilbert norm", p. 15.

11.4. **An explicit norm formula for the multiplicative Hilbert operator**

Can $\|T_\alpha\|$ be computed explicitly for $\alpha > \alpha_*$?

Ma, Lu, and Zu, *On the spectra of multiplicative Hilbert operator on Bergman spaces of Dirichlet series*, arXiv:2608.22882v3, §6.2, Question 2, p. 74 (version 3, 9 October 2026). Here T_α acts on Dirichlet series $f(s) = \sum_{n\geq 2} a_n n^{-s}$ with $\|f\|^2_{\mathcal{A}^2_{\alpha,0}} = \sum_{n\geq 2} |a_n|^2/d_\alpha(n)$, where $\zeta(s)^\alpha = \sum_{n\geq 1} d_\alpha(n) n^{-s}$, by

$$(T_\alpha f)(s) = \int_{1/2}^{\infty} f(t)(\zeta(s+t)^\alpha - 1)(t-1/2)^{\alpha-1}\,\mathrm{d}t.$$

The band edge is $\Lambda_\alpha = \Gamma(\alpha/2)^2/\Gamma(\alpha)$, and $\alpha_* \in (1,2)$ is the unique parameter above which eigenvalues occur. The spectral description is Theorem 1.1 of the linked paper. Its proof factors T_α through a weighted integral Hankel operator and compares its boundary kernel with a homogeneous kernel diagonalized by the Mellin transform.

Open. Evidence: arXiv.

The paper shows that this norm is the largest eigenvalue above the threshold, but gives no explicit formula for that eigenvalue.

11.5. **An explicit diagonalization below the spectral threshold**

For $1 < \alpha \leq \alpha_*$, can one construct explicitly a unitary operator $U_\alpha : \mathcal{A}^2_{\alpha,0} \longrightarrow L^2((0,\Lambda_\alpha), \mathrm{d}\lambda)$ such that $(U_\alpha T_\alpha U_\alpha^{-1} g)(\lambda) = \lambda g(\lambda)$?

Ma, Lu, and Zu, *On the spectra of multiplicative Hilbert operator on Bergman spaces of Dirichlet series*, arXiv:2608.22882v3, §6.3, Question 3, p. 74 (version 3, 9 October 2026). Here T_α acts on Dirichlet series $f(s) = \sum_{n\geq 2} a_n n^{-s}$ with $\|f\|^2_{\mathcal{A}^2_{\alpha,0}} = \sum_{n\geq 2} |a_n|^2/d_\alpha(n)$, where $\zeta(s)^\alpha = \sum_{n\geq 1} d_\alpha(n) n^{-s}$, by

$$(T_\alpha f)(s) = \int_{1/2}^{\infty} f(t)(\zeta(s+t)^\alpha - 1)(t-1/2)^{\alpha-1}\,\mathrm{d}t.$$

The band edge is $\Lambda_\alpha = \Gamma(\alpha/2)^2/\Gamma(\alpha)$, and $\alpha_* \in (1,2)$ is the unique parameter above which eigenvalues occur. Theorem 1.1 of the linked paper obtains the spectral interval by Mellin comparison and scattering; this determines the spectral type without an explicit transform for the exact zeta kernel.

Open. Evidence: arXiv.

The paper identifies the spectrum abstractly below α_*, but does not construct an explicit unitary map to multiplication on $L^2((0,\Lambda_\alpha))$.

11.6. **Closability and maximal realization in infinitely many variables**

Under the Carleman condition, closability of h_0 is equivalent to

Ma, Lu, and Zu, *Unbounded Helson forms with a moment representation*, arXiv:2610.12252, Theorems 1.1–1.2. The inferred extension asks whether the closability criterion $\sqrt{n}\,\alpha(n) \to 0$ and the maximal-realization conclusion extend to product moment measures on $(-1,1)^\infty$. This is our question, not one stated by the authors. Here $h_0[a,b] = \sum_{m,n\geq 1} \alpha(mn) a_n \overline{b_m}$ on finitely supported sequences, with $\alpha(n) = \int_{\mathbb{R}} n^{-s}\,\mathrm{d}\mu(s)$ for a positive measure μ. The source's Carleman condition is $\sum_{k\geq 1} \alpha(4^k)^{-1/(2k)} = \infty$.

Partial. Evidence: Sketch.

For independent coordinates, a parity-block calculation gives an exact closability criterion and product norm. In a natural class the maximal map is closed, but equality of its domain with the polynomial graph closure remains open; a support-only criterion is insufficient.

11.7. **Spectral information from unbounded Helson block profiles**

Moreover, H_μ is trace class if and only if $\sum_{n\geq 1} \alpha(n^2) < \infty$.

Ma, Lu, and Zu, *Unbounded Helson forms with a moment representation*, arXiv:2610.12252, §3. The inferred question is whether spectral type and compact-resolvent behavior can be classified from block profiles of the representing measure. This extension is not stated as a question in the source. Here $\alpha(n) = \int_{\mathbb{R}} n^{-s}\,\mathrm{d}\mu(s)$, and H_μ is the positive operator associated with the closure of $h_0[a,b] = \sum_{m,n\geq 1} \alpha(mn) a_n \overline{b_m}$.

Partial. Evidence: Sketch.

Separated-atom criteria give discrete eigenvalue bounds, but block masses alone do not characterize compact resolvent, even within atomic measures. A useful criterion must retain within-block separation or Gram information.

12. Coefficient inequalities

12.1. **Real-power divisor reverse estimate**

> "It seems reasonable to conjecture that (20) should hold whenever j is a real number larger than 1."

Source: Bondarenko, Heap, and Seip, "An Inequality of Hardy–Littlewood Type for Dirichlet Polynomials," arXiv:1405.6516v3, concluding remarks, p. 14, following equation (20). Here $d(n)$ is the number of positive divisors of n. In (20), $q = 2^j$ and j is a positive integer; the question here is the stated real-j extension with

$$\left(\sum_{n=1}^{N} |a_n|^2 d(n)^{j-1}\right)^{1/2} \geq \left\|\sum_{n=1}^{N} a_n n^{-s}\right\|_{\mathcal{H}^{2j}}.$$

Answered: yes. Evidence: Published.

Llinares, "Contractive Inequalities between Dirichlet and Hardy Spaces," Revista Matemática Iberoamericana 40 (2024), doi:10.4171/RMI/1418, Theorem 3.1 and Corollary 3.2, proves the reciprocal generalized-divisor weight estimate. Put $p = 2^j$, $b = j-1$, and $\alpha = 2/p = 2^{-b}$. The prime-power comparison $1/d_\alpha(n) \leq d(n)^b$ gives the requested estimate for every real $j > 1$. Llinares proves the one-variable contraction by dilating a polynomial extremizer and applying the Hardy–Stein identity; tensorization and the Bohr lift give the Dirichlet-series inequality.

13. Contractivity and hypercontractivity

13.1. **Contractive Bergman inclusions $A^p_\alpha \subset A^q_\beta$**

> "For any $0 < p \leq q$ and $\alpha, \beta \geq 1$ such that $q/\beta \leq p/\alpha$, does the contractive inequality $\|f\|_{A^q_\beta(\mathbb{D})} \leq \|f\|_{A^p_\alpha(\mathbb{D})}$ hold?"

The source uses the probability measure $(\alpha-1)(1-|z|^2)^{\alpha-2}\,\mathrm{d}m$ for $\alpha > 1$ and sets $A^p_1 = H^p$. Source: Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, "Contractive inequalities for Bergman spaces and multiplicative Hankel forms," arXiv:1701.06897v2, §2.2, pp. 9–10.

Answered: yes. Evidence: Published.

Llinares, *Contractive inequalities between Dirichlet and Hardy spaces*, doi:10.4171/RMI/1418, Corollary 1.1, characterizes contractive inclusions of normalized weighted Bergman spaces. Its parameter criterion includes this disc range.

The proof applies Kulikov's equal-ratio contraction at the critical target exponent, then uses monotonicity of normalized weighted integrals to reach the requested larger weight.

13.2. **Sharp Hardy to weighted Bergman inequality**

> "The inequality (3) holds for every $0 < p \leq 2$."

Here equation (3) is $\left(\sum_{n=0}^{\infty} |a_n|^2/c_{2/p}(n)\right)^{1/2} \leq \|f\|_{H^p}$, and $c_\alpha(n)$ is defined by $(1-z)^{-\alpha} = \sum_{n=0}^{\infty} c_\alpha(n) z^n$. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 1, equation (3), p. 1.

Answered: yes. Evidence: Published.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Theorem 1.1 and Corollary 1.3, proves the sharp Hardy-to-weighted-Bergman coefficient inequality for $0 < p \leq 2$, answering this conjecture affirmatively.

The proof uses the hyperbolic isoperimetric inequality and coarea to control the superlevel distribution of $|f|^p(1-|z|^2)$; integrating this bound with the appropriate power gives the weighted Bergman norm.

Part 3. Where Nehari's theorem fails: Schatten classes

14. Paley–Wiener spaces

14.1. Bounded symbols for Schatten-class Hankel operators

> "whether the conclusion of the Nehari theorem holds when $H_\varphi \in S_p$, $1 < p < \infty$"

The question in §4, printed p. 20, of Konstantinos Bampouras and Karl-Mikael Perfekt, *Besov spaces and Schatten class Hankel operators for Hardy and Paley–Wiener spaces in higher dimensions*, arXiv:2409.04184v6. For a bounded convex domain Ω, the reconstructed target is to classify the pairs (Ω, p) for which every S_p Hankel operator on $PW(\Omega)$ has a bounded symbol.

Here $PW(\Omega) = \{f \in L^2(\mathbb{R}^n) : \operatorname{supp} \widehat{f} \subset \overline{\Omega}\}$, S_p denotes operators with p-summable singular values, and a bounded symbol is an $L^\infty(\mathbb{R}^n)$ function generating the same Hankel operator.

Partial. Evidence: Sketch.

Bampouras and Perfekt, *Besov spaces and Schatten class Hankel operators for Hardy and Paley–Wiener spaces in higher dimensions*, Theorem 4.3, arXiv:2409.04184v6, settle trace-class symbols; Carlsson and Perfekt, *Nehari's theorem for convex domain Hankel and Toeplitz operators in several variables*, Theorem 1.1, arXiv:1709.01843, settle the bounded-operator case for simple convex polytopes. A recent arXiv paper claims the Hilbert–Schmidt case for convex domains without affine lines (Bampouras, *Helson inequality, Hankel operators, and weak factorization on Paley–Wiener spaces of convex domains*, Theorem 1.2, arXiv:2609.10730v1), while curved domains with a positive-curvature patch fail for $p > 4$ by Bampouras, *Nehari's theorem and Hardy's inequality for Paley–Wiener spaces*, Theorem 1.1, doi:10.1016/j.jfa.2026.111408. The general curved-domain range $2 < p \leq 4$ remains open. The disc endpoint note explains the remaining S_4 estimate.

14.2. Bounded symbols for Hankel operators on the disc

> Does every bounded Hankel operator on $PW(\mathbb{D})$ have a bounded symbol?

The first question in the introduction, printed p. 2, of Ole Fredrik Brevig and Karl-Mikael Perfekt, *The Nehari problem for the Paley–Wiener space of a disc*, doi:10.1007/s12220-022-01063-2, J. Geom. Anal. 33 (2023), Article 16.

Here $PW(\mathbb{D}) = \{f \in L^2(\mathbb{R}^2) : \operatorname{supp} \widehat{f} \subset \overline{\mathbb{D}}\}$ for the unit disc; a bounded symbol is $\psi \in L^\infty(\mathbb{R}^2)$ generating the same Hankel operator, equivalently agreeing with the original symbol in Fourier variables on $2\mathbb{D}$.

Answered: no. Evidence: Published.

Their Theorem 1 constructs a bounded Hankel operator on the disc Paley–Wiener space that has no bounded symbol. The Hilbert–Schmidt case is a separate question. They place many symbol pieces near disjoint boundary caps, so the Hankel operator is an orthogonal sum with controlled norm. A dual L^1 test forces every bounded symbol norm to grow with the number of caps, contradicting the uniform bound that a Nehari theorem would imply.

14.3. Hilbert–Schmidt symbols for Hankel operators on the disc

> Does every Hankel operator on $PW(\mathbb{D})$ belonging to S_2 have a bounded symbol?

The second question in the introduction, printed p. 2, of Brevig and Perfekt, *The Nehari problem for the Paley–Wiener space of a disc*, doi:10.1007/s12220-022-01063-2.

Here $PW(\mathbb{D}) = \{f \in L^2(\mathbb{R}^2) : \operatorname{supp} \widehat{f} \subset \overline{\mathbb{D}}\}$ for the unit disc; a bounded symbol is $\psi \in L^\infty(\mathbb{R}^2)$ generating the same Hankel operator, equivalently agreeing with the original symbol in Fourier variables on $2\mathbb{D}$.

Answered: yes. Evidence: arXiv.

Bampouras and Perfekt's *Boundary-weighted Fourier inequalities for convex domains*, Corollary 1.8, arXiv:2608.19806v1, proves the result for every open ball in $\mathbb{R}^n$, $n \geq 2$, including the disc. The proof is in their latest arXiv preprint. The boundary-weighted Fourier estimate puts PW^1 into the negative-order Besov space dual to Hilbert–Schmidt Hankel symbols. Hahn–Banach then extends the symbol functional from PW^1 to L^1, producing an L^∞ symbol (Lemma 7.2 and Proposition 7.3).

15. Hankel forms and Schatten classes

15.1. AAK theory beyond the circle

To extend the AAK theorem beyond its classical $H(T)$-setting (see e.g. [N], App. 4).

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Notes and Comments §4.1, p. 137. The classical Adamyan–Arov–Krein theorem identifies the distance from a compact Hardy-space Hankel operator to Hankel operators of rank at most r with its $(r+1)$st singular value. The source's [N] is Nikolski, *Treatise on the shift operator*, Appendix 4, doi:10.1007/978-3-642-70151-1.

Partial. Evidence: Published.

Extensions exist under shift hypotheses: Andersson, Carlsson, and Perfekt, *AAK-type theorems for Hankel operators on weighted spaces*, doi:10.1016/j.bulsci.2014.08.008, Theorem 1.5 and Example 4.1. Their proof uses singular vectors to construct a finite-codimensional shift-invariant subspace; the Dirichlet example shows that the rank-one distance identity need not survive. The broad extension program across their analytic-space framework is not settled.

15.2. Hankel Schatten theory below one

It would be interesting to have any general results for $0 < p < 1$ too.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Notes and Comments §4.2, p. 137. For an operator H, membership in S_p means $\sum_n s_n(H)^p < \infty$, where s_n are its singular values. Peetre–Zhang's §6 proof uses lattice expansions of the symbol into reproducing-kernel atoms for sufficiency, then separates the positive and negative Laurent parts for necessity.

Partial. Evidence: Published.

Wallstén, *The S^p-criterion for Hankel forms on the Fock space,* $0 < p < 1$, doi:10.7146/math.scand.a-12251, gives a substantial special case. Peetre and Zhang, *Projective structures on an annulus and Hankel forms*, doi:10.1017/S0017089500008314, the unnumbered theorem in §6, cover $1/(\alpha+2) < p < 1$ on an annulus. A general theory for the analytic spaces and hypotheses of the paper remains open.

15.3. A Bergman analogue of the metaplectic group

Is there an analogue of the metaplectic group in the case of Bergman space (cf. [P6])?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Appendix II, Problem 4.4, p. 137. The metaplectic group is the double cover of the real symplectic group, acting on Fock space through the oscillator representation. The source's [P6] is Peetre, *Hankel forms in weaker assumptions*, Appendix I of the same paper, doi:10.4171/RMI/46, which develops invariant-space Hankel criteria; the preceding Shale–Weil

discussion instead refers to [P7], *Some calculations related to Fock space and the Shale–Weil representation*, doi:10.1007/BF01199757.

Partial. Evidence: Published.

Weighted Bergman realizations of the one-mode metaplectic representations are recorded by Chailuek and Hall, *Toeplitz operators on generalized Bergman spaces*, arXiv:0903.0651, §1.1 and Theorem 4: the two summands have reproducing kernels $(1 - z\overline{w})^{-3/2}$ and $(1 - z\overline{w})^{-1/2}$. Their construction defines the latter space by derivative norms when an ordinary weighted integral is unavailable. Caspers and van Velthoven, *Density conditions with stabilizers for lattice orbits of Bergman kernels on bounded symmetric domains*, arXiv:2109.14498, equation (1.1) and §5, give projective Bergman actions. The original question specifies neither the domain nor the covariance or symbol-space structure required of an analogue.

15.4. **Hankel forms on infinite-dimensional Fock space**

> What happens on the infinite-dimensional Fock space for $p \neq 2, 4$ (in particular for $p = \infty$)?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, §7, Problem after Theorem 7.8, p. 103. Here p indexes the Schatten class of the Hankel form, not a Gaussian L^p norm. The infinite-dimensional Fock Hilbert space has orthogonal monomials $z^\alpha/\sqrt{\alpha!}$ indexed by finitely supported nonnegative multiindices; $p = \infty$ asks for boundedness of the form.

Open. Evidence: Heuristic.

The source derives formal identities at $p = 2$ and $p = 4$ while explicitly setting aside definition issues. The bounded search found no general classification for other exponents in the commutative infinite-dimensional Fock realization.

15.5. **Same-rank approximation by point evaluations**

> We conjecture that it is possible to take N_m as the rank of H, i.e. that the set of continuous Hankel forms of rank $\leq r$ coincides with the closure (in any reasonable topology) of the set $\{(f, g) \mapsto \sum_{j=1}^{r} c_j f(z_j) g(z_j) : c_1, \dots, c_r \in \mathbb{C}, z_1, \dots, z_r \in \Omega\}$. (For $n = 1$, this follows easily from (14.3), but we have been unable to find a proof in higher dimension.)

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Remark 14.2, p. 126. In context, the conjecture includes the Fock-space case and claims the corresponding approximation for all finite-rank Hankel forms. Here $\Omega \subset \mathbb{C}^n$ is the analytic domain and $H(f, g)$ depends only on the product fg. Formula (14.3) describes a one-variable finite-codimension polynomial ideal as $J = \{f : f^{(\nu)}(z_j) = 0,\ 0 \leq \nu < k_j,\ 1 \leq j \leq N\}$; coalescing point evaluations approximate these derivative evaluations.

Answered: no. Evidence: Sketch.

A continuous rank-fourteen Hankel form on a six-dimensional Fock space has a neighborhood, detected by finitely many polynomial moments, containing no sum of at most fourteen evaluation forms. The one-variable assertion is not affected.

Answers, "Rank-fourteen obstruction to same-rank evaluation approximation", p. 15.

Rank-fourteen obstruction to same-rank evaluation approximation

15.6. **Hankel forms concentrated on analytic subvarieties**

> What can be said about the boundedness or smoothness of such forms?

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Remark 14.3, p. 126. The forms in question are $H(f, g) = \int_V f(z) g(z)\, d\sigma(z)$, with $V \subset \Omega$ an analytic subvariety and σ a measure on V. The kernel ideal is $J = \{f : H(f, g) = 0 \text{ for every } g\}$ on the polynomial algebra; its common zero set is the associated variety.

Partial. Evidence: Sketch.

Examples on a linear subvariety in Gaussian Fock space and one criterion in $\mathbb{C}^2$ address special cases. The source asks for a general description of the kernel ideal, the corresponding variety, and regularity, which remains open.

15.7. **Schatten classes without bounded symbols**

For every $p > 2$ there exists a multiplicative Hankel form H_φ in $S_p(\mathscr{H}^2)$ without a bounded symbol.

Brevig and Perfekt, *Weak product spaces of Dirichlet series*, arXiv:1510.02019 (2016), Conjecture 1 following Theorem 6, p. 10. The question concerns the full $\mathscr{H}^2$ space, with the constant row and column included.

Partial. Evidence: arXiv.

Conditional on: arXiv:2610.06728v1.

Brevig and Perfekt construct examples for $p > 5.7388\ldots$ in their Theorem 6; Bampouras, *Failure of Nehari's Theorem for Multiplicative Hankel Forms in Schatten Classes II*, arXiv:2610.06728v1, Theorem 1, extends failure to every $p > 4$. The proof finds a finite-dimensional symbol whose bounded-symbol norm exceeds its Schatten norm, then tensors copies on disjoint coordinate blocks. The range $2 < p \leq 4$ remains open.

15.8. **Existence of a trace limit for multiplicative Toeplitz matrices**

Let $\sigma_N = \{1, \ldots, N\}$; does the limit in (3.4) exist?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, arXiv:2001.01474, §3.8(a), p. 16. Equation (3.4) is the normalized trace $N^{-1} \operatorname{Tr} f(T_{\sigma_N}(\varphi))$ for bounded real φ and continuous f. Here $T_{\sigma_N}(\varphi) = (\widehat{\varphi}(j/k))_{j,k\in\sigma_N}$; $\widehat{\varphi}(j/k)$ denotes the Fourier coefficient indexed by the prime exponent vector of the positive rational j/k.

Answered: yes. Evidence: Published.

Guo, Li, and Zhou prove existence for these sections in *The first Szegő limit theorem on multi-dimensional torus*, doi:10.1016/j.aim.2024.109744, Theorem 1.11. They first decompose finite-coordinate symbols into prime-supported blocks, then approximate a general symbol by conditional expectations and control normalized traces through Hilbert–Schmidt estimates.

15.9. **A closed expression for the multiplicative trace limit**

Can it be expressed in a closed form in terms of f and φ?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, arXiv:2001.01474, §3.8(a), p. 16. The requested expression is for $N^{-1} \operatorname{Tr} f(T_{\sigma_N}(\varphi))$, with bounded real φ and continuous f. Here $T_{\sigma_N}(\varphi) = (\widehat{\varphi}(j/k))_{j,k\in\sigma_N}$; $\widehat{\varphi}(j/k)$ denotes the Fourier coefficient indexed by the prime exponent vector of the positive rational j/k.

Partial. Evidence: Sketch.

Our construction represents the limit as an expectation of a vector state of a random compression. Whether the question requires an elementary scalar formula remains open.

15.10. **The determinant formula for finite multiplicative measures**

Let $\varphi \geq 0$, $\varphi \in L^1(\mathbb{T}^\infty)$. Does the relation (3.9) necessarily hold in this generality? Furthermore, does it hold (similarly to the additive case) for multiplicative Toeplitz matrices of the form $\{\widehat{\mu}(j/k)\}_{j,k=1}^{\infty}$, where μ is a finite measure on $\mathbb{T}^\infty$ with $d\mu = \varphi\, dm + \mu_{\text{sing}}$?

Nikolski and Pushnitski, *Szegő-type limit theorems for " multiplicative Toeplitz" operators and non-Følner approximations*, arXiv:2001.01474, §3.8(b), p. 16. Relation (3.9) is

$$\lim_{N\to\infty} \det(T_{\sigma_N}(\varphi))^{1/|\sigma_N|} = \exp \int_{\mathbb{T}^\infty} \log \varphi \, \mathrm{d}m,$$

where $T_{\sigma_N}(\varphi) = (\widehat{\varphi}(j/k))_{j,k\in\sigma_N}$ and $|k\sigma_N \cap \sigma_N|/|\sigma_N| \to 1$ for every integer $k \geq 1$. For the measure version, φ is the density of the absolutely continuous part with respect to Haar measure m.

Answered: yes. Evidence: arXiv.

Conditional on: arXiv:2412.13751v4.

Guo, Li, and Zhou, *The first Szegő limit theorem on multi-dimensional torus*, doi:10.1016/j.aim.2024.109744, Theorem 1.9, establish (3.9) for nonnegative L^1 symbols. Austin, *Entropy and determinants for unitary representations*, arXiv:2412.13751v4, Theorem A, extends it to finite positive measures. His proof combines a determinant lower bound from Jensen's inequality with a Følner trace upper bound; a variational principle suppresses the singular component.

15.11. Operator ideals with the Berger–Coburn phenomenon

Are there nontrivial operator ideals in $B(H^2(\mathbb{T}), L^2(\mathbb{T}))$ or in $B(A^2(\mathbb{B}_n), L^2(\mathbb{B}_n))$ for which the Berger-Coburn phenomenon holds?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, arXiv:2304.09716, Open Problem 1, p. 4. Here $H_f = (I - P)M_f$, with P the orthogonal projection onto the analytic space and M_f multiplication by f. The Berger–Coburn phenomenon means $H_f \in \mathcal{I}$ if and only if $H_{\overline{f}} \in \mathcal{I}$ for every bounded f.

Answered: no. Evidence: Sketch.

Any ideal with the phenomenon must contain every bounded operator, so the only such ideal is the whole operator space. The proof is in our public note.

Answers, "Operator ideals with the Berger–Coburn phenomenon", p. 16.

Three open problems of Hu and Virtanen

15.12. Schatten exponents for the Berger–Coburn phenomenon on Fock spaces

Let F^2_ϕ be a doubling Fock space. For which values of p does the Berger-Coburn phenomenon hold for $S_p(F^2_\phi, L^2_\phi)$?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, arXiv:2304.09716, Open Problem 2, p. 9. Here $H_f = (I-P)M_f$, with P the orthogonal projection onto the analytic space and M_f multiplication by f. The Berger–Coburn phenomenon means $H_f \in \mathcal{I}$ if and only if $H_{\overline{f}} \in \mathcal{I}$ for every bounded f. A doubling Fock space is the entire subspace of $L^2_\phi = L^2(\mathbb{C}, e^{-2\phi}\,\mathrm{d}A)$, where the positive measure $\Delta\phi$ satisfies $\Delta\phi(D(z,2r)) \leq C\Delta\phi(D(z,r))$.

Answered: $p = 2$ on every space; some spaces for each $p > 1$. Evidence: Sketch.

For $0 < p \leq 1$ it fails on every doubling Fock space; for $1 < p < 2$ and $p > 2$ there are counterexample weights, while constant-Laplacian weights retain the phenomenon for all $1 < p < \infty$. The proof is in our public note.

Answers, "Schatten exponents for the Berger–Coburn phenomenon on Fock spaces", p. 17.

Three open problems of Hu and Virtanen

15.13. Compactness and conjugate Fock Hankel operators

Let F^2_ϕ be a doubling Fock space and $f \in L^\infty$. Is it true that if $H_f : F^2_\phi \to L^2_\phi$ is compact, then $H_{\bar{f}}$ is also compact?

Hu and Virtanen, *On the Berger-Coburn phenomenon*, arXiv:2304.09716, Open Problem 3, p. 9. Here $H_f = (I-P)M_f$, with P the orthogonal projection onto the analytic space and M_f multiplication by f. The Berger–Coburn phenomenon means $H_f \in \mathcal{I}$ if and only if $H_{\overline{f}} \in \mathcal{I}$ for every bounded f. A doubling Fock space is the entire subspace of $L^2_\phi = L^2(\mathbb{C}, e^{-2\phi}\,\mathrm{d}A)$, where the positive measure $\Delta\phi$ satisfies $\Delta\phi(D(z,2r)) \leq C\Delta\phi(D(z,r))$.

Answered: yes. Evidence: Sketch.

Compactness forces the local distance from f to holomorphic functions to vanish at infinity. Rescaling by the doubling radius and applying Liouville's theorem yields vanishing mean oscillation, which gives compactness of the conjugate Hankel operator.

Answers, "Compactness and conjugate Fock Hankel operators", p. 17.

Three open problems of Hu and Virtanen

15.14. Unbounded symbols and the Berger–Coburn phenomenon

Characterize unbounded symbols f for which BCP holds for compact operators or for Schatten class operators H_f on Fock spaces.

Hu and Virtanen, *On the Berger-Coburn phenomenon*, arXiv:2304.09716, Open Problem 4, p. 11. Here $H_f = (I - P)M_f$, with P the orthogonal projection onto the analytic space and M_f multiplication by f. The Berger–Coburn phenomenon means $H_f \in \mathcal{I}$ if and only if $H_{\overline{f}} \in \mathcal{I}$ for every bounded f. For an unbounded symbol the question is posed when the corresponding Hankel operators are densely defined; BCP abbreviates that equivalence.

Open. Evidence: Heuristic.

The source gives examples on both sides, but no characterization. The compact bounded-symbol result and the Schatten results for doubling spaces do not settle this unbounded-symbol problem.

15.15. Logarithmic-BMO symbols on weighted Bergman A^1

For $1 < p < \infty$, the problem whether $H^\omega_f : A^1_\omega \to L^1_\omega$ is bounded

Duan, Guo, Hong, and Yan, *Toeplitz and Hankel operators on Bergman spaces with doubling weights*, arXiv:2609.23402, final paragraph before the references. The source asks whether this holds for every $f \in \mathrm{BMO}^p_{\omega,\log}$, even in the classical Bergman space; its hypotheses use radial two-sided doubling weights $\omega \in \mathcal{D}$. Here $L^1_\omega = L^1(\mathbb{D}, \omega\,\mathrm{d}A)$, A^1_ω is its analytic subspace, and $H^\omega_f g = (I - P_\omega)(fg)$ for the weighted Bergman projection P_ω. The question assumes $f \in \mathrm{BMO}^p_{\omega,\log}$: for a fixed sufficiently large Bergman radius r,

$$\sup_{z\in\mathbb{D}} \log\frac{e}{1-|z|}\left(\frac{\int_{D(z,r)} |f - \widehat{f}_{r,\omega}(z)|^p \omega\,\mathrm{d}A}{\int_{D(z,r)} \omega\,\mathrm{d}A}\right)^{1/p} < \infty,$$

where $\widehat{f}_{r,\omega}(z)$ is the weighted average on $D(z,r)$. The radial class $\mathcal{D}$ means the tail $\widehat{\omega}(t) = \int_t^1 \omega(s)\,\mathrm{d}s$ satisfies upper and reverse doubling inequalities.

Answered: yes. Evidence: Sketch.

For every such weight, the logarithmic local-oscillation condition implies the $A^1_\omega \to L^1_\omega$ Hankel bound. The full proof is in our public note.

Answers, "Hankel operators on weighted Bergman A^1", p. 18.

Hankel operators with logarithmic-BMO symbols

15.16. Why the threshold $p = 4$ appears in two settings

It would be interesting to understand whether the appearance of the same threshold in these two settings reflects some connection between them. In particular, the endpoint case $p = 4$ seems especially interesting, since its Schatten norm can be explicitly computed.

Bampouras, *Failure of Nehari's Theorem for Multiplicative Hankel Forms in Schatten Classes II*, arXiv:2610.06728v1, end of §1. In particular, the authors single out the computable S_4 endpoint; the question here is whether the matching threshold has a structural explanation.

Answered: the mechanisms differ. Evidence: Sketch.

On the torus the S_4 identity has a positive constant-term contribution, which yields the L^4 bound; in the Paley-Wiener setting the same bound fails. The planar cap-counting method reaches the same threshold through packing and square-root cancellation. The endpoint Nehari question itself remains open.

Answers, "Why the threshold $p = 4$ appears in two settings", p. 19.

Two mechanisms for the Schatten threshold four

15.17. **Finite-rank forms on a punctured spectral-algebra surface**

For the one complex dimensional Riemann surface of Example 3.4 it seems reasonable to conjecture that every finite rank symbol is of the form $b(\cdot) = \sum_{k=1}^{N} a_k Z^{n_k} K_{z_k}(\cdot)$ for some constants a_k, z_k, n_k.

Deck and Gross, *Hankel forms on complex manifolds*, doi:10.2140/pjm.2002.205.43, Remark 6.10, p. 94, for the one-complex-dimensional Riemann surface in Example 3.4. Example 3.4 is the n-sheeted cover of $\mathbb{C} \setminus \{0\}$ for $z^{1/n}$, with the lifted Gaussian measure. Its spectral algebra is spanned by $z^{k/n}$, $k \geq 0$. Here K_{z_k} is the reproducing kernel at z_k, and Z is the lift of the Gaussian Euler vector field $a^{-1} z \partial_z$ in the covering coordinate; $Z^{n_k} K_{z_k}$ represents a derivative evaluation.

Answered: no. Evidence: Sketch.

On the punctured surface the constant symbol gives a bounded rank-one Hankel form, but it cannot be a finite sum of the proposed point-jet symbols: the spectral algebra contains only nonnegative powers, and the missing branch point prevents a finite jet representation. This counterexample addresses the literal example, not the broader manifold program.

Answers, "A finite-rank form missed by point jets on a punctured surface", p. 21.

A finite-rank Hankel form missed by point jets

15.18. **Real interpolation of Gaussian Fock scales**

It is an interesting (open) question to determine the spaces which arise by real interpolation from this scale.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Introduction, p. 70; for fixed p this is related to spectral analysis of the dilation operator. Here F^p_α is the space of entire functions with norm

$$\|f\|^p_{F^p_\alpha} = \left(\frac{p\alpha}{2\pi}\right)^n \int_{\mathbb{C}^n} |f(z)|^p e^{-p\alpha|z|^2/2} \, \mathrm{d}A(z),$$

with the weighted supremum norm at $p = \infty$; both p and $\alpha > 0$ vary.

Partial. Evidence: Published.

For fixed interior p, Peetre, Thangavelu, and Wallin, *Generalized Fock spaces, interpolation, multipliers, circle geometry*, doi:10.4171/RMI/195, §3, Theorems 1–2, identify the varying-Gaussian couple in one variable by dilation resolvents and disprove the uniform estimate in Example 0.2. They transfer a bounded-variation Fourier multiplier from the circle to Fock space to bound the dilation resolvents uniformly for $1 < p < \infty$; at $p = 1$, a coefficient-cutoff test has a logarithmically growing norm. A classification across mixed exponents and Gaussian scales remains open.

15.19. **Higher-weight Hankel forms in several variables**

To extend the theory of higher weight Hankel forms in [JP2] to more general cases, for instance the unit ball in $\mathbb{C}^d$. This is basically a question of Invariant Theory.

Janson, Peetre, and Rochberg, *Hankel forms and the Fock space*, doi:10.4171/RMI/46, Notes and Comments §4.3, p. 137. The source's [JP2] is Janson and Peetre, *A new generalization of Hankel operators (the case of higher weights)*, doi:10.1002/mana.19871320121. Higher-weight forms pair a symbol with bilinear differential expressions in f and g that transform covariantly, rather than with their ordinary product.

Partial. Evidence: Published.

Invariant-theory decompositions are given by Peng and Zhang, *Tensor products of holomorphic representations and bilinear differential operators*, doi:10.1016/j.jfa.2003.09.006, the tensor-product decomposition theorem and bilinear intertwining construction. Sundhäll, *Trace class criteria for bilinear Hankel forms of higher weights*, doi:10.1090/S0002-9939-06-08583-2, the main theorem, gives the ball criteria for $1 \leq p \leq \infty$. Peetre, *Hankel kernels of higher weight for the ball*, doi:10.1017/S0027763000004499, §§2–3 and §5, constructs the

invariant kernels and transvectants: mixed derivatives of $f \otimes g$ are restricted to the diagonal, producing differential expressions with the required transformation law. The full analytic theory on more general domains, including Cartan domains, remains only partly developed.

15.20. **The Wilf eigenvalue-counting asymptotic**

> We have the following corollary, which was conjectured by Wilf. For the Hilbert matrix $((j+k+\alpha)^{-1})$, $j,k = 0,1,\dots,N-1$, with $\alpha \neq 0,-1,-2,\dots$, the number of eigenvalues which exceed πa $(0 < a < 1)$ is asymptotically $2/\pi \log N \operatorname{sech}^{-1} a$.

Widom, *Hankel matrices*, doi:10.1090/S0002-9947-1966-0187099-X, Corollary on printed pp. 30–31. The question attributed there to Wilf is whether this counting asymptotic holds for each fixed admissible α and $0 < a < 1$.

Answered: the stated asymptotic. Evidence: Published.

Widom attributes the formula to Wilf and proves it as a corollary to Theorem 4.3(ii). His proof computes trace moments of the finite sections, compares them with a convolution kernel $\operatorname{sech}(x-y)$, and approximates the interval indicator to pass from moments to eigenvalue counts.

15.21. **The Paley-Wiener L^1–S_4 endpoint**

> Let Ω be bounded, convex, with C^2 boundary of positive curvature. Is there C_Ω with $|\langle h, \varphi\rangle| \le C_\Omega \|h\|_1 \|H_\varphi\|_{S_4}$ for all Schwartz h, φ with Fourier support in $2\overline{\Omega}$?

Søvik, Paley-Wiener boundary-cap questions, endpoint question after the S_p-counting discussion. Here H_φ is the Fourier-side integral operator with kernel $\widehat{\varphi}(\xi+\eta)$ on $L^2(\Omega)$, and $\langle h, \varphi\rangle = \int_{\mathbb{R}^n} h\overline{\varphi}$. Bampouras, *Nehari's Theorem and Hardy's inequality for Paley–Wiener spaces*, arXiv:2504.05986v3, Theorem 1.1, gives the failure above 4; its proof packs boundary patches with disjoint input lenses and compares Schatten norms with the dual analytic L^1 norm.

Partial. Evidence: arXiv.

Conditional on: arXiv:2504.05986v3.

The S_2 estimate is known and curved-patch constructions give failure above $p = 4$. Reviewed packet estimates handle restricted families, but the full L^1–S_4 inequality is open.

Part 4. The borderline: the multiplicative Hilbert matrix

16. Nehari theorems and Hankel forms

16.1. **Strict weak-product inclusion after differentiation**

> The inclusion between the standard weak product and its skew counterpart is strict, $H^2 \odot H^2 \subsetneq \partial^{-1}H^2 \odot \partial H^2$.

Brevig and Perfekt, *Weak product spaces of Dirichlet series*, arXiv:1510.02019, Conjecture 2, p. 14. Here $\partial(n^{-s}) = -(\log n)n^{-s}$ and the full Hardy space includes constants. Here $H^2 \odot H^2$ is the completion of sums $\sum_j f_j g_j$ with norm $\inf \sum_j \|f_j\|_2 \|g_j\|_2$, and the skew space consists of F with $F(+\infty) = c$ and $F\prime = \sum_j f_j g_j\prime$, with norm $|c| + \inf \sum_j \|f_j\|_2 \|g_j\|_2$.

Open. Evidence: Heuristic.

Brevig and Perfekt establish the zero-constant-term comparison in Theorem 12 of the cited paper; see also their *Correction to: Weak product spaces of Dirichlet series*, arXiv:2205.03244. It leaves open whether constants make the full-space inclusion strict or whether equality holds.

16.2. **Fractional zeta primitives on $\mathscr{H}^p$**

> For $0 < p \le 2$, does $\varphi_{1/p}$ defined in equation (19) induce a bounded linear functional on $\mathscr{H}^p$ exactly when $0 < p \le 1$?

Bondarenko, Brevig, Saksman, and Seip, *Linear space properties of H^p spaces of Dirichlet series*, arXiv:1801.06515, §3, unnumbered conjecture. Here (19) defines $\varphi_\beta(s) = 1 + \sum_{n\geq 2} n^{-1/2-s}(\log n)^{-\beta}$, and its coefficient functional on $f(s) = \sum_n a_n n^{-s}$ is $a_1 + \sum_{n\geq 2} a_n/(\sqrt{n}(\log n)^\beta)$.
Answered: yes, exactly for $0 < p \leq 1$. Evidence: Sketch.

The completed proof establishes boundedness through $p = 1$ and failure above it. The case $p = 1$ was proved first by Perfekt, arXiv:2609.03855; the argument here is independent and covers every p. The full argument is in the public book draft. The proof

16.3. The sharp multiplicative Hilbert functional constant

Is $C = \pi$, the norm of the multiplicative Hilbert matrix on $\mathscr{H}^2$?

Søvik, *Where Nehari Survives*, Chapter 9, the sharp-constant problem, p. 34. Here $C = \sup_{\|f\|_{\mathscr{H}^1}=1,\ a_1=0} |\ell(f)|$, where $\ell(f) = \sum_{n\geq 2} a_n/(\sqrt{n}\log n)$ for $f(s) = \sum_n a_n n^{-s}$.
Partial. Evidence: Sketch.

Søvik proves $\pi \leq C \leq 5.515$ in the cited chapter. Exact evaluation and a proof that $C > \pi$ remain open; numerical optimization alone does not establish a strict gap. Book discussion

16.4. The absolute real-curve Fejér–Riesz constant

Is the sharp absolute real-prime-curve embedding constant equal to π?

Søvik, the question following the real-curve Fejér–Riesz estimates. Here the constant is $\sup_{\|F\|_{H^1}=1,\ F(0)=0} \int_0^\infty |F(r(\sigma))|\,d\sigma$, where F is the Bohr lift and $r(\sigma) = (p_j^{-1/2-\sigma})_j$.
Partial. Evidence: Sketch.

Homogeneous and finite-degree families have constants strictly below π, but the global supremum is undetermined. The existing estimates do not give either a universal upper bound π or a strict lower bound above it. Book discussion

17. Hilbert-type forms

17.1. Large-parameter asymptotics for Hilbert-type forms

What is the asymptotic decay of $\|B_\alpha\| - 1$ as $\alpha \to \infty$?

Brevig, *Sharp norm estimates for composition operators and Hilbert-type inequalities*, arXiv:1705.01316, §5.2(b), p. 14. The source notes bounds of orders $4^{-\alpha}$ and $2^{-\alpha}$ and suggests the former as the correct one; the form has kernel $(mn)^{\alpha-1/2}/\max(m,n)^{2\alpha}$. Here $B_\alpha(a,b) = \sum_{m,n\geq 1}(mn)^{\alpha-1/2}\max(m,n)^{-2\alpha}a_m b_n$ on $\ell^2 \times \ell^2$, and $\|B_\alpha\|$ is its bilinear operator norm.
Answered: order $4^{-\alpha}$. Evidence: Sketch.

The Schur complement at the first coordinate reduces the excess norm to the squared norm of its off-diagonal row. The remaining block has norm bounded strictly below one, giving $\|B_\alpha\| - 1 \sim 4^{-\alpha}$.

18. Coefficient inequalities

18.1. Divisor-power weighted small-p pseudomoments

Determine the precise asymptotic growth of $\left\|\sum_{n=1}^N [d(n)]^\gamma n^{-1/2-s}\right\|_{H^p}$ when $N \to \infty$ for $p \leq 1$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.4, p. 6. The printed problem does not specify a range for γ; here $d(n)$ counts positive divisors and $[d(n)]^\gamma$ means ordinary divisor powers, distinguished from generalized divisor coefficients.
Partial. Evidence: Sketch.

Gerspach–Lamzouri, "Low Pseudomoments of Euler Products," arXiv:2103.03870, *Quarterly Journal of Mathematics* 73 (2022), Theorem 1.2, gives the norm order for $\gamma < 0$; Gerspach, "Low Pseudomoments of the Riemann Zeta Function and Its Powers," doi:10.1093/imrn/rnaa159, *International Mathematics Research Notices* (2022), Corollary

1, gives it for $\gamma = 0$. Our proof gives the matching order for $0 < \gamma < 1$ and $p_0 < p \le 1$, with upper bounds below p_0. The precise $\log\log N$ factors and the unrestricted positive-γ range remain open. Here $p_0 = 4(2^\gamma - 1)/4^\gamma$.

18.2. Optimal power-weight coefficient contraction

> "The problem raised in the preceding paragraph is to find the largest exponent $\gamma = \gamma(p)$ for which the contractive inequality
>
> $$\left(\sum_{n=0}^{\infty} |c_n|^2 (n+1)^\gamma\right)^{1/2} \le \|f\|_{H^q(\mathbb{T})}$$
>
> holds for every $f \in H^q(\mathbb{T})$, $0 < q < 2$."

Source: Bondarenko, Heap, and Seip, "An Inequality of Hardy–Littlewood Type for Dirichlet Polynomials," arXiv:1405.6516v3, concluding remarks, p. 14. Here $f(z) = \sum_{n\ge 0} c_n z^n$; we use the variable q throughout.

Answered: $\gamma(q) = \min\{\log_2(q/2), 1 - 2/q\}$. Evidence: Published.

Kulikov, "Functionals with Extrema at Reproducing Kernels," doi:10.1007/s00039-022-00608-5, Geometric and Functional Analysis 32 (2022), Section 5, equation (5.1), proves the binomial-weight coefficient inequality for every $0 < q < 2$. Comparing the binomial weights gives $\gamma(q) = \min\{\log_2(q/2), 1 - 2/q\}$. Kulikov obtains the weighted contraction from a distribution-function comparison based on the hyperbolic isoperimetric inequality. Linear perturbations and singular functions show sharpness. The value is $1 - 2/q$ for $0 < q \le 1$ and $\log_2(q/2)$ for $1 \le q < 2$.

18.3. H^1 partial-sum norm growth

> Determine the precise asymptotic growth of the norm of the partial sum operator $S_N : \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ when $N \to \infty$ for $p = 1$ (or more generally, for $p \le 1$ or $p = \infty$).

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.5, p. 6; the present answer addresses the $p = 1$ case.

Answered: $\asymp \log N / \log\log N$. Evidence: Sketch.

Bondarenko, Brevig, Saksman, and Seip, "Linear Space Properties of H^p Spaces of Dirichlet Series," doi:10.1090/tran/7898, Theorem 5.2, gives the upper bound $O(\log N/\log\log N)$; our entropy argument supplies a matching lower bound, so the H^1 order is $\log N/\log\log N$. The leading constant remains open.

Answers, "H¹ partial-sum norm growth", p. 22.

Public proof.

18.4. Higher moments of homogeneous Dirichlet sums

> Assume k is an integer larger than 1. For which m (depending on N) will the L^{2k} norms of m-homogeneous Dirichlet polynomials of length N be comparable to their L^2 norms?

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.2, p. 5. The coefficient-one interpretation is a restricted interpretation of the printed question: the canonical sum is $\sum_{n\le N,\,\Omega(n)=m} n^{-s}$, where $\Omega(n)$ counts prime factors with multiplicity; the source itself says "m-homogeneous Dirichlet polynomials."

Partial. Evidence: Sketch.

Bondarenko and Seip, "Helson's Problem for Sums of a Random Multiplicative Function," arXiv:1411.6388v2, Lemma 2 and equations (12) and (16), determine the sharp fourth-moment threshold for the canonical sums. Mahatab, "6-th Norm of a Steinhaus Chaos," arXiv:1710.08201v2, Theorem 1 (published in *International Journal of Number Theory*, doi:10.1142/S1793042120500670 16 (2020), 1295–1306), proves sixth-moment comparability when $m \ll (\log\log N)^{1/3}$. Our written proof characterizes comparability for every fixed

$2k$ only in the upper-degree cone $m \geq \lfloor \log(N/2^m)/\log(3/2) \rfloor$; the sparse and intermediate thresholds remain open.

18.5. **Small-p critical-line polynomial norm**

Determine the asymptotic behavior of $\left\| \sum_{n=1}^{N} n^{-1/2-s} \right\|_{H^p}$ when $N \to \infty$ for $0 < p \leq 1$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.3, p. 5.

Answered: order determined. Evidence: Published.

Gerspach, "Low Pseudomoments of the Riemann Zeta Function and Its Powers," doi:10.1093/imrn/rnaa159, Corollary 1, determines the matching asymptotic order for every fixed positive p, hence throughout the source range $0 < p \leq 1$. The Bohr correspondence turns the norm into a random multiplicative moment; a reduction to integrals of random Euler products and a split into dyadic height ranges yield matching upper and lower bounds.

18.6. **Endpoint partial-sum operator norms**

Determine the precise asymptotic growth of the norm of the partial sum operator $S_N : \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ when $N \to \infty$ for $p = 1$ (or more generally, for $p \leq 1$ or $p = \infty$).

This entry concerns $0 < p < 1$ and $p = \infty$; the $p = 1$ case is answered separately above. Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.5, p. 6.

Partial. Evidence: Sketch.

Our proof establishes the $p = 1$ order $\log N/\log\log N$; see the public proof. Bondarenko, Brevig, Saksman, and Seip, "Linear Space Properties of H^p Spaces of Dirichlet Series," *Transactions of the American Mathematical Society*, doi:10.1090/tran/7898 372 (2019), Theorems 5.2–5.3, give the published H^1 upper bound and the logarithmic scale for each fixed $0 < p < 1$, with unmatched leading constants; Chen and Fu, "A Note on the Partial Sum of Bounded Dirichlet Series," arXiv:2608.17989v1, Theorem 1.1, prove the $\mathcal{H}^\infty$ order $\log N$. The sharp fixed-p rate for $0 < p < 1$ remains open.

18.7. **Absolute-coefficient Hardy inequality**

"The validity of the Hardy inequality, featuring absolute values on the coefficients, $|a_1| + \sum_{n\geq 2} \frac{|a_n|}{\sqrt{n}\log n} \lesssim \|f\|_{\mathcal{H}^1}$, remains an open problem."

Source: Perfekt, "A Fejér–Riesz inequality for Dirichlet series," arXiv:2609.03855v2, equation (7), p. 4.

Partial. Evidence: Sketch.

Perfekt's Theorem 1.1 and Corollary 1.2 prove the signed H^1 functional inequality, while the absolute-value inequality remains explicitly open. Our checked work gives restricted-support cases but no unrestricted conclusion.

18.8. **Bohr coefficient inequality at the critical endpoint**

For every $a_n n^{-s} \in \mathcal{H}^\infty$ and every $0 \leq c < 1/\sqrt{2}$, $\sum_{n=1}^{\infty} |a_n| e^{c\sqrt{\log n \log\log n}}/\sqrt{n} < \infty$. Moreover, the constant $1/\sqrt{2}$ cannot be improved.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, Corollary 9.2, p. 206. The endpoint question is inferred from the displayed strict range and the sharpness statement; the cited corollary does not decide $c = 1/\sqrt{2}$.

Partial. Evidence: Sketch.

The subcritical inequality holds for $c < 1/\sqrt{2}$ and fails for $c > 1/\sqrt{2}$. The endpoint remains open; the recorded second-order criteria leave an interval of possible weights.

18.9. **Sidon constant of Dirichlet polynomials in $\mathcal{H}^1$**

"For every $f \in H^1(\mathbb{T}^\infty)$, we have

$$\left(\sum_{\alpha \in \mathbb{N}_0^{(\mathbb{N})}} \left(\frac{|\widehat{f}(\alpha)|}{\sqrt{\alpha+1}} \right)^2 \right)^{1/2} \leq \int_{\mathbb{T}^\infty} |f(w)| \, \mathrm{d}w.$$

"

Here $\alpha + 1 = \prod_j (\alpha_j + 1)$ under the Bohr lift. The sharp order of $S_1(x) = \sup\{\sum_{n \leq x} |a_n| : \|\sum_{n \leq x} a_n n^{-s}\|_{\mathcal{H}^1} \leq 1\}$ is inferred from this Sidon-type inequality and its surrounding discussion; the cited theorem does not explicitly pose that growth question. Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," Theorem 12.14, p. 299.

Partial. Evidence: Sketch.

The checked bounds are $\sqrt{x}\exp(c(\log\log x)^{1/3})$ below and $(1+o(1))\sqrt{x\log x}$ above. The exact order is unknown.

18.10. **Rectangular factorization in the disk algebra (Queffélec)**

Has any function $f \in A(\mathbb{D})$ a rectangular factorization $f = g * h$ with both g and h in $A(\mathbb{D})$?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, doi:10.4171/OWR/2014/06, Problem Session, §3.2, Question 2, p. 386 (reported by O. F. Brevig). Here $A(\mathbb{D})$ is the disc algebra and $g * h = \sum_{n \geq 0} g_n h_n z^n$ is the coefficientwise product; W denotes its subalgebra of absolutely summable Taylor coefficients.

Answered: no. Evidence: Sketch.

If $g, h \in A(\mathbb{D})$, then their Taylor coefficients lie in ℓ^2, so the coefficients of $g * h$ lie in ℓ^1 by Cauchy–Schwarz. Since $A(\mathbb{D})$ strictly contains the Wiener algebra, the proposed factorization fails for some $f \in A(\mathbb{D})$. The source's separate factorization question for $f \in W$ remains open.

19. Dirichlet series and frequencies

19.1. **The logarithm in partial-sum bounds**

Question., arXiv:1902.02073v1 Defant and Schoolmann ask: "Determine optimal bounds for $\|S_N : \mathscr{D}_\infty((\log n)) \to \mathscr{D}_\infty((\log n))\|$." Problem 2.5, p. 8 of the arXiv and published surveys.

Chen and Fu, *A note on the partial sum of bounded Dirichlet series*, Theorem 1.1, arXiv:2608.17989, encode triangular matrix truncation by products $i\lfloor N/j \rfloor$: the cutoff is exactly $i \leq j$ on a block of size comparable to $\sqrt{N}$, transferring the logarithmic triangular-projection lower bound.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

Partial. Evidence: arXiv.

Chen and Fu claim the sharp order $\log N$ in an arXiv preprint. The source asks for optimal bounds, and the exact leading constant remains open.

Part 5. Related function-theoretic questions

20. Extremal coefficient problems

20.1. **Suffridge singular-inner-factor conjecture**

"It is conjectured on [13, p. 187] that this singular inner function is trivial when $0 < p < 1$."

Source: Brevig and Saksman, "Coefficient Estimates for H^p Spaces with $0 < p < 1$," arXiv:1905.09547, §5.2, p. 12, reporting Suffridge's conjecture. Here [13] is T. J. Suffridge, "Extremal Problems for Non-vanishing H^p Functions," doi:10.1007/BFb0087907; its extremal

quantity is $\widetilde{C}(k,p) = \sup\{\mathrm{Re}(f^{(k)}(0)/k!) : f \in H^p,\ \|f\|_{H^p} = 1,\ f(z) \neq 0 \text{ in } \mathbb{D}\}$. In context, the function is the singular inner factor of an extremal for the nonvanishing-coefficient problem.

Answered: yes. Evidence: Sketch.

The public outer-function theorem rules out every nontrivial inner factor for extremals of nonzero bounded functionals, which implies Suffridge's singular-inner-factor assertion.

Public proof.

20.2. Brevig–Saksman nonvanishing conjecture

> "For $0 < p < 1$ any extremal f for $C(k,p)$ does not vanish in $\mathbb{D}$."

Source: Brevig and Saksman, "Coefficient Estimates for H^p Spaces with $0 < p < 1$," arXiv: 1905.09547, §5.1, Conjecture 1, p. 12. Here $C(k,p) = \sup\{\mathrm{Re}(f^{(k)}(0)/k!) : f \in H^p,\ \|f\|_{H^p} = 1\}$, for integers $k \geq 1$.

Answered: yes. Evidence: Sketch.

Our public theorem proves the stronger result that every extremal for a nonzero bounded functional on H^p, $0 < p < 1$, is outer, hence zero-free in the disc.

Public proof.

20.3. Brevig–Saksman strict coefficient monotonicity

> "For $0 < p < 1$ the sequence $C(k,p)$ is strictly increasing."

Source: Brevig and Saksman, "Coefficient Estimates for H^p Spaces with $0 < p < 1$," arXiv: 1905.09547, §5.1, Conjecture 2, p. 12. Here $C(k,p) = \sup\{\mathrm{Re}(f^{(k)}(0)/k!) : f \in H^p,\ \|f\|_{H^p} = 1\}$, for integers $k \geq 1$.

Answered: yes. Evidence: Sketch.

The same outer-function theorem implies strict increase of $C(k,p)$ in k for every $0 < p < 1$.

Public proof.

20.4. BGI Question 1: at most one zero

> Suppose that $0 < p < 1$. Is it true that the extremal for $\Phi_k(p,t)$ has at most one zero in $\mathbb{D}$?

Source: Ole Fredrik Brevig, Sigrid Grepstad, and Sarah May Instanes, "F. Wiener's trick and an extremal problem for H^p," doi:10.1007/s40315-022-00469-x, §6, Question 1, p. 16. The paper defines $\Phi_k(p,t)$ in (1), with $k \geq 1$ and $0 \leq t \leq 1$. Here $\Phi_k(p,t) = \sup\{\mathrm{Re}(f^{(k)}(0)/k!) : f \in H^p,\ \|f\|_{H^p} \leq 1,\ f(0) = t\}$.

Answered: yes. Evidence: Sketch.

Our proof establishes that every extremal has at most one zero in the disc, counted with multiplicity, for all $k \geq 1$, $0 < p < 1$, and $0 \leq t \leq 1$.

Public proof.

20.5. BGI Question 2: terminal equality interval

> Fix $k \geq 2$ and $0 < p < 1$. Is there some t_0 such that $\Phi_k(p,t) = \Phi_1(p,t)$ holds for every $t_0 \leq t \leq 1$?

Source: Ole Fredrik Brevig, Sigrid Grepstad, and Sarah May Instanes, "F. Wiener's trick and an extremal problem for H^p," doi:10.1007/s40315-022-00469-x, §6, Question 2, p. 17. Here $\Phi_k(p,t) = \sup\{\mathrm{Re}(f^{(k)}(0)/k!) : f \in H^p,\ \|f\|_{H^p} \leq 1,\ f(0) = t\}$.

Partial. Evidence: Sketch.

The authors give numerical and analytical evidence for $k = 2$, with one extremal case excluded only numerically. Our conditional proof establishes the exact terminal equality interval for $k \geq 3$; the $k = 2$ case remains open.

Conditional on: the canonical-factorization and positive-parameter extremal results stated in the public proof.

Public proof.

21. Hardy and Bergman extremals

21.1. **Characterizing zero sets of Bergman spaces**

Characterise the zero sets for A^p, or at least find some non-trivial converse to (8.1).

Problem 8.6 and Update 8.6, printed p. 190, of Hayman and Lingham's *Research Problems in Function Theory, Fiftieth Anniversary Edition*, arXiv:1809.07200. The question asks for a characterization, or a nontrivial converse to Horowitz's growth bound, for zero sequences of the unweighted Bergman space A^p, $p > 0$. Here (8.1) is $\prod_{k=1}^{n} |z_k|^{-1} = O(n^{1/p+\varepsilon})$ for every $\varepsilon > 0$, with the nonzero zeros ordered by modulus.

Answered: yes. Evidence: Published.

Korhonen and Rättyä's weighted-Bergman zero-set criterion specializes to an exact characterization for A^p in "Zero Sequences, Factorization and Sampling Measures for Weighted Bergman Spaces," doi:10.1007/s00209-019-02243-7, Theorem 1 and Corollary 2 (2019). They divide by a canonical zero product and prove comparability with the associated nonvanishing weighted function; this reduces zero sets to integrability of an explicit zero potential minus a harmonic function.

21.2. **Separation of simplest-fraction sets in A^2_1**

In particular, are pairwise distances between these sets bounded away from 0 in the space A^2_1?

Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, arXiv:2009.01898, Math. Ann. 379 (2021), 1507–1532. Here SF_n denotes the set of sums of n Cauchy kernels with poles on the unit circle, counted with multiplicity, and A^2_1 has its standard normalized weighted-area norm.

Answered: yes. Evidence: Sketch.

A Fourier-coefficient argument proves the uniform lower bound $\operatorname{dist}_{A^2_1}(SF_n, SF_{n+k}) \geq \pi/\sqrt{6}$ for all integers $n, k \geq 1$.

Answers, "Separation of simplest-fraction sets in A^2_1", p. 22.

Full proof note.

21.3. **Power weights in the simplest-fraction problem**

Does Theorem 1 hold for larger classes of g? For instance, for $g(t) = t^\alpha$, $\alpha > 1$?

Question 1, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, arXiv:2009.01898, Math. Ann. 379 (2021), 1507–1532. Theorem 1 concerns minimizers for the radial weight g in the simplest-fraction problem: among sums $\sum_{j=1}^{n}(z-a_j)^{-1}$ with $a_j \in \mathbb{T}$, the weighted norm $\|\cdot\|_{(g)}$ is minimized by equispaced poles, uniquely up to rotation.

Partial. Evidence: arXiv.

Georgia Corbett, "Two New Proofs of Chui's Conjecture in Weighted Bergman Spaces," arXiv:2609.21121v1, Theorem 8, proves equispacing for every nonzero nonnegative integrable weight g with $\int_0^1 g(t)/t \, \mathrm{d}t < \infty$, hence every power weight $g(t) = t^\alpha$, $\alpha > 0$. Her proof writes the squared norm as a pairwise interaction energy whose kernel is completely monotonic in squared chordal distance, then applies Cohn and Kumar, "Universally Optimal Distribution of Points on Spheres," doi:10.1090/S0894-0347-06-00546-7, Theorem 1.2, to the regular polygon. The No extension beyond these integrability hypotheses is verified here.

Conditional on: Corbett's unrefereed arXiv:2609.21121v1, Theorem 8.

21.4. **The conjectured distance formula for simplest-fraction sets**

Our conjecture is that the answer is positive, and, moreover, if $\alpha > 0$, $n, k \geq 1$, then

$$\operatorname{dist}_{A^2_\alpha}(SF_n, SF_{n+k}) = \|\Psi_k\|_\alpha.$$

The conjectured formula in Question 2, p. 9, of Evgeny Abakumov, Alexander Borichev and Konstantin Fedorovskiy, *Chui's conjecture in Bergman spaces*, arXiv:2009.01898, Math. Ann. 379 (2021), 1507–1532. The separate question there about positive separation in A^2_1 is listed in the preceding entry. Here $SF_n = \{\sum_{j=1}^n (z-a_j)^{-1} : a_j \in \mathbb{T}\}$, $\Psi_k(z) = \sum_{j=0}^{k-1}(z - e^{2\pi ij/k})^{-1}$, and $\|f\|^2_\alpha = (\alpha+1)\int_{\mathbb{D}} |f|^2(1-|z|^2)^\alpha \, \mathrm{d}m_2$, with $\mathrm{d}m_2$ normalized area.

Answered: no. Evidence: Sketch.

For every $\alpha > 2$, a second-variation calculation shows that one pole is not a nearest point to the two-pole set, contradicting the proposed formula. The exact distances for $0 < \alpha \le 2$ remain undetermined.

21.5. Two-factor division in weighted Bergman spaces

> The norm computation "constitutes an interesting open problem in our opinion."

The final paragraph of Section 3.2, p. 6, of Adrián Llinares and Dragan Vukotić, *Extremal problems for vanishing functions in Bergman spaces* (arXiv:2012.05091v2). The inferred target is the sharp norm of $f \mapsto f/(\varphi_a\varphi_b)$ on the subspace of A^p_α functions vanishing at $a, b \in \mathbb{D}$, for $\alpha > -1$, $p > 0$, including repeated zeros and the dependence on the separation of the points. Here $\varphi_a(z) = (a-z)/(1-\overline{a}z)$ and $\|f\|^p_{A^p_\alpha} = (\alpha+1)\int_{\mathbb{D}} |f|^p(1-|z|^2)^\alpha \, \mathrm{d}m_2$, with normalized area $\mathrm{d}m_2$.

Partial. Evidence: Sketch.

Llinares and Vukotić's Theorem 3 solves the one-factor problem sharply; the authors note that iterating it does not give a sharp two-factor bound. Our Hilbert-space calculation gives a local expansion as the two zeros coalesce, but the general norm remains open.

21.6. Norms and extremals of the backward shift on H^p

> "We hope this note may inspire further work for $p \neq 1, 2, \infty$, including the case $0 < p < 1$."

The last paragraph of the introduction, p. 2, of Ole Fredrik Brevig and Kristian Seip, *The norm of the backward shift on H^1 is $2/\sqrt{3}$*, arXiv:2309.11360, Pure Appl. Funct. Anal. 9 (2024), no. 4, 991–994. The reconstructed question is to determine $\|B\|_{H^p \to H^p}$ and its extremals for the exponents not covered by the exact cases $p = 1, 2, 4, \infty$. Here $Bf(z) = (f(z) - f(0))/z$, with its removable value at zero. Bampouras and Llinares give the $p = 4$ value and extremal family in *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$*, arXiv:2605.10469v1, Theorem 1.1, where $\varphi = (1+\sqrt{5})/2$.

Partial. Evidence: Sketch.

We prove quadratic-order growth of the norm near $p = 2$ and refute the inner-translate candidate at $p = 6$; the exact norm outside the known cases remains open.

21.7. Best constant in a radial H^1 difference inequality

> "It would be interesting to know what the best constant C in the estimate appearing Lemma 2 is."

The paragraph after Lemma 2, p. 2, of Ole Fredrik Brevig, *Two footnotes to the F. and M. Riesz theorem* (arXiv:2407.06843v2). The estimate in Lemma 2 is $\|f_r - f_\varrho\|_1 \le C(\|f_\varrho\|_1^2 - \|f_r\|_1^2)^{1/2}$ for analytic f on the unit disc and $0 \le r \le \varrho < 1$; here $f_r(e^{i\theta}) = f(re^{i\theta})$ and the norm uses normalized arc measure.

Answered: $\sqrt{2}$. Evidence: Sketch.

The last Cauchy–Schwarz step in the published proof can be sharpened to give the sharp constant $\sqrt{2}$ for every fixed pair of radii; the lower bound is attained asymptotically by a first-order perturbation of a constant function.

21.8. The inner-translate candidate for the backward-shift norm

> the extremal functions are determined by inner functions with a prescribed value at the origin

This is an implicit question raised by the passage in Section 1, p. 2, of Konstantinos Bampouras and Adrián Llinares, *The norm of the backward shift on H^4 is $\sqrt[4]{\varphi}$* (arXiv:2605.10469v1). The inferred candidate asks whether, for $2 < p < \infty$, $\|B\|_{H^p \to H^p}$ equals the supremum over inner translates $I - c$ with $I(0) = a$, $0 < a < 1$, and $c \in \mathbb{C}$; here $Bf(z) = (f(z) - f(0))/z$.
Answered: no. Evidence: Sketch.

At $p = 6$ the candidate supremum is attained inside its parameter range, but the full variational equation for an operator extremizer forces an impossible third-order pole cancellation. *Answers*, "The inner-translate candidate for the backward-shift norm", p. 23.

Full proof note.

21.9. Largest majorant-comparison radius at fixed constant term

It would be interesting to determine the largest r

The last paragraph, p. 3, of Ole Fredrik Brevig, *A bootstrap proof of Bohr's theorem*, arXiv:2609.26850v1. Here S is the Schur class and $\varphi(z) = (w - z)/(1 - \overline{w}z)$ with $w = f(0)$; the question is to find the largest radius as a function of $0 < \varrho < 1$. More precisely, with $M_f(r) = \sum_{n \geq 0} |a_n| r^n$ for $f(z) = \sum_{n \geq 0} a_n z^n$, require $M_f(r) \leq M_\varphi(r)$ for every $f \in S$ with $|f(0)| = \varrho$; S consists of analytic functions bounded by one in the disc.
Partial. Evidence: Sketch.

Brevig's Corollary proves the lower bound $r(\varrho) \geq \max(\varrho, 1/3)$; our reviewed example gives $r(1/2) \leq 11/16$, but the exact threshold is open.

21.10. Bombieri's Bohr radius for a prescribed constant term

"Find the best constant $B(\lambda)$"

Problem 5.18 and Update 5.18, printed p. 91, of W. K. Hayman and E. F. Lingham, *Research Problems in Function Theory, Fiftieth Anniversary Edition*, arXiv:1809.07200 (Springer, 2019). The source asks for the sharp radius $B(\lambda)$ such that the majorant $F(r) = \lambda + \sum_{n \geq 1} |a_n| r^n$ is at most one for every analytic $f(z) = \lambda + \sum_{n \geq 1} a_n z^n$ bounded by one on the disc. For the present range $0 < \lambda < 1/2$, the exact value remains unknown; the source's update reports no progress.
Partial. Evidence: Sketch.

Our proof sheet gives a sharp expansion at $\lambda = 0$ and improves the lower bound for small positive λ, but it does not determine the interior formula.

22. Contractivity and hypercontractivity

22.1. Defant–Pérez conjecture on hypercontractive constants for m-homogeneous polynomials

"Conjecture 5.1. For every $1 \leq p < q < \infty$ and $m \in \mathbb{N}$ we have that

$$H_m^{q,p} \leq m^{1/(2q) - 1/(2p)} \left(\frac{q}{p}\right)^{m/2}.$$

"

Source: Andreas Defant and Alejandro Pérez, "Optimal Comparison of p-Norms of Dirichlet Polynomials," arXiv:1603.02128v1, §5, Conjecture 5.1, p. 10. Here

$$H_m^{q,p} = \sup_{P \neq 0} \|P\|_{L^q(\mathbb{T}^n)} / \|P\|_{L^p(\mathbb{T}^n)},$$

taking all m-homogeneous analytic polynomials and all n.
Partial. Evidence: Sketch.

Defant and Pérez, *Optimal comparison of p-norms of Dirichlet polynomials*, arXiv:1603.02128v1, Conjecture 5.1 and the paragraph following it, prove the bound when $p < q$ are powers of two. Our proof gives the exact constant for $p = 2\ell$ and $q = 2\ell s$ with integers $\ell \geq 1$, $s \geq 2$. The proposed bound for general real exponents remains open.

22.2. All-weight Bergman dilation hypercontractivity

"As in the previous subsection, we conjecture that Theorem 6 is true for all values of $\alpha \geq 1$."

In Theorem 6 the dilation $f(z) \mapsto f(rz)$ is conjectured to contract A^p_α into A^q_α precisely for $r \leq \sqrt{p/q}$, when $0 < p \leq q < \infty$. The source proves the criterion for $\alpha = (n+1)/2$, $n \in \mathbb{N}$. Source: Bayart, Brevig, Haimi, Ortega-Cerdà, and Perfekt, "Contractive Inequalities for Bergman Spaces," arXiv:1701.06897v2, §2.3, p. 11. Here $\|f\|^p_{A^p_\alpha} = (\alpha-1)\int_{\mathbb{D}} |f(z)|^p (1-|z|^2)^{\alpha-2}\,\mathrm{d}A(z)$ for $\alpha > 1$, where $\mathrm{d}A = \mathrm{d}x\,\mathrm{d}y/\pi$, and $A^p_1 = H^p$.

Partial. Evidence: Sketch.

The sharp dilation criterion is proved for the stated discrete range of weights. Whether it holds for every $\alpha \geq 1$ remains open.

22.3. Hyperbolic superlevel estimate

"Suppose that $f \in H^2$. Then

$$\mu\big(\{z \in \mathbb{D} : |f(z)|^2(1-|z|^2) > \lambda \|f\|^2_{H^2}\}\big) \leq 1/\lambda - 1.$$

"

Here $0 < \lambda < 1$ and $\mathrm{d}\mu = \pi^{-1}(1-|z|^2)^{-2}\,\mathrm{d}x\,\mathrm{d}y$ is hyperbolic measure. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 2, p. 7.

Answered: yes. Evidence: Published.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Theorem 3.1, proves the estimate for every norm-one $f \in H^2$, which gives the stated scale-invariant form.

The coarea formula and hyperbolic isoperimetry show that $t(\mu\{u > t\} + 1)$ decreases for $u = |f|^2(1-|z|^2)$. The Hardy boundary normalization controls its limit at zero and yields the stated bound.

22.4. Reverse coefficient contractivity

"The inequality

$$\|f\|_{H^q} \leq \left(\sum_{n=0}^{\infty} |a_n|^2 c_{q/2}(n)\right)^{1/2}$$

holds for every $2 \leq q < \infty$."

Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 3, p. 9. Here $f(z) = \sum_{n\geq 0} a_n z^n$ and $c_\alpha(n) = \Gamma(n+\alpha)/(\Gamma(\alpha)n!)$, the coefficients of $(1-z)^{-\alpha}$.

Answered: yes. Evidence: Published.

Llinares, *Contractive inequalities between Dirichlet and Hardy spaces*, doi:10.4171/RMI/1418, Theorem 2.1, proves $\|f\|_{H^q} \leq \|f\|_{D_{q/2}}$ for $q > 2$; at $q = 2$ the two norms agree. This is the stated coefficient inequality.

For each finite polynomial degree the norm ratio attains a maximum. Radial dilation, the Hardy–Stein identity, and Kulikov's sharp Bergman inequality force every maximizing polynomial to be constant; polynomial truncation gives the full statement.

22.5. One-dimensional critical Riesz exponent

"The Riesz projection P is a contraction from $L^r(\mathbb{T})$ to H^q with $q = 4(1-1/r)$ for $1 < r \leq \infty$. If f is in $L^1(\mathbb{T})$, then

$$\exp\left(\int_0^{2\pi} \log|Pf(e^{i\theta})|\,\mathrm{d}\theta/(2\pi)\right) \leq \|f\|_{L^1}.$$

"

Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Conjecture 4 and equation (19), p. 11. Here $P(\sum_{n\in\mathbb{Z}} \widehat{f}(n)z^n) = \sum_{n\geq 0} \widehat{f}(n)z^n$ is the analytic Riesz projection.
Answered: yes. Evidence: arXiv.

Fu and Li, *On the contractivity of the Riesz projection*, arXiv:2609.08603v1, Theorem 1.1, prove the exact critical exponent for every finite $1 < r < \infty$. The endpoint cases $r = 1$ and $r = \infty$ were settled in earlier published work; the current interior proof remains an arXiv result.

The proof constructs a subharmonic minorant of the difference between the input and output powers and applies the submean inequality after analytic composition. The endpoints are Marzo–Seip, *L^∞ to L^p constants for Riesz projections*, arXiv:1005.1842, Theorem 1, and Brevig–Llinares–Seip, *Critical exponents of the Riesz projection*, doi:10.1007/s40315-024-00566-z, Theorem 1.

22.6. Monotonicity of normalized Bergman power norms

"Suppose that f is an outer function with $\|f\|_{H^2} = 1$. Is it true that

$$\frac{\partial}{\partial\alpha} U_f(\alpha) \leq 0$$

for all $\alpha > 1$?"

Here $U_f(\alpha) = \int_{\mathbb{D}} |f|^{2\alpha}(\alpha - 1)(1 - |z|^2)^\alpha \, \mathrm{d}\mu$. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Question 1, p. 6. Here $\mathrm{d}\mu = (1 - |z|^2)^{-2} \, \mathrm{d}x \, \mathrm{d}y/\pi$ is hyperbolic measure.
Answered: yes. Evidence: Published.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Corollary 1.3, proves contractivity when $p/\alpha = q/\beta = 2$. This implies monotonicity of $U_f(\alpha)$ and answers the question.

Kulikov's superlevel-set comparison gives the equal-ratio contraction $\|f\|_{A^{2\beta}_\beta} \leq \|f\|_{A^{2\alpha}_\alpha} \leq 1$ for $\beta > \alpha > 1$. Raising to 2β also uses that these norms are at most one, and yields $U_f(\beta) \leq U_f(\alpha)$.

22.7. Radial level-set maximal integral

"Is it true that

$$\int_0^{2\pi} \max\{|f(re^{i\theta})|^2 : |f(re^{i\theta})|^2(1 - r^2) = \lambda\} \, \frac{\mathrm{d}\theta}{2\pi} \leq \|f\|_{H^2}^2$$

holds whenever $0 < \lambda < |f(0)|$?"

The source first normalizes $\|f\|_{H^2} = 1$ and, by Möbius invariance, assumes that $u(re^{i\theta}) = |f(re^{i\theta})|^2(1 - r^2)$ attains its maximum at the origin. Source: Brevig, Ortega-Cerdà, Seip, and Zhao, "Contractive Inequalities for Bergman Spaces," arXiv:1706.00738v2, Question 2, equation (16), p. 8. For each angle the maximum is over radii $0 \leq r < 1$ solving the displayed level equation; the source's normalization makes this set nonempty in the stated level range.
Partial. Evidence: Sketch.

No complete resolution is recorded. Available weak-type and layer-cake estimates address related level-set quantities but leave the maximal radial integral open.

22.8. Infinite-torus Riesz critical exponent

"Question. What is the largest $p = p_\infty$ such that the Riesz projection (1) is bounded from $L^\infty(\mathbb{T}^\infty)$ to $L^p(\mathbb{T}^\infty)$?"

Here (1) is the Riesz projection $Pf = \sum_{\alpha\in\mathbb{N}_0^{(\mathbb{N})}} c_\alpha z^\alpha$. Source: Ole Fredrik Brevig, "Linear Functions and Duality on the Infinite Polytorus," arXiv:1806.10849v2, Introduction, p. 1.
Answered: yes. Evidence: Published.

Konyagin, Queffélec, Saksman, and Seip, *Riesz projection and bounded mean oscillation for Dirichlet series*, doi:10.4064/sm200601-22-5, Theorem 2.1 and Corollary 2.2, identify the largest exponent as $p_\infty = 2$ and prove unboundedness for every $p > 2$.

The proof transfers polytope Fourier multipliers to analytic Riesz projections by adjoining coordinates. The Euclidean ball-multiplier obstruction then provides finite-dimensional witnesses for every exponent above two; tensor powers turn a norm greater than one into unboundedness.

22.9. **Lieb–Solovej majorization conjecture for** $\mathrm{SU}(1,1)$

> "Conjecture 6.1 (Majorization of Φ_L). For all density matrices ρ on D_K the eigenvalues of $\Phi_L(\rho)$ will be majorized by the eigenvalues of $\Phi_L(|z\rangle_K\langle z|)$. "

Here $D_K = \overline{\mathrm{span}}\{|j+K, j\rangle : j \geq 0\}$ is the two-mode Fock subspace of number difference K. The source writes $\Phi_L(\rho) = cP(\rho \otimes I_L)P$, with P the projection onto its designated lowest representation component of $D_K \otimes D_L$ and c chosen to preserve trace; $|z\rangle_K = (1-|z|^2)^{(K+1)/2} \sum_{j\geq 0} \binom{j+K}{K}^{1/2} z^j |j+K, j\rangle$. Source: Elliott H. Lieb and Jan Philip Solovej, "Wehrl-Type Coherent State Entropy Inequalities for $\mathrm{SU}(1,1)$ and its $ax+b$ Subgroup," arXiv:1906.00223v2, §6, Conjecture 6.1, p. 12.

Partial. Evidence: Sketch.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Theorem 1.2, and Frank, *Sharp inequalities for coherent states and their optimizers*, arXiv:2210.14798, Theorem 6, prove related coherent-state convex-functional and entropy inequalities, but not this channel eigenvalue majorization. Van Haastrecht, *Functional calculus of quantum channels for the holomorphic discrete series of* $\mathrm{SU}(1,1)$, doi:10.1016/j.jfa.2025.111036, Proposition 4.4 and the discussion on p. 49, explicitly leaves lowest-component majorization open; the source conjecture remains unresolved.

22.10. **Outer-function Bergman norm monotonicity**

> Question 3. Suppose that f is an outer function. For $\alpha > 1$, denote
>
> $$N_f(\alpha) = \|f\|_{A^{2\alpha}_\alpha} = \left(\int_{\mathbb{D}} |f(z)|^{2\alpha} (\alpha - 1)(1-|z|^2)^\alpha \, \mathrm{d}\mu(z) \right)^{1/(2\alpha)}.$$
>
> Is it true that
>
> $$\frac{\partial}{\partial \alpha} N_f(\alpha) \leq 0$$
>
> for all $\alpha > 1$?

Here $\mathrm{d}\mu(z) = (1-|z|^2)^{-2} \, \mathrm{d}x \, \mathrm{d}y/\pi$. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 3, p. 3.

Answered: yes. Evidence: Published.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Corollary 1.3, and Llinares, *Contractive inequalities between Dirichlet and Hardy spaces*, doi:10.4171/RMI/1418, Corollary 1.1, gives $N_f(\beta) \leq N_f(\alpha)$ for $\beta > \alpha > 1$. Differentiation under the integral then gives the source's derivative inequality.

The comparison follows by integrating the sharp hyperbolic superlevel distribution for equal exponent-to-weight ratios. Local differentiation is justified by domination using nearby exponents in the same contractive family.

22.11. **Kernel-matrix monotonicity conjecture and equivalent formulation**

> Conjecture 4. Suppose $\alpha > 0$ and $(\mathbf{c}, \mathbf{w}) \in \Lambda_\alpha$. Then $D_\alpha(\mathbf{c}, \mathbf{w})$ defined as in (3.2) and (3.4) is non-positive.

The source states that this is equivalent to the following reformulation:

> Conjecture 6. Suppose $\alpha > 0$, $\mathbf{w} \in (-1,1)^k$ and $\mathbf{f} \in \mathbb{H}^k$. Then $\widehat{D}_\alpha(\mathbf{f}, \mathbf{w}) \leq 0$.

Here $\mathbb{H} = \{z \in \mathbb{C} : \operatorname{Re} z > 0\}$; Λ_α consists of real nodes $\mathbf{w} \in (-1,1)^k$ and coefficients $\mathbf{c}$ for which the samples of $\mathbf{f}_\alpha = \mathbf{c}W_\alpha$, with $W_\alpha = [(1-\overline{w_i}w_j)^{-\alpha}]$, lie in $\mathbb{H}$. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Conjecture 4, p. 19, and equivalent Conjecture 6, p. 20. Write $f_i = (\mathbf{c}W_\alpha)_i$, $N = \mathbf{c}W_\alpha\mathbf{c}^*$ and $a_{ij} = (W_\alpha)_{ij}$. The referenced formulas define

$$D_\alpha = 2\operatorname{Re}\sum_i c_i\overline{f_i \operatorname{Log} f_i} - \sum_{i,j} c_i\overline{c_j}a_{ij}\operatorname{Log} a_{ij} - N\log N.$$

For real coefficients the first sum is $2\sum_i c_i f_i \log|f_i|$, with $0\log 0 = 0$; $\widehat{D}_\alpha(\mathbf{f},\mathbf{w})$ is this expression with $\mathbf{c} = \mathbf{f}W_\alpha^{-1}$ for distinct nodes, and the principal logarithm.

Partial. Evidence: Sketch.

Our construction gives a three-node rational example at $\alpha = 1$ with $D_1(\mathbf{c},\mathbf{w}) > 268.1785$, contradicting both formulations. The counterexample has not received an independent review.

22.12. **Conditional norm monotonicity below the reproducing-kernel range**

> Question 5. Suppose Conjecture 4 holds. Does it imply that the norms $\|f^\alpha\|_{A^2_\alpha}^{1/\alpha}$ are non-increasing for $\alpha > 0$, for some set of functions f?

The question is conditional on the source's kernel-matrix Conjecture 4; the phrase "some set of functions" is left unspecified in the source. Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 5, p. 20.

Conditional on: Conjecture 4 of Dan, Guo, and Wang.

Here the kernel-matrix conjecture asserts $D_\alpha(\mathbf{c},\mathbf{w}) \le 0$ for real nodes and samples $\mathbf{c}W_\alpha$ in the right half-plane; A^2_α for $\alpha > 0$ is the Hilbert space with reproducing kernel $(1-\overline{w}z)^{-\alpha}$.

Open. Evidence: Heuristic.

The source's premise is contradicted by the three-node example above. No specific function class is given for the proposed below-one monotonicity consequence.

22.13. **Entropy inequality for arbitrary positive semidefinite matrices**

> Question 7. Suppose $\mathbf{A} = [a_{ij}]$ is semi-positive definite and $\operatorname{Re} a_{ij} \ge 0$, for all $i,j = 1,\dots,k$. Suppose $\mathbf{c} \in \mathbb{C}^k$ is such that $\mathbf{c}\mathbf{A} = 0$. Do we have
>
> $$\widetilde{D}(\mathbf{c},\mathbf{A}) = -\sum_{i,j=1}^{k} c_i\overline{c_j}a_{ij}\operatorname{Log} a_{ij} \le 0?$$

Source: Hui Dan, Kunyu Guo, and Yi Wang, "A Sharp Inequality of Hardy–Littlewood Type via Derivatives," arXiv:1908.01320v1, Question 7, p. 20. This is stated for arbitrary positive semidefinite matrices, not only kernel matrices. Here Log is the principal complex logarithm and the continuous convention $0\operatorname{Log} 0 = 0$ is used.

Partial. Evidence: Sketch.

Our construction gives a four-node positive semidefinite counterexample. Its archived proof received an independent review, but the remaining matrix sizes are still open.

22.14. **Asymptotic norm of the Bloch-to-Bergman inclusion**

> "Conjecture 3.9. For any $\alpha > -1$,
>
> $$\lim_{p\to\infty} \frac{C_\alpha(p)}{p/(2e)} = \frac{1}{\sqrt{(\alpha+1)(\alpha+2)}}.$$
>
> "

Here $C_\alpha(p)$ denotes the inclusion norm from the Bloch space with norm $|f(0)| + \sup_{z\in\mathbb{D}}(1-|z|^2)|f'(z)|$ to the normalized weighted Bergman space A^p_α. Source: Adrián Llinares, "Norms of Inclusions between Some Spaces of Analytic Functions," arXiv:2107.11315v2, §3, Conjecture 3.9, p. 9.

Answered: no. Evidence: Sketch.

A uniform hyperbolic-average deficit gives constants $\eta < 1$ and $\varepsilon_0 > 0$ for which the limsup is below the conjectured constant whenever $-1 < \alpha < -1 + \varepsilon_0$. The actual limit and its existence remain undetermined.

Answers, "Asymptotic norm of the Bloch-to-Bergman inclusion", p. 24.

Full proof note.

22.15. **Sharp weighted Bergman contraction**

"Conjecture 1.3 (Lieb, Solovej). Let $\alpha > -1$. If $p > 2$, then

$$\|f\|_{A^p_{(\alpha+2)p/2-2}} \le \|f\|_{A^2_\alpha}, \qquad \forall f \in A^2_\alpha.$$

In addition, equality is possible if and only if f is a constant multiple of the Bergman kernel."

Source: Adrián Llinares, "On a Conjecture about Contractive Inequalities for Weighted Bergman Spaces," arXiv:2112.09962v1, Conjecture 1.3, p. 3. Here $\|f\|^r_{A^r_a} = (a+1)\int_{\mathbb{D}} |f(z)|^r (1-|z|^2)^a \, dA(z)$ for $a > -1$, and a kernel is a multiple of $(1 - \overline{w}z)^{-(a+2)}$.

Answered: yes. Evidence: Published.

Kulikov, *Functionals with extrema at reproducing kernels*, doi:10.1007/s00039-022-00608-5, Theorem 1.2, Corollary 1.3 and Remark 4.3, prove the sharp weighted Bergman contraction for all weights above the Hardy endpoint and classifies equality by reproducing kernels, answering Conjecture 1.3.

Coarea and hyperbolic isoperimetry compare the distribution of the weighted modulus with that of a normalized kernel. Integration gives contraction, and equality in the geometric comparison forces the kernel family.

22.16. **Higher-dimensional critical Riesz exponents**

"Critical exponent conjecture ($d > 1$).

$$p_d(q) = \begin{cases} a_d(q), & \dfrac{2d}{d+1} \le q \le \infty, \\ -1, & 1 \le q < \dfrac{2d}{d+1}. \end{cases}$$

"

Here $a_d(q) = 2 + 2/(d + 2/(q-2))$, with endpoint values interpreted as limits. Source: Ole Fredrik Brevig, Adrián Llinares, and Kristian Seip, "Critical Exponents of the Riesz Projection," arXiv:2402.09787v1, pp. 4–5. Here $p_d(q) = \sup\{p > 0 : \|P_+\|_{L^q(\mathbb{T}^d)\to L^p(\mathbb{T}^d)} = 1\}$, or -1 if no positive exponent works, and P_+ keeps exactly the nonnegative multi-indices.

Partial. Evidence: Sketch.

Brevig, Llinares, and Seip, *Critical exponents of the Riesz projection*, doi:10.1007/s40315-024-00566-z, Theorem 3, establish sharp bounds in selected dimensions and endpoint ranges. The full d-dimensional critical-exponent formula remains open.

22.17. **Equality cases at the critical Riesz exponent**

"Conjecture. Suppose that $1 < q < 2$ or $2 < q \le \infty$. If there is a nontrivial function ψ in L^q such that

$$\|P_+\psi\|_{p(q)} = \|\psi\|_q,$$

then there is a constant $C \neq 0$ and an inner function I such that $\psi = CI$."

Here $p(q)$ is the critical exponent for the Riesz projection. Source: Ole Fredrik Brevig, Adrián Llinares, and Kristian Seip, "Critical Exponents of the Riesz Projection," arXiv:2402.09787v1, §3, p. 10. Here P_+ retains nonnegative Fourier modes and $p(q) = \sup\{p > 0 : \|P_+\|_{L^q\to L^p} = 1\}$; nontrivial means nonzero.

Answered: yes. Evidence: Sketch.

Our proof establishes the equality cases for all finite $q \neq 2$, conditional on Fu–Li's strict subharmonic minorant; the $q = \infty$ case is Brevig–Llinares–Seip, *Critical exponents of the Riesz projection*, doi:10.1007/s40315-024-00566-z, Theorem 6.

Conditional on: Fu and Li, *On the contractivity of the Riesz projection*, arXiv:2609.08603v1, Lemma 2.1.

Public proof.

22.18. **Sharp boundary growth of point evaluation**

> "This result does not only extend the already known lower estimates given in [2, p. 26] from the range $\alpha \in (-1, 0)$ to any value of $\alpha > -1$, but it also provides an affirmative answer to Question 2 from [8]."

The cited Question 2 asks for the order of the evaluation norm near the boundary. The inferred target is whether, for $1 \le p < \infty$ and $\alpha > -1$, $\|\delta_s\|_{(A^p_\alpha)^*} \asymp (\operatorname{Re} s/(\operatorname{Re} s - 1/2))^{(\alpha+2)/p}$ for $\operatorname{Re} s > 1/2$. Source: Carlos Gómez-Cabello, Pascal Lefèvre, and Hervé Queffélec, "Integration Type Operators and Point Evaluation on Weighted Bergman Spaces of Dirichlet Series," arXiv:2402.12523v2, p. 3, referring to Fu, Guo, and Yan, Question 2. The original question text was not locally available. Here $\delta_s(f) = f(s)$ and $\mathcal{A}^p_\alpha$ uses the probability density $2^{\alpha+1}\sigma^\alpha e^{-2\sigma}/\Gamma(\alpha+1)$ to average $\|f(\sigma+\cdot)\|^p_{\mathcal{H}^p}$. In the quote [2] is Bailleul–Lefèvre, *Some Banach spaces of Dirichlet series*, doi:10.4064/sm226-1-2, p. 26 and [8] is Fu–Guo–Yan, *Generalized multiplicative Hilbert operators and Volterra operators on McCarthy–Bergman spaces of Dirichlet series*, https://www.researchgate.net/publication/374029212, Question 2.

Answered: yes. Evidence: Published.

Gómez-Cabello, Lefèvre, and Queffélec, *Integration type operators and point evaluation on weighted Bergman spaces of Dirichlet series*, doi:10.1090/proc/17060, Theorem 5.2, proves the global point-evaluation estimate in equation (1.1), including the stated boundary growth up to constants.

Fractional integration shifts the Bergman weight and transfers the known lower estimate for negative weights to all $\alpha > -1$. Splitting its integral near the evaluation point and absorbing the far tail yields the lower bound; the established upper estimate and testing constants finish the comparison.

22.19. **Tight Bergman inclusions on the unit ball**

> Conjecture 22(a). Suppose $0 < p, q < \infty$ and $\alpha, \beta > -1$. Then $A^p_\alpha \subset A^q_\beta$ is a tight fitting if and only if
>
> $$p < q \quad \text{and} \quad (n+1+\alpha)q = (n+1+\beta)p.$$
>
> Furthermore, $\|f\|_{A^p_\alpha} = \|f\|_{A^q_\beta}$ if and only if $f(z) = c/(1-\langle z, a\rangle)^{2(n+1+\alpha)/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

Source: Guanlong Bao, Pan Ma, Fugang Yan, and Kehe Zhu, "Embedding and Compact Embedding between Bergman and Hardy Spaces," arXiv:2502.08406v1, Conjecture 22(a), p. 19. Here A^p_α is the probability-normalized weighted Bergman space on $\mathbb{B}^n$ with density proportional to $(1-|z|^2)^\alpha$, H^p has normalized sphere means, and tight fitting means a proper, contractive, noncompact inclusion.

Answered: yes. Evidence: arXiv.

Kalaj, *The Gromov–Ros conjecture for rank-one symmetric spaces of noncompact type*, arXiv:2607.20055v3, Theorem 7.2 and Corollary 7.4, proves the norm-one contraction under $0 < p < q$, $n < \gamma < \delta$, and $p/\gamma = q/\delta$, with equality only for the stated kernel functions. The result is an arXiv claim.

The claimed proof combines coarea with complex-hyperbolic isoperimetry to compare weighted-modulus superlevel volumes, then integrates the comparison. Bao–Ma–Yan–Zhu, *Embedding and compact embedding between Bergman and Hardy spaces*, arXiv:2502.08406v1, Theorems A and B, give properness and noncompactness on the critical line.

22.20. **Tight Hardy-to-Bergman inclusions on the unit ball**

> Conjecture 22(b). Then $H^p \subset A^q_\alpha$ is a tight fitting if and only if
>
> $$p < q \quad \text{and} \quad nq = (n+1+\alpha)p.$$

Furthermore, $\|f\|_{A^q_\alpha} = \|f\|_{H^p}$ if and only if $f(z) = c/(1-\langle z,a\rangle)^{2n/p}$, where $c \in \mathbb{C}$ and $a \in \mathbb{B}^n$.

The hypotheses are $0 < p, q < \infty$, $\alpha > -1$, and $n \geq 1$, as in the preceding sentence of Conjecture 22. Source: Guanlong Bao, Pan Ma, Fugang Yan, and Kehe Zhu, "Embedding and Compact Embedding between Bergman and Hardy Spaces," arXiv:2502.08406v1, Conjecture 22(b), p. 19. Here A^p_α is the probability-normalized weighted Bergman space on $\mathbb{B}^n$ with density proportional to $(1-|z|^2)^\alpha$, H^p has normalized sphere means, and tight fitting means a proper, contractive, noncompact inclusion.

Answered: yes. Evidence: Sketch.

Conditional on: the complex-hyperbolic isoperimetric hypothesis stated in our proof.

Our proof establishes the full Hardy-to-Bergman statement conditional on the complex-hyperbolic isoperimetric hypothesis, with exact equality kernels. Kalaj, *The Gromov–Ros conjecture for rank-one symmetric spaces of noncompact type*, arXiv:2607.20055v3, Theorem 1.1, claims the geometric input; the equality bridge remains conditional.

Answers, "Tight Hardy-to-Bergman inclusions on the unit ball", p. 27.

Full proof note.

22.21. Contractive projections on H^p for $0 < p < 1$

"It seems that the assumption $Pz^n \in H^\infty(\mathbb{T})$ in Theorem 3.8 arises from technical reasons. We conjecture that this assumption can be dropped. Specifically, we conjecture that if P is a contractive projection on $H^p(\mathbb{T})$ with $0 < p < 1$ satisfying $P1 = 1$, then it automatically maps bounded functions to bounded functions, and therefore $Pf = f(0)$ or P is the identity."

Source: Xiang Fu, Kunyu Guo, and Dilong Li, "Contractive Projections, Conditional Expectations, and Idempotent Coefficient Multipliers on H^p Spaces $(0 < p < 1)$," arXiv:2503.06615v2, paragraph following Theorem 3.8, pp. 15–16. Here a projection is a linear idempotent of norm at most one. Theorem 3.8 gives the evaluation-or-identity conclusion under the additional assumption $Pz^n \in H^\infty$ for every $n \geq 0$.

Answered: yes. Evidence: Sketch.

Conditional on: Fu, Guo, and Li, arXiv:2503.06615v2, Theorem 3.8.

Our argument proves the conjectured dichotomy conditional on Fu, Guo, and Li's Theorem 3.8. The reduction to the source theorem is written and independently checked.

Answers, "Contractive projections on H^p for $0 < p < 1$", p. 28.

Full proof note.

22.22. Linearity in the Dirichlet range

"Fix $0 < \alpha < 1$ and $p \neq 2$. Is A^p_α a linear space?"

Source: Brevig, Kulikov, Seip, and Zlotnikov, "Contractive Inequalities in the Dirichlet Range," arXiv:2510.14333v1, Problem 1, p. 4. Here A^p_α consists of analytic f for which $\|f\|^p_{\alpha,p} = |f(0)|^p + \int_0^1 (\mathrm{d}/\mathrm{d}r) M_p(r,f)^p (1-r^2)^{\alpha-1}\,\mathrm{d}r < \infty$, where M_p is the normalized circular p-mean.

Partial. Evidence: arXiv.

For $0 < \alpha < 1$, $p > 0$, $p \neq 2$, and $\alpha + p > 1/2$, Dayan, Llinares, and Monsalve-López, *A^p_α classes in the Dirichlet range: inner-outer factorization, Carleson measures and weak products*, arXiv:2604.21347v2, Theorem 1.3, give counterexamples to linearity. The triangle $0 < p < 1/2$, $\alpha \leq 1/2 - p$ is not covered by that cited result.

The proof uses an equivalent norm expressed by the difference between the Poisson integral of the boundary modulus and the interior modulus, then constructs summands whose combination fails the required boundary integrability.

22.23. Monotonicity in p below the Hardy endpoint

"Fix $0 < \alpha < 1$ and let f be an analytic function in $\mathbb{D}$. Is the function $p \mapsto \|f\|_{A^p_\alpha}$ increasing?"

Source: Brevig, Kulikov, Seip, and Zlotnikov, "Contractive Inequalities in the Dirichlet Range," arXiv:2510.14333v1, Problem 2, p. 8. Here A^p_α consists of analytic f for which $\|f\|^p_{\alpha,p} = |f(0)|^p + \int_0^1 (\mathrm{d}/\,\mathrm{d}r) M_p(r,f)^p (1-r^2)^{\alpha-1}\,\mathrm{d}r < \infty$, where M_p is the normalized circular p-mean.

Answered: no. Evidence: arXiv.

Dayan, Llinares, and Monsalve-López, *A^p_α classes in the Dirichlet range: inner-outer factorization, Carleson measures and weak products*, arXiv:2604.21347v2, Introduction, give a monomial counterexample for every $0 < \alpha < 1$, so the proposed monotonicity fails throughout the stated range.

For $f(z) = z^n$ the pth power of the norm is $\Gamma(\alpha)\Gamma(np/2+1)/\Gamma(np/2+\alpha)$. For suitable n this norm exceeds one at finite p but tends to one as $p \to \infty$, contradicting increase.

22.24. Exact norm of a weighted Bergman multiplier

> Question 1.6. For a multiplier g with $\|g\|_{B_{(\beta-\alpha)/2}} = 1$, what is its exact multiplier norm?

Here $\beta > \alpha > -1$, $M_g : A^2_\alpha \to A^2_\beta$ is multiplication by g, and $\|g\|_{B_\delta} = \sup_{z\in\mathbb{D}} |g(z)|(1-|z|^2)^\delta$. Source: Jianjun Jin, "On the Norms of the Multiplication Operators between Weighted Bergman Spaces," arXiv:2603.16170v9, Question 1.6, p. 3.

Partial. Evidence: Sketch.

Our calculation gives exact norms for normalized monomial multipliers and an explicit lower bound for the general supremum. The exact norm for an arbitrary analytic multiplier remains open.

22.25. Sharp supremum of weighted Bergman multiplier norms

> Question 1.7. What is the exact value of
>
> $$M(\alpha,\beta) := \sup_{\|g\|_{B_{(\beta-\alpha)/2}}=1} \|g\|_{M_{\alpha,\beta}}?$$

Here $\beta > \alpha > -1$ and $M_{\alpha,\beta}$ denotes the norm of the multiplication operator $M_g : A^2_\alpha \to A^2_\beta$. Source: Jianjun Jin, "On the Norms of the Multiplication Operators between Weighted Bergman Spaces," arXiv:2603.16170v9, Question 1.7, p. 3. The source computes the norm for $g_0(z) = (1-z^2)^{-(\beta-\alpha)/2}$ in Theorem 1.8; the separate Question 5.17 is outside this target.

Partial. Evidence: Sketch.

The monomial calculation supplies an explicit lower bound, but no interaction-sensitive upper bound determines the supremum.

22.26. Does every Banach space of cotype q have hypercontractive polynomial cotype q?

> The question we now ask is whether or not every Banach space with cotype q has hypercontractive polynomial cotype q.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, §25.4, p. 623. Here hypercontractive polynomial cotype q means that a constant C satisfies

$$\Big(\sum_{|\alpha|\le m} \|x_\alpha\|_X^q\Big)^{1/q} \le C^m \int_{\mathbb{T}^n} \Big\|\sum_{|\alpha|\le m} x_\alpha z^\alpha\Big\|_X \,\mathrm{d}m_n(z)$$

for all degrees m, dimensions n, and finite coefficient families in X.

Answered: yes. Evidence: Published.

Carando, Marceca, and Sevilla-Peris, *Hausdorff–Young type inequalities for vector-valued Dirichlet series*, doi:10.1090/tran/8147, Theorem 3.1 and Remark 3.3, prove that cotype q is equivalent to exponential-degree polynomial cotype q, including an L^1 norm on the right. This gives the affirmative answer for every Banach space of cotype q.

Cotype is first iterated on tetrahedral homogeneous polynomials. Transfer to Walsh polynomials and a decomposition of general monomials extend the estimate with only exponential degree loss; polynomial Kahane inequalities replace the right-hand norm by L^1.

22.27. **Haagerup sharp Steinhaus Khintchine constants**

> "Our main result is a step towards substantiating the Haagerup conjecture (cf. Pełczyński [8], Section 4)."

The inferred target asks whether the optimal lower Steinhaus constant is $A_p = \min\{\gamma_p, d_p\}$ for $0 < p < 2$, and the optimal upper constant is $B_p = \gamma_p$ for $p > 2$, where $\gamma_p = \Gamma((p+2)/2)^{1/p}$ and $d_p = \sqrt{2}(\Gamma((p+1)/2)/(\sqrt{\pi}\Gamma((p+2)/2)))^{1/p}$. Sawa's §4.3 defines these candidates; the added-in-proof result gives the lower bound with constant γ_p for $p_0 < p \le 2$, where $p_0 = 0.47562\ldots$, and the sharp upper bound γ_p for $p \ge 2$. Source: Jacek Sawa, "Some Remarks on the Khintchine Inequality for Complex Steinhaus Variables," doi:10.4064/sm-81-1-107-126, §4.3, pp. 124–125 and Added in proof, p. 125 (scan pp. 10–11). The quoted sentence motivates but does not itself state the inferred sharp-constant question. Here $A_p\|a\|_2 \le (\mathbb{E}|\sum_j a_j\xi_j|^p)^{1/p} \le B_p\|a\|_2$ for independent uniform-circle variables ξ_j; [8] refers to Pełczyński, *Norms of classical operators in function spaces*, https://numdam.org/item/AST_1985__131__137_0/, Section 4.

Partial. Evidence: arXiv.

Rapaport, Tkocz, and Wu, *Negative moments of Steinhaus sums*, arXiv:2512.09077v2, Section 3, identify and repair the induction gap in König's lower-constant proof for $0 < p < 1$. This supplies a preprint result for part of Sawa's proposed constants; the full two-sided conjecture for all $0 < p < 2$ and $p > 2$ is not settled here.

The repair uses a global induction, with separate estimates for small maximal coefficients and for a dominant coefficient, so that the inductive hypothesis remains valid after merging variables. It repairs König, *On the best constants in the Khintchine inequality for Steinhaus variables*, doi:10.1007/s11856-013-0006-y, main theorem, rather than assuming the invalid restricted induction.

22.28. **Sphere orthogonal-polynomial hypercontractivity**

> Conjecture. If μ is the uniform measure on a sphere in $\mathbb{R}^d$, then $\|T_\omega\|_{p,q} = 1$ for $-\sqrt{(p-1)/(q-1)} \le \omega \le \sqrt{(p-1)/(q-1)}$.

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," doi:10.1007/BF02384302, Section 6, Conjecture, pp. 107–108. Here $1 < p \le q < \infty$ and T_ω multiplies the degree-k orthogonal-polynomial component by ω^k; Nelson's real-parameter condition is $|\omega| \le \sqrt{(p-1)/(q-1)}$.

Answered: yes. Evidence: Published.

Beckner, *Sobolev inequalities, the Poisson semigroup, and analysis on the sphere S^n*, doi:10.1073/pnas.89.11.4816, Theorem 1, proves the sharp logarithmic Sobolev inequality on spheres; its hypercontractive consequence gives the Nelson range $|\omega| \le \sqrt{(p-1)/(q-1)}$ for the degree multiplier. This settles Janson's conjecture in the stated probability normalization.

Differentiating the $L^{q(t)}$ norm along the classical Poisson semigroup reduces the contraction to Beckner's sharp logarithmic Sobolev bound for the degree generator. Antipodal reflection gives the negative-ω case.

22.29. **Gegenbauer probability-measure hypercontractivity**

> Conjecture. The same is true for the measures $c_\alpha(1-x^2)^\alpha\,\mathrm{d}x$ on $(-1,1)$, $\alpha > -1$ (Gegenbauer polynomials).

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," doi:10.1007/BF02384302, Section 6, Conjecture, p. 108. Here $1 < p \le q < \infty$ and T_ω multiplies the degree-k orthogonal-polynomial component by ω^k; Nelson's real-parameter condition is $|\omega| \le \sqrt{(p-1)/(q-1)}$.

Answered: no. Evidence: Sketch.

A degree-two polynomial and an exact rational moment computation disprove the universal conjecture for α sufficiently close to -1.

Answers, "Gegenbauer probability-measure hypercontractivity", p. 29.

Full proof note.

22.30. Measures with sharp orthogonal-polynomial hypercontractivity

> "The general and still unsolved problem is to decide when the operators are contractions and, in particular, for which measures this holds with Nelson's conditions on ω."

Source: Svante Janson, "On hypercontractivity for multipliers on orthogonal polynomials," doi:10.1007/BF02384302, Introduction, p. 97. Here the operators are Janson's orthogonal-polynomial degree multipliers. Here $1 < p \le q < \infty$ and T_ω multiplies the degree-k orthogonal-polynomial component by ω^k; Nelson's real-parameter condition is $|\omega| \le \sqrt{(p-1)/(q-1)}$.
Partial. Evidence: Sketch.

The source supplies sufficient conditions for several classes of measures and shows the Nelson condition is necessary. A general characterization of measures remains open.

23. Bohr radii and monomial convergence

23.1. Second-order asymptotics of the polydisc Bohr radius

$$\lim_{n\to\infty} \frac{K_n}{\sqrt{(\log n)/n}} = 1.$$

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, Theorem 10.23, p. 265. The next-order question is inferred from this leading asymptotic; the theorem does not state a second-order problem. Here $K_n = K(\mathbb{D}^n)$ and $K(B)$ is the largest r for which $\sum_\alpha |a_\alpha z^\alpha| \le \sup_B |f|$ for every bounded analytic $f(z) = \sum_\alpha a_\alpha z^\alpha$ and every $z \in rB$.
Partial. Evidence: Sketch.

The leading equivalent is known. Pellegrino–Teixeira, *Polynomial growth of complex polynomial Bohnenblust–Hille constants*, arXiv:2608.16584v3, Theorem D and equations (9.5)–(9.6), claim an $O(\log\log n/\log n)$ remainder and bounds for the normalized liminf and limsup, but not the true second-order term.

23.2. Critical monomial-convergence boundary on c_0

> For $c < 1$, $(c/\sqrt{n})_{n\in\mathbb{N}} \in \operatorname{mon} H^\infty(B_{c_0})$, while for $c > 1$ the corresponding sequence is not in $\operatorname{mon} H^\infty(B_{c_0})$. In the case $c > 1$, the sequence is understood as constantly equal to $1/c$ for $1 \le n \le c^2$ and $c/\sqrt{n}$ for $n > c^2$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, §10.6, formula (10.54), pp. 260–261. The endpoint question asks whether the critical $n^{-1/2}$ sequence, with its first coordinate clipped to lie in the open ball, belongs to the set. This is inferred from the strict threshold and the source's finite-coordinate observation; it is not the same as the displayed $c > 1$ sequence. Here $\operatorname{mon} H^\infty(B_X)$ consists of points where the monomial series of every bounded holomorphic function on B_X converges absolutely; $\operatorname{mon} P_m(X)$ is the corresponding set for all continuous m-homogeneous polynomials, and z^* is the decreasing rearrangement of $|z_j|$.
Partial. Evidence: Sketch.

The membership criterion is known when the normalized limsup is below 1 and fails above 1. The critical sequence remains unresolved.

23.3. Exact asymptotics of the Bohr radius of ℓ_p^n balls

> "Whether or not the equality from (19.2) holds for $1 < p < 2$ is unknown."

Equation (19.2) in the source is the proposed limit

$$\lim_{n\to\infty} \frac{K(B_p^n)}{((\log n)/n)^{1-1/\min(p,2)}} = 1.$$

For $1 < p < 2$, this asks whether the normalized Bohr radius tends to 1 with exponent $1-1/p$. Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, Chapter 19, after Theorem 19.1, equations (19.1)–(19.2), p. 487. Here $B_p^n = \{z \in \mathbb{C}^n : \sum_j |z_j|^p < 1\}$, and $K(B_p^n)$ is the largest r for which every bounded analytic $f = \sum a_\alpha z^\alpha$ satisfies $\sum |a_\alpha z^\alpha| \le \sup_{B_p^n} |f|$ on rB_p^n. The source's (19.2) is the displayed normalized limit with denominator $((\log n)/n)^{1-1/p}$ in this range.

Answered: no. Evidence: Sketch.

Our public note proves a strict upper constant below 1 for each fixed $1 < p < 2$, disproving the proposed limit. It does not determine whether the normalized Bohr radius converges.

Answers, "Exact asymptotics of the Bohr radius of ℓ_p^n balls", p. 30.

Public proof.

23.4. Exact set of bounded holomorphic monomial convergence on ℓ_p

For $1 < p < \infty$, define q by $1/q = 1/2 + 1/\max\{p, 2\}$. Then $B_{\ell_p} \cap \ell_q \subset \operatorname{mon} H^\infty(B_{\ell_p})$. Conversely, if $z \in \operatorname{mon} H^\infty(B_{\ell_p})$, then $z_n^* \lesssim (\log n)^{1-1/\min\{p,2\}}/n^{1/2+1/\max\{p,2\}}$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, Theorem 20.23, p. 523. The exact-set target is inferred from the logarithmic gap between the theorem's sufficient and necessary conditions. Here $\operatorname{mon} H^\infty(B_X)$ consists of points where the monomial series of every bounded holomorphic function on B_X converges absolutely; $\operatorname{mon} P_m(X)$ is the corresponding set for all continuous m-homogeneous polynomials, and z^* is the decreasing rearrangement of $|z_j|$.

Partial. Evidence: Sketch.

The book gives strict inclusions and rearrangement bounds for the set of monomial convergence. Its exact description remains open.

23.5. Exact monomial-convergence sets for homogeneous polynomials on ℓ_p

For $1 < p < \infty$ and $m \ge 2$, $\ell_{2m/(m-1),\infty} \cdot \ell_p \subset \operatorname{mon} P_m(\ell_p) \subset \ell_{((m-1)/(2m)+1/p)^{-1},\infty}$ for $2 \le p < \infty$, and $\ell_{mp,1} \subset \operatorname{mon} P_m(\ell_p) \subset \ell_{mp,\infty}$ for $1 < p \le 2$.

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, Theorem 20.7, p. 510. Determining the exact universal convergence set between these inclusions is an inferred target; the theorem itself gives the stated bounds. Here $\operatorname{mon} H^\infty(B_X)$ consists of points where the monomial series of every bounded holomorphic function on B_X converges absolutely; $\operatorname{mon} P_m(X)$ is the corresponding set for all continuous m-homogeneous polynomials, and z^* is the decreasing rearrangement of $|z_j|$. For $s < \infty$, $\ell_{r,s}$ has $\sum_{j\ge1} j^{s/r-1}(z_j^*)^s < \infty$, while $\ell_{r,\infty}$ has $\sup_j j^{1/r} z_j^* < \infty$; the product of sequence spaces means coordinatewise products.

Partial. Evidence: Sketch.

Known inclusions characterize several large subsets of the universal convergence set, but the exact set for each degree and $1 < p < \infty$ is unknown.

23.6. Exact values of multidimensional Bohr radii

Bohr "gives the exact value of K_1 (actually the only exact value known so far)."

Source: Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, §8.1, p. 183. The question for K_n with $n \ge 2$ or for other Reinhardt domains is inferred from this statement, not explicitly posed in the quoted sentence. Here $K_n = K(\mathbb{D}^n)$ and $K(B)$ is the largest r for which $\sum_\alpha |a_\alpha z^\alpha| \le \sup_B |f|$ for every bounded analytic $f(z) = \sum_\alpha a_\alpha z^\alpha$ and every $z \in rB$.

Open. Evidence: Heuristic.

The exact multidimensional Bohr radius is unknown beyond the classical one-variable and special arithmetic-domain cases.

23.7. Optimal growth of the complex multilinear Bohnenblust–Hille constants

> "The optimal exponent here so far seems not to be relevant in our context."

This sentence concerns the optimal growth exponent for the complex multilinear Bohnenblust–Hille constants B_m. The natural question of their optimal growth is inferred from this discussion; boundedness is stronger than merely asking whether the exponent is zero. Source: Defant, García, Maestre, and Sevilla-Peris, *Dirichlet Series and Holomorphic Functions in High Dimensions*, doi:10.1017/9781108691611, §6.7 Notes, p. 152. Here B_m is the dimension-uniform constant bounding the coefficient $\ell_{2m/(m+1)}$ norm of a complex m-linear form by its supremum norm on products of unit polydiscs, and γ is Euler's constant.

Partial. Evidence: Published.

Bayart, Pellegrino, and Seoane-Sepúlveda, *The Bohr radius of the n-dimensional polydisk is equivalent to* $\sqrt{(\log n)/n}$, arXiv:1310.2834v2, Proposition 3.1 and Corollary 3.2, prove $B_m \le Cm^{(1-\gamma)/2}$. Their exact exponent $(1-\gamma)/2$ is approximately 0.21139217; the textbook's displayed decimal 0.21392 is inconsistent. The optimal growth and dimension-free boundedness remain open.

Mixed-norm interpolation and the sharp Steinhaus Khintchine inequality give a recursive bound for B_m in terms of B_{m-1}. Expanding the logarithms of the resulting gamma-function product yields the power $(1-\gamma)/2$.

23.8. Growth of the complex polynomial Bohnenblust–Hille constants

> "There are two ways to improve this constant: to find a better constant in the multilinear inequality and/or to use better polarization constants."

The passage concerns the polynomial Bohnenblust–Hille constants B_m^{pol}. The question of their optimal growth, in particular whether $\sup_m B_m^{\mathrm{pol}} < \infty$, is inferred from the discussion. Source: Defant, García, Maestre, and Sevilla-Peris, *Dirichlet Series and Holomorphic Functions in High Dimensions*, doi:10.1017/9781108691611, §6.7, p. 152. Theorem 8.19 and Theorem 10.22 give the cited subexponential controls and $\lim_m (B_m^{\mathrm{pol}})^{1/m} = 1$. Here B_m^{pol} is the dimension-uniform constant in $(\sum_{|\alpha|=m} |a_\alpha|^{2m/(m+1)})^{(m+1)/(2m)} \le B_m^{\mathrm{pol}} \sup_{\mathbb{D}^n} |\sum a_\alpha z^\alpha|$.

Partial. Evidence: Sketch.

The source's Theorems 8.19 and 10.22 give subexponential upper bounds and the exact root limit $\lim_m (B_m^{\mathrm{pol}})^{1/m} = 1$. These results do not determine whether the constants are bounded or their optimal subexponential growth.

23.9. Lempert's problem on monomial convergence and ℓ_1

> "Does a Banach sequence space X equal ℓ_1 whenever $rB_X \subset \operatorname{mon} H^\infty(B_X)$ for some $0 < r < 1$?"

Here X has a 1-unconditional basis and $\ell_1 \subset X \subset c_0$. Defant, García, Maestre, and Sevilla-Peris, Lemma 20.27, identify equivalent formulations using finite sections and the Bohr radii. Source: Lempert's question as quoted by Defant, García, Maestre, and Sevilla-Peris, "Dirichlet Series and Holomorphic Functions in High Dimensions," doi:10.1017/9781108691611, §20.5, pp. 526–527, Question 20.26 and Lemma 20.27. Here $\operatorname{mon} H^\infty(B_X)$ consists of points where the monomial series of every bounded holomorphic function on B_X converges absolutely; $\operatorname{mon} P_m(X)$ is the corresponding set for all continuous m-homogeneous polynomials, and z^* is the decreasing rearrangement of $|z_j|$.

Answered: no. Evidence: arXiv.

Wang, *Domains of monomial convergence in Banach sequence spaces*, arXiv:2610.08873v1, Theorem 4.16 with Proposition 4.5 and Corollary 4.7, claims a complex Banach sequence space Y strictly between ℓ_1 and c_0 for which every Reinhardt domain has the asserted monomial convergence. This gives a negative answer conditional on the unrefereed proof.

The construction uses a symmetric sequence-space norm with rapidly separated finite support scales. An exponential-degree estimate for absolute coefficient forms then gives normal monomial convergence on every Reinhardt domain through Taylor expansion; the strict inclusions exclude $Y = \ell_1$.

23.10. Polynomial Bohnenblust–Hille constants on the Boolean cube

> "The aim is to show that $BH(d)$ is at most of the polynomial growth or is even bounded by some universal constant."

Here $BH(d)$ is the Bohnenblust–Hille constant for d-homogeneous polynomials on the Boolean cube. The inferred question is whether $BH(d)$ has polynomial growth in d, or is bounded independently of d; the original AimPL problem-list numbering is unavailable, so no Problems 3.1–3.2 wording is claimed. Source: AIM workshop report, "Hypercontractivity and Quantum Information Theory," https://aimath.org/pastworkshops/hypercubequantumrep.pdf, Bohnenblust–Hille section, p. 2.

Answered: yes. Evidence: arXiv.

Ivanisvili, *Polynomial growth of Bohnenblust–Hille constants on the Hamming cube*, arXiv:2609.12427, Theorem 1.1, and Slote, Tseng, and Volberg, *An $m^{2.943}$ Bohnenblust–Hille bound on the Boolean cube*, arXiv:2609.21144, Theorem 2.1, prove polynomial-in-degree Bohnenblust–Hille bounds on the Boolean cube for degree-at-most-d polynomials; restriction to homogeneous polynomials gives polynomial growth for $BH(d)$. The optimal power is unresolved.

Both proofs combine random bicoloring of square-free monomials with mixed-norm coefficient interpolation and Bonami–Beckner hypercontractivity. Weighted sums of all degree levels satisfy a strict contraction, while derivative estimates handle the finitely many low levels.

23.11. Asymptotics of the Bohr radius for degree-N polynomials (Khavinson, Fournier)

> "What is the rate of convergence of R_N to $1/3$ as $N \to \infty$? In other words, what is the asymptotic of $R_N - 1/3$?"

Here R_N is the Bohr radius for polynomials of degree at most N bounded by one on $\mathbb{D}$. The source says Fournier suggested the asymptotic $R_N - 1/3 \sim \pi^2/(3N^2)$. Source: Dmitry Khavinson, after Guadarrama and Fournier, "Problems for Polynomials of One and Several Variables," Oberwolfach Report 06/2014, doi:10.4171/OWR/2014/06, Problem Session, §3, item 3, p. 382.

Answered: yes. Evidence: Published.

Chu, *Asymptotic Bohr radius for the polynomials in one complex variable*, doi:10.1090/conm/638/12804, Theorem 1.2, proves $R_N = 1/3 + \pi^2/(3N^2) + o(N^{-2})$, confirming Fournier's proposed asymptotic.

Fournier's determinant criterion identifies R_N as the smallest root of a Toeplitz determinant. Elementary row operations give a second-order determinant recurrence, and the location of its first trigonometric root yields the N^{-2} asymptotic.

23.12. Are the perfect squares stationary? (Queffélec)

> Is the set S of perfect squares stationary?

Source: Hervé Queffélec, "Dirichlet Series and Function Theory in Polydiscs," Oberwolfach Report 06/2014, doi:10.4171/OWR/2014/06, Problem Session, §3.1, Question 1, pp. 385–386 (reported by O. F. Brevig). Here stationary means that Pisier's randomized supremum norm is bounded by a constant times the ordinary supremum norm.

Answered: no. Evidence: Sketch.

Our explicit polynomials supported on odd squares have Pisier norm larger than their supremum norm by a factor of order $\sqrt{\log N}$, so the full square set is not stationary.

Answers, "Are the perfect squares a stationary set?", p. 31.

Full proof note.

24. Random multiplicative questions

24.1. Farmer–Gonek–Hughes global zeta maximum

"Conjecture A.

$$\max_{t\in[0,T]}\left|\zeta\left(\frac{1}{2}+it\right)\right| = \exp\left((1+o(1))\sqrt{\tfrac{1}{2}\log T\log\log T}\right).$$

"

Source: David W. Farmer, S. M. Gonek, and C. P. Hughes, "The Maximum Size of L-Functions," arXiv:math/0506218v3, Conjecture A, equation (1.3), p. 2; published in *Journal für die reine und angewandte Mathematik*, doi:10.1515/CRELLE.2007.064 609 (2007), 215–236.

Partial. Evidence: Sketch.

No proof of the global maximum conjecture is known in the bounded search. Our conditional coefficient-spread result is an obstruction for one resonator family only.

24.2. Sharp logarithmic exponent of Gál sums for $0 < \alpha < 1/2$

"For every α, $0 < \alpha < 1/2$, there exist positive constants $a(\alpha)$ and $b(\alpha)$ such that

$$N^{1-2\alpha}(\log N)^{a(\alpha)} \le \Gamma_\alpha(N) \le N^{1-2\alpha}(\log N)^{b(\alpha)}$$

for sufficiently large N."

Here $\Gamma_\alpha(N) = \sup_{|M|=N}|M|^{-1}\sum_{m,n\in M}\big((m,n)/[m,n]\big)^\alpha$ is the normalized Gál sum over arbitrary finite sets of positive integers; (m,n) and $[m,n]$ denote their greatest common divisor and least common multiple. The source does not ask for a single sharp logarithmic exponent; the target here is the inferred question whether $\Gamma_\alpha(N) = N^{1-2\alpha}(\log N)^{c(\alpha)+o(1)}$ for some $c(\alpha)$. Source: Andriy Bondarenko, Titus Hilberdink, and Kristian Seip, "Gál-Type GCD Sums beyond the Critical Line," arXiv:1512.03758v2, Theorem 1, p. 2.

Partial. Evidence: Sketch.

Bondarenko, Hilberdink, and Seip give matching polynomial-scale bounds with nonmatching logarithmic exponents. The exact exponent function remains open.

24.3. Gál sums for $0 < \alpha < 1/2$ over square-free sets: can the log power be removed?

> "One may notice that it would not be possible to construct a similar example if we required the set to consist only of square-free numbers. Hence it remains an open problem to prove the analogue of Theorem 1 in the square-free case. More specifically, we may ask whether the logarithmic power can be discarded in this case as well."

For $0 < \alpha < 1/2$, let $\Gamma_\alpha^{\mathrm{sf}}(N)$ be the normalized Gál maximum over square-free N-element sets. The question is whether the logarithmic factor in the unrestricted estimate can be removed, that is, whether $\Gamma_\alpha^{\mathrm{sf}}(N) \asymp N^{1-2\alpha}$. Source: Andriy Bondarenko, Titus Hilberdink, and Kristian Seip, "Gál-Type GCD Sums beyond the Critical Line," arXiv:1512.03758v2, §1, p. 3.

Partial. Evidence: Sketch.

The known unrestricted estimates include logarithmic factors, while the square-free lower-bound construction does not settle whether they can be removed.

24.4. Exact a.s. growth of random multiplicative sums

Determine the almost sure growth rate (in N) of the character sum $\sum_{n=1}^{N}\chi(n)$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 4.6, p. 6. Immediately before the problem, $\chi(p)$ are defined as independent uniform random variables on $\mathbb{T}$ and extended multiplicatively.

Partial. Evidence: Sketch.

Harper, *International Mathematics Research Notices*, doi:10.1093/imrn/rnab299 (2023), "Almost Sure Large Fluctuations of Random Multiplicative Functions," Theorem 1, proves arbitrarily large fluctuations at the 1/4 exponent, with any prescribed diverging loss. Durkan and Pearce-Crump, "A Sharp Almost Sure Upper Bound for Partial Sums of Random Multiplicative Functions," arXiv:2607.29429v2, Theorem 1 and Corollary 1.1, prove the matching upper exponent; Verreault, "Almost Sure Upper Bound for Sums of Random Multiplicative Functions and Critical Chaos," arXiv:2608.21354v2, Theorem 1.1, gives an upper bound with a $\log_3 x$ factor. Thus the exponent is determined, but the exact almost-sure order and normalized limsup remain open.

24.5. GCD sums and Hardy operator theory

Link the estimates for GCD sums to the function and operator theory of the spaces H^p.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 5.1, p. 7. Here the source uses normalized sums

$$\Gamma_\alpha(N) = \sup_{|M|=N} N^{-1} \sum_{m,n\in M} ((m,n)^2/(mn))^\alpha,$$

where (m,n) is the greatest common divisor.

Partial. Evidence: Sketch.

Aistleitner, Berkes, and Seip, "GCD Sums from Poisson Integrals and Systems of Dilated Functions," doi:10.4171/JEMS/537, equation (23), Theorem 5 and the following discussion, identify GCD sums with Poisson integrals and give H^2 spectral consequences. Further operator-theoretic applications remain unspecified and open.

24.6. GCD sums and zeta applications

Develop further applications to and links with the Riemann zeta function.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 5.2, p. 7. In context, the requested applications are of the GCD-sum estimates $N^{-1}\sum_{m,n\in M}((m,n)^2/(mn))^\alpha$, with (m,n) the greatest common divisor.

Partial. Evidence: Sketch.

De la Bretèche and Tenenbaum, "Sommes de Gál et applications," doi:10.1112/plms.12224, Theorems 1.1 and 1.4, give sharp Gál-sum estimates and zeta resonance applications; a separate square-free divisor-form question is answered below. The broader programme has no single completion criterion.

24.7. Fyodorov–Hiary–Keating tightness

Consider $[0,T]$ as a probability space, with normalised Lebesgue measure, and denote the corresponding variable by $\omega \in [0,T]$. Then, as $T \to \infty$, one has $\max_{h\in[0,1]} \log|\zeta(1/2 + ih + i\omega)| = \log\log T - \frac{3}{4}\log\log\log T + E$, where the error term E is bounded in probability as $T \to \infty$.

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 6.1, pp. 9–10. The later Arguin–Bourgade–Radziwiłł result uses τ uniform on $[T,2T]$ and $|h| \le 1$, so its interval and shift window do not literally match this statement.

Partial. Evidence: arXiv.

Arguin, Bourgade, and Radziwiłł, "The Fyodorov–Hiary–Keating Conjecture. I," arXiv:2007.00988, Theorem 1, and "The Fyodorov–Hiary–Keating Conjecture. II," arXiv:2307.00982, Theorem 1 and Corollary 1, state the Fyodorov–Hiary–Keating tightness claim. Their proof is written for τ uniform on $[T,2T]$ and $|h| \le 1$, whereas the source formulation uses ω uniform on $[0,T]$ and $h \in [0,1]$; the literal one-unit lower-tail transfer is not verified here. They constrain prime-sum trajectories between upper and lower barriers, estimate the resulting events by Dirichlet-polynomial moments, and compare high points of the prime sums with zeta through a finite-Euler-product approximation.

24.8. Multiplicative chaos of the random prime field: supercritical maximum and freezing

> Let the field X be defined as in (8). Study the properties non-Gaussian chaos "$\exp(\beta X(x))$."

Source: Saksman and Seip, "Some open questions in analysis for Dirichlet series," arXiv:1601.01616, Problem 6.2, p. 10. The wording above preserves the source's printed phrase, which omits "of the"; the random prime field is defined by equation (8): $X(x) = \sum_p p^{-1/2}(\cos(x\log p)\cos\theta_p + \sin(x\log p)\sin\theta_p)$, understood through prime cutoffs, with independent θ_p uniform on $[0, 2\pi]$.

Partial. Evidence: arXiv.

Saksman and Webb, "Multiplicative Chaos Measures for a Random Model of the Riemann Zeta Function," arXiv:1604.08378, Theorems 4 and 5, construct subcritical and critical chaos for the direct random-zeta prime model. They couple the prime field to a Gaussian field with a uniformly convergent continuous error; Gaussian multiplicative-chaos convergence then transfers after multiplication by the exponential of that error. These results give a concrete partial analogue, while the source's broader programme about the random prime field remains open.

24.9. Subcritical zeta-chaos convergence

> "For $\beta \in (0, 2)$ the random densities $(\log T)^{-\beta^2/4}|\zeta(1/2 + ix + iT)|^\beta$, $x \in [0, 1]$, converge in distribution to a constant multiple of the multiplicative chaos measure described in Theorem 1.8."

The source's printed formula has iT; the intended randomly shifted interpretation takes T to be ωT with ω uniform on $[0, 1]$. Source: Saksman and Webb, "The Riemann Zeta Function and Gaussian Multiplicative Chaos: Statistics on the Critical Line," arXiv:1609.00027v3, Conjecture 1.10, p. 14. Here Theorem 1.8 constructs the weak limit η_β of $|\zeta_{N,\mathrm{rand}}(1/2 + ix)|^\beta \, \mathrm{d}x/\mathbb{E}|\zeta_{N,\mathrm{rand}}(1/2 + ix)|^\beta$, where $\zeta_{N,\mathrm{rand}}(s) = \prod_{j=1}^N (1 - p_j^{-s} e^{2\pi i\theta_j})^{-1}$ where p_j is the jth prime and the θ_j are independent uniforms on $[0, 1]$. The source's Theorem 1.8 concerns this random model.

Open. Evidence: Heuristic.

Saksman and Webb's Theorems 1.7–1.8 couple the random Euler product to a Gaussian field and establish its multiplicative-chaos limit, but the actual randomly shifted zeta measure in this conjecture remains open.

24.10. Critical zeta-chaos convergence

> "The previous conjecture holds for $\beta = \beta_c = 2$ as soon as one adds the normalizing factor $(\log\log T)^{1/2}$."

Source: Saksman and Webb, "The Riemann Zeta Function and Gaussian Multiplicative Chaos: Statistics on the Critical Line," arXiv:1609.00027v3, Conjecture 1.11, p. 14. The preceding conjecture is convergence in distribution of $(\log T)^{-1}|\zeta(1/2 + i\omega T + ix)|^2 \, \mathrm{d}x$ on $[0, 1]$ after multiplication by $\sqrt{\log\log T}$, with ω uniform on $[0, 1]$; the target is the critically rescaled random-Euler-product chaos of Theorem 1.9.

Expected yes. Evidence: Heuristic.

Harper announced a proof of the critical case in "The Riemann Zeta Function, Random Multiplicative Functions, and Gaussian Multiplicative Chaos" (MPIM talk, 2025), https://www.mpim-bonn.mpg.de/node/14389, but no public proof manuscript was found; it remains an announcement rather than a checked resolution.

24.11. Mesoscopic real zeta-chaos convergence

> "There are mesoscopic analogues of the above conjectures and they can be formulated in a similar way as in Section 5."

The target is the pair of real positive-measure conjectures for the actual zeta function on the shrinking windows described in Section 5. Source: Saksman and Webb, "The Riemann

Zeta Function and Gaussian Multiplicative Chaos: Statistics on the Critical Line," arXiv:1609.00027v3, Conjecture 1.12, p. 14. Section 5 studies windows $x \mapsto \zeta(1/2 + i\omega T + i\delta_T x)$ with $\delta_T \downarrow 0$ sufficiently slowly; its complex-field limit has a divergent random scalar factor e^{Y_T}, with Y_T distributed as $\sqrt{\log(1/\delta_T)}Z + R$, where Z is standard complex Gaussian and R has exponential moments. The question is the corresponding normalized real modulus-power measures, both below and at the critical exponent 2.

Open. Evidence: Heuristic.

Saksman and Webb's Theorems 1.2 and 1.5 prove mesoscopic complex-chaos convergence, while the actual zeta-function analogues remain open.

24.12. Helson better-than-square-root first moment

> "For example, Helson [15] conjectured in the Steinhaus case one should have
>
> $$\mathbb{E}|\sum_{n\le x} f(n)| = o(\sqrt{x}),$$
>
> and observed that if this is true then a certain generalisation of Nehari's theorem from harmonic analysis is false."

Here f is a Steinhaus random multiplicative function; the source's reference [15] is Henry Helson, "Hankel Forms," doi:10.4064/sm198-1-5 (2010). Source: Harper, "Moments of Random Multiplicative Functions, I: Low Moments," arXiv:1703.06654v1, p. 2.

Answered: yes. Evidence: Published.

Harper, "Moments of Random Multiplicative Functions, I: Low Moments," doi:10.1017/fmp.2019.7, *Forum of Mathematics, Pi* 8 (2020), Theorem 1, proves that the Steinhaus first moment has order $\sqrt{x}/(\log\log x)^{1/4}$, which implies Helson's prediction. The proof conditions on small primes and reduces the moment to a fractional moment of integrated random Euler products; barrier estimates for the resulting approximately Gaussian multiplicative chaos produce the log-log saving.

24.13. Square-root low-moment counter-conjecture

> "They also conjectured, in opposition to Helson's conjecture, that $\mathbb{E}|\sum_{n\le x} f(n)|^{2q} \sim C(q)x^q$ as $x \to \infty$, for each fixed $0 \le q \le 1$."

"They" refers to Harper, Nikeghbali, and Radziwiłł; the source discusses the Steinhaus model and then the analogous Rademacher model. Source: Harper, "Moments of Random Multiplicative Functions, I: Low Moments," arXiv:1703.06654v1, p. 3.

Answered: no. Evidence: Published.

Harper, "Moments of Random Multiplicative Functions, I: Low Moments," doi:10.1017/fmp.2019.7, *Forum of Mathematics, Pi* 8 (2020), Theorems 1 and 2, proves that for each fixed $0 < q < 1$ the Steinhaus and Rademacher $2q$ moments are $o(x^q)$ by a diverging log-log factor, refuting the positive-constant prediction. The same conditioning and Euler-product reduction identifies a critical-chaos loss in both models, rather than the square-root moment scale predicted by the counter-conjecture.

24.14. Non-Gaussianity of normalized Rademacher sums

> "does one have a central limit theorem for $\sum_{n\le x} f(n)/\sqrt{\mathbb{E}|\sum_{n\le x} f(n)|^2}$?"

The source attributes the failure-of-CLT conjecture in the Rademacher case to Chatterjee. Source: Harper, "Moments of Random Multiplicative Functions, I: Low Moments," arXiv:1703.06654v1, p. 2, question (i), with the Rademacher attribution on p. 3.

Answered: no. Evidence: Published.

Harper, "Moments of Random Multiplicative Functions, I: Low Moments," doi:10.1017/fmp.2019.7, *Forum of Mathematics, Pi* 8 (2020), Corollary 1, proves that the mean-square-normalized sum tends to zero in probability in both the Steinhaus and Rademacher models, which precludes a nondegenerate Gaussian limit. The negative answer to the CLT question

verifies Chatterjee's non-Gaussianity conjecture. Corollary 1 follows from the fractional-moment saving by Markov's inequality: every fixed nonzero threshold has probability tending to zero.

24.15. **A limiting law for long random multiplicative sums**

> "One now has an obvious and interesting follow-up question, namely what is the distribution of the properly renormalised sum $\sum_{n\le x} f(n)/(\sqrt{x}/(\log\log x)^{1/4})$?"

The source paragraph discusses both the Steinhaus and Rademacher models. Source: Harper, "Moments of Random Multiplicative Functions, I: Low Moments," arXiv:1703.06654v1, p. 10.

Partial. Evidence: Sketch.

Gorodetsky and Wong, "On the Limiting Distribution of Sums of Random Multiplicative Functions," arXiv:2508.12956v3, Theorem 1.3, claim the normalized Steinhaus limit; the corresponding full Rademacher limit is not covered, so the two-model question is only partial.

24.16. **Sharp tail probabilities for random multiplicative sums**

> "It seems another interesting open question to determine the exact magnitude of the probabilities in Corollary 2."

Here Corollary 2 concerns $\mathbb{P}(|\sum_{n\le x} f(n)| \ge \lambda\sqrt{x}/(\log\log x)^{1/4})$, for $\lambda \ge 2$; Harper gives a fixed-λ prediction and a separate large-λ scale. Source: Harper, "Moments of Random Multiplicative Functions, I: Low Moments," arXiv:1703.06654v1, §1.2, p. 11.

Partial. Evidence: Sketch.

Gorodetsky and Wong, "On the Limiting Distribution of Sums of Random Multiplicative Functions," arXiv:2508.12956v3, Theorem 1.3, prove the limiting tail probability at each fixed threshold for the normalized Steinhaus sum. Harper, "Moments of Random Multiplicative Functions, II: High Moments," arXiv:1804.04114, *Annales de l'Institut Henri Poincaré, Probabilités et Statistiques* 56 (2020), Corollary 1.3, gives upper estimates in a high-threshold range. Uniform sharp estimates across fixed and growing thresholds, and the corresponding Rademacher limit, remain open.

24.17. **Weighted circular GMC moment formula**

> "Conjecture 1. Let $\gamma \in (0,2)$. For all real p such that $p < 4/\gamma^2$ we expect to have
>
> $$\mathbb{E}\left[\left(\frac{1}{2\pi}\int_0^{2\pi} |1-e^{i\theta}|^2 e^{\gamma X(e^{i\theta})/2}\,\mathrm{d}\theta\right)^p\right] = \frac{\Gamma(1-p\gamma^2/4)\Gamma(1+8/\gamma^2)\Gamma(1+4/\gamma^2-p)}{\Gamma(1-\gamma^2/4)^p\Gamma(1+4/\gamma^2)\Gamma(1+8/\gamma^2-p)}.$$
>
> and we can write again
>
> $$\frac{1}{2\pi}\int_0^{2\pi} |1-e^{i\theta}|^2 e^{\gamma X(e^{i\theta})/2}\,\mathrm{d}\theta \overset{\text{law}}{=} Y_\gamma X_2^{-1},$$
>
> with Y_γ, X_2 independent and $X_2 \sim B(1+4/\gamma^2, 4/\gamma^2)$. "

Here X is the circular Gaussian field with covariance $\mathbb{E}[X(e^{i\theta})X(e^{i\phi})] = 2\log|e^{i\theta}-e^{i\phi}|^{-1}$; exponentials mean regularized Gaussian multiplicative chaos. The variable Y_γ is its normalized unweighted total mass, Γ is the gamma function, and $\mathrm{Beta}(a,b)$ is the standard beta law. Source: Guillaume Rémy, "The Fyodorov-Bouchaud Formula and Liouville Conformal Field Theory," arXiv:1710.06897v3, Conjecture 1, equations (1.36)–(1.37), p. 10.

Answered: yes. Evidence: Published.

Rémy and Zhu, "Integrability of Boundary Liouville Conformal Field Theory," doi:10.1007/s00220-022-04455-1, Communications in Mathematical Physics 395 (2022), Theorem 1.7 and Corollary 1.12, prove the inserted circular Gaussian multiplicative chaos moment formula and limiting law. Their normalization matches the source after the parameter change $\beta = -4/\gamma$. They derive two shift equations for a boundary Liouville correlation using differential equations for degenerate insertions; the shifts determine the gamma-function expression uniquely, and its Mellin transform gives the independent-factor law.

24.18. **Divisor-restricted Gál form over square-free sets**

> « Le résultat suivant, relatif aux éléments de $\mathcal{A}$, laisse supposer que la taille maximale de $Q(M)$ lorsque M décrit les sous-ensemble de $\mathbb{N}_1$ de taille N vaut $\Gamma^*(N)^{1/2+o(1)}$. Cette question demeure ouverte en l'état. Rappelons la définition des normes $Q(M)$ en $(1 \cdot 6)$. »

The text immediately preceding this passage defines the $Q(M)$ in §5 as the divisor-restricted quadratic form, and asks about all square-free sets of size N, not only $M \in \mathcal{A}$. We write $\mathcal{Q}(M)$ for that restricted form in the answer to distinguish it from the full Gál-form norm $Q(M)$ in the introduction. Here $\mathbb{N}_1$ denotes the square-free integers and $\Gamma^*(N)$ is their maximal normalized Gál sum. Source: Régis de la Bretèche and Gérald Tenenbaum, "Sommes de Gál et applications," §5.1, before Lemma 5.2, version 8, p. 18 (published in *Proceedings of the London Mathematical Society*, arXiv:1804.01629 119 (2019), doi:10.1112/plms.12224). Here $\mathcal{A}$ consists of finite square-free sets M whose prime factors are at most the $|M|$th prime. Equation $(1 \cdot 6)$ defines the full norm $Q(M) = \sup_{\|c\|_2=1} |\sum_{m,n\in M} c_m c_n (m,n)/\sqrt{mn}|$, while the form under discussion is $\mathcal{Q}(M) = \sup_{\|c\|_2=1} |\sum_{m,n\in M,\, n|m} c_m c_n \sqrt{n/m}|$.

Answered: yes. Evidence: Sketch.

Our proof establishes the conjectured square-root exponent for the divisor-restricted form on square-free sets, using de la Bretèche and Tenenbaum's Theorem 1.2, equations (1.5), (1.7) and (5.5), for matching bounds.

Answers, "Divisor-restricted Gál form over square-free sets", p. 31.

Full proof note.

24.19. **Erdős/Halász failure of a.s. square-root bound**

> "On the other hand, Erdős [4] conjectured (in the Rademacher case) that almost surely $\sum_{n\le x} f(n) \neq O(\sqrt{x})$, whilst writing that 'As far as I know nobody has a plausible guess for the true order of magnitude of $|\sum_{n\le x} f(n)|$'."

The preceding paragraph states Halász's corresponding question for Steinhaus functions. Here [4] is Paul Erdős, "Some Applications of Probability Methods to Number Theory," Mathematical Statistics and Applications, volume B (1985), pp. 1–18. Source: Harper, "Almost Sure Large Fluctuations of Random Multiplicative Functions," arXiv:2012.15809, p. 3.

Answered: yes. Evidence: Published.

Harper, "Almost Sure Large Fluctuations of Random Multiplicative Functions," doi:10.1093/imrn/rnab299, *International Mathematics Research Notices* (2023), Theorem 1, proves that for each prescribed $V(x) \to \infty$ there are almost surely arbitrarily large x with $|\sum_{n\le x} f(n)| \ge \sqrt{x}(\log\log x)^{1/4}/V(x)$ in both models. In particular the proposed almost-sure $O(\sqrt{x})$ bound fails. Harper conditions on all but the largest primes and proves a multivariate Gaussian approximation; controlling the conditional covariances supplies enough roughly independent large fluctuations.

24.20. **Harper almost-sure logarithmic exponent**

> "As we will explain, it now seems plausible that a bound of the form
>
> $$\sqrt{x}(\log\log x)^{1/4+o(1)}$$
>
> is sharp in this problem, and if so Theorem 1 is quite close to best possible."

Source: Harper, "Almost Sure Large Fluctuations of Random Multiplicative Functions," arXiv:2012.15809, p. 3. Here Theorem 1 says that for each $V(x) \to \infty$, almost surely $|\sum_{n\le x} f(n)| \ge \sqrt{x}(\log\log x)^{1/4}/V(x)$ for arbitrarily large x, in either the Steinhaus or Rademacher model. The quote predicts a matching upper exponent.

Answered: yes. Evidence: arXiv.

Harper, "Almost Sure Large Fluctuations of Random Multiplicative Functions," doi:10.1093/imrn/rnab299, *International Mathematics Research Notices* (2023), Theorem 1, proves arbitrarily large lower fluctuations at the 1/4 exponent for each prescribed diverging

loss. Durkan and Pearce-Crump, arXiv:2607.29429v2, Theorem 1 and Corollary 1.1, prove the matching almost-sure upper exponent for both Steinhaus and Rademacher sums. Their proof stops a tilted Euler-product energy supermartingale and bounds higher conditional moments on thin prime blocks, making the exceptional probabilities summable. Verreault, "Almost Sure Upper Bound for Sums of Random Multiplicative Functions and Critical Chaos," arXiv:2608.21354v2, Theorem 1.1, proves another upper bound with a $\log_3 x$ factor. The exponent is determined, while the exact normalized limsup remains open.

24.21. **Smooth-number deterministic/random first-mean comparison for $x \le T^A$, $A \le 2$**

"For any fixed $A > 0$ and small $\epsilon > 0$, we have, very roughly speaking," followed by

$$\frac{1}{T}\int_{\substack{|t|\le T\\ t\not\approx 0}}\Bigg|\sum_{\substack{n\le x\\ P(n)\le x^{\epsilon}}} n^{-it}\Bigg|\,\mathrm{d}t \ll \mathbb{E}\Bigg|\sum_{\substack{n\le x\\ P(n)\le x^{\epsilon}}} f(n)\Bigg|, \qquad x \le T^A.$$

The source labels (1.2) a rough conjecture and does not quantify the exclusion $t \not\approx 0$. Our literal interpretation has no main-term subtraction and is false for every $A > 2$ and fixed $0 < \epsilon < 1$; that range is refuted by our written counterexample. Whether a precise version holds for $0 < A \le 2$ remains open. Source: Seth Hardy and Max Wenqiang Xu, "Helson's Conjecture for Smooth Numbers," arXiv:2511.03430v3, equation (1.2), p. 4. Here $P(n)$ is the largest prime factor, with $P(1) = 1$, and f is Steinhaus.
Partial. Evidence: Sketch.

24.22. **Smooth numbers in square-root-length intervals**

"coming close to solving an old problem that, for any fixed $\epsilon > 0$, there exists an x^{ϵ}-smooth number in $[x, x + \sqrt{x}]$ when x is sufficiently large."

This is the older assertion mentioned by the authors after noting that Matomäki and Radziwiłł prove existence in intervals of length $h \gg_{\epsilon} \sqrt{x}$; the question asks for the exact coefficient one. Source: Seth Hardy and Max Wenqiang Xu, "Helson's Conjecture for Smooth Numbers," arXiv:2511.03430v3, Introduction, p. 3.
Partial. Evidence: arXiv.

Harman, "Numbers with a Large Prime Factor," doi:10.4064/aa-91-3-279-289, *Acta Arithmetica* 91 (1999), Theorem 1, proves the exact interval assertion for $\epsilon > 1/(4\sqrt{e})$. Matomäki and Radziwiłł, "Multiplicative Functions in Short Intervals," doi:10.4007/annals.2016.183.3.6, *Annals of Mathematics* 183 (2016), Corollary 1, prove existence in $[x, x + C(\epsilon)\sqrt{x}]$ for every fixed $\epsilon > 0$. Neither result establishes existence in every exact interval $[x, x + \sqrt{x}]$ for all $\epsilon > 0$. Matomäki and Radziwiłł count products of two smooth factors in a bilinear short-interval average; their uniform estimate requires a sufficiently large multiple of $\sqrt{x}$.

24.23. **Critical-chaos law of the iterated logarithm**

"Conjecture 1.4. Let f be a Steinhaus random multiplicative function, and let V_∞ be the associated random variable in [GW25, Theorem 1.3]. Almost surely,

$$\limsup_{x\to\infty} \frac{|M_f(x)|}{\sqrt{x}(\log_2 x)^{1/4}} = \sqrt{V_\infty}.$$

"

Here $M_f(x) = \sum_{n\le x} f(n)$ and $\log_2 x = \log\log x$. Source: William Verreault, "Almost Sure Upper Bound for Sums of Random Multiplicative Functions and Critical Chaos," arXiv:2608.21354v2, Conjecture 1.4, §1.6, p. 5. Here [GW25] is Gorodetsky and Wong, "On the Limiting Distribution of Sums of Random Multiplicative Functions," arXiv:2508.12956v3, Theorem 1.3: V_∞ is the positive critical-chaos mass for which $\sum_{n\le x} f(n)/(\sqrt{x}/(\log_2 x)^{1/4})$ converges in law to $\sqrt{V_\infty}G$, with G an independent standard complex Gaussian.
Partial. Evidence: arXiv.

Verreault's Theorem 1.1 proves an upper bound with an extra $(\log_3 x)^{1+\epsilon}$ factor; the exact limsup identity remains open. The upper bound stops the normalized Parseval energy in each block; an ℓ^2-valued Doob inequality controls consecutive prime layers without treating them as independent.

24.24. **Keating–Snaith zeta moment leading constant**

> "Then, using a random matrix analogy (now called the 'Keating-Snaith philosophy'), they make the following conjecture for the moments of the Riemann zeta function (see [16],[18]):
>
> $$\lim_{T\to\infty} \frac{1}{T(\log T)^{\lambda^2}} \int_0^T \left|\zeta\left(\frac{1}{2}+it\right)\right|^{2\lambda} \mathrm{d}t = M(\lambda)A(\lambda),$$
>
> where
>
> $$M(\lambda) = \frac{G(1+\lambda)^2}{G(1+2\lambda)}, \qquad A(\lambda) = \prod_p \left(1-\frac{1}{p}\right)^{\lambda^2} \sum_{m=0}^{\infty} \left(\frac{\Gamma(\lambda+m)}{m!\,\Gamma(\lambda)}\right)^2 p^{-m}.$$
>
> "

The displayed conjecture is stated for real $\lambda \geq 0$; the paper attributes it to Keating and Snaith; G is the Barnes function normalized by $G(1) = 1$ and $G(z+1) = \Gamma(z)G(z)$. Source: Ashkan Nikeghbali and Marc Yor, "The Barnes G-Function and Its Relations with Sums and Products of Generalized Gamma Convolution Variables," *Electronic Communications in Probability*, doi:10.1214/ecp.v14-1488 14 (2009), paragraph following Proposition 1.1, p. 397. Here [16] is Keating and Snaith, "Random Matrix Theory and $\zeta(1/2+it)$," doi:10.1007/s002200000261; [18] is Mezzadri and Snaith (editors), "Recent Perspectives in Random Matrix Theory and Number Theory," doi:10.1017/CBO9780511550492 (2005).
Partial. Evidence: Sketch.

The exact cases $\lambda = 0, 1, 2$ follow respectively from the normalization, the second moment, and A. E. Ingham, "Mean-Value Theorems in the Theory of the Riemann Zeta-Function," doi:10.1112/plms/s2-27.1.273, for the fourth moment. The conjectured leading constant for general real $\lambda \geq 0$ remains open.

24.25. **HLS almost-sure square-root logarithmic bound**

> "Can the above be sharpened to the statement that, almost surely, $\sum_{n=1}^{N} \chi(n) = O(\sqrt{N}\log N)$ as $N \to +\infty$?"

Here χ is a Haar-random multiplicative character; the preceding text gives the $O_\epsilon(N^{1/2+\epsilon})$ bound. Source: Håkan Hedenmalm, Peter Lindqvist, and Kristian Seip, "A Hilbert Space of Dirichlet Series and Systems of Dilated Functions in $L^2(0,1)$," §4.8, question after Corollary 4.8, p. 19; *Duke Mathematical Journal*, doi:10.1215/S0012-7094-97-08601-4 86 (1997).
Answered: yes. Evidence: Published.

Rachid Caich, "Almost Sure Upper Bound for Random Multiplicative Functions," *Acta Arithmetica* 224 (2026), 297–325, doi:10.4064/aa250225-20-2, Theorem 1.1, proves almost surely

$$\sum_{n\leq x} f(n) \ll_\epsilon \sqrt{x}(\log_2 x)^{3/4+\epsilon}$$

for both models. At $\epsilon = 1/4$ this is $o(\sqrt{x}\log x)$, answering the cited question affirmatively for Steinhaus sums. Caich reveals largest prime factors to form a martingale, smooths its quadratic variation, and bounds the resulting Euler-product energy before applying exponential martingale estimates.

24.26. **Bounded partial sums for square-summable character coefficients**

> "The following conjecture seems reasonable: for almost all χ,
>
> $$\sum_{n=1}^{N} a_n \chi(n) = O(1) \qquad (N \to +\infty).$$

Here $\sum_{n=1}^{\infty}|a_n|^2<\infty$ is fixed and almost all characters are with respect to Haar measure. Source: Håkan Hedenmalm, Peter Lindqvist, and Kristian Seip, "A Hilbert Space of Dirichlet Series and Systems of Dilated Functions in $L^2(0,1)$," Duke Mathematical Journal 86 (1997), doi:10.1215/S0012-7094-97-08601-4, Section 6, p. 30.

Answered: yes. Evidence: Published.

Hedenmalm and Saksman, "Carleson's Convergence Theorem for Dirichlet Series," Pacific Journal of Mathematics 208 (2003), doi:10.2140/pjm.2003.208.85, Theorem 1.5, proves that $\sum_{n=1}^{\infty}a_n\chi(n)$ converges for almost every character whenever $(a_n)\in\ell^2$. Convergence implies bounded partial sums, so this proves the conjecture in its full stated scope. The proof transfers the one-variable Carleson–Hunt maximal inequality to finitely many prime coordinates with a constant independent of their number; polynomial approximation then gives almost-everywhere convergence.

24.27. Montgomery's Dirichlet-polynomial moment conjecture

> "One of Montgomery's conjectures states that for every $\varepsilon>0$ and $p\in(2,4)$ there exists a constant $C=C(\varepsilon)$ such that for all finite Dirichlet polynomials $f(s)=\sum_{n=1}^{N}a_m n^{-s}$ with $|a_n|\le 1$ one has
>
> $$\int_0^T |f(it)|^p\,\mathrm{d}t \le CN^{p/2+\varepsilon}(T+N^{p/2}) \quad \text{for } T>1.$$
>
> "

The summand in this reported passage prints a_m where the coefficient is indexed by n; the formula uses the standard indexed expansion $f(s)=\sum_{n=1}^{N}a_n n^{-s}$. Søvik attributes the conjecture to Hugh L. Montgomery, "Ten Lectures on the Interface between Analytic Number Theory and Harmonic Analysis," CBMS Regional Conference Series in Mathematics 84, doi:10.1090/cbms/084, p. 129; the original book page was not independently checked. The quoted passage is from Øistein Søvik, "Hankel Forms and Nehari's Theorem," section "The Hardy Space $\mathscr{H}^p$," discussion following the Dirichlet-series embedding question.

Partial. Evidence: arXiv.

Guth, "Large Value Estimates in Number Theory, Harmonic Analysis, and Computer Science," arXiv:2503.07410, §3.4, Conjecture 3.4, states a related discrete bounded-coefficient ℓ^q estimate and says it is beyond current methods. Guth and Maynard, "New Large Value Estimates for Dirichlet Polynomials," doi:10.4007/annals.2026.203.2.6, *Annals of Mathematics* 203 (2026), Theorem 1.1, improve large-value bounds in selected ranges, while their Conjecture 1.5 remains open; neither proves the continuous L^p bound above. Guth and Maynard reduce large values to estimates for the singular values of a matrix of Dirichlet sums, exploiting cancellation in its higher trace moments; their theorem covers selected large-value scales.

25. Composition operators

25.1. Compactness of composition operators on $\mathscr{H}^2$

Question., arXiv:1601.01616 Saksman and Seip ask: "Characterize the compact composition operators on H^2." Problem 3.3, p. 5. Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616.

The characteristic-zero classification is Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, Theorem 1.4, arXiv:2005.05094.

Partial. Evidence: Sketch.

The characteristic-zero case is known, and the general positive-characteristic case remains open. New sufficient classes include contact sets whose boundary mass has sufficiently high decay and certain mixed-cusp symbols; the known averaged counting necessity does not give an equivalence.

25.2. **Is the higher-dimensional target exponent optimal?**

Source passage., arXiv:1602.03446 Bayart and Brevig write: "For $\dim(\varphi) \geq 3$, we conjecture that $(2^\alpha - 1)/d$ is not optimal." See the remark after Theorem 21. The inferred question concerns characteristic-zero Dirichlet polynomials of degree one or two with unrestricted range in dimension $d \geq 3$.

Here $d = \dim(\varphi)$ counts independent prime coordinates in the Bohr lift, and $\mathcal{D}_\alpha$ has squared norm $\sum_n |a_n|^2/d(n)^\alpha$, where $d(n)$ is the divisor function; the quoted exponent is the target weight β for $C_\varphi : \mathcal{D}_\alpha \to \mathcal{D}_\beta$.

Bayart and Brevig, *Composition operators and embedding theorems for some function spaces of Dirichlet series*, arXiv:1602.03446v2.

Partial. Evidence: Sketch.

The public sharpness note proves a strictly better target, so the stated exponent is not optimal. The exact best universal exponent is not determined by that improvement.

Answers, "Nonoptimality of a target exponent", p. 33.

Sharpness of composition-operator estimates

25.3. **Do Volterra boundedness classes depend on the exponent?**

Question., arXiv:1602.04729 "Is $\mathbf{T}_g$ bounded on $\mathcal{H}^2$ if and only if it is bounded on $\mathcal{H}^p$ for every $p < \infty$?" Brevig–Perfekt–Seip, Question 1, p. 3.

Here $\mathbf{T}_g f(s) = -\int_s^{+\infty} f(w) g'(w)\, dw$ is the Volterra operator on the Hardy space of ordinary Dirichlet series.

The linear-symbol result is Brevig, Perfekt, and Seip, *Volterra operators on Hardy spaces of Dirichlet series*, Theorem 5.5, arXiv:1602.04729.

Partial. Evidence: Sketch.

The exponent-independence question remains open for general symbols. The paper proves it for linear symbols; current work gives several additional sufficient classes but does not cover the logarithmic one-prime example or arbitrary symbols.

25.4. **Volterra symbols and the dual of $\mathscr{H}^1$**

Question., arXiv:1602.04729 "Do we have $\mathcal{X}_2 \subset (\mathcal{H}^1)^*$?" Brevig–Perfekt–Seip, Question 2, p. 3.

Here $\mathcal{X}_2$ is the space of symbols g for which $\mathbf{T}_g$ is bounded on $\mathcal{H}^2$, with the pairing $\langle f, g \rangle = \sum_n a_n \overline{b_n}$.

The known inclusion is Brevig, Perfekt, and Seip, *Volterra operators on Hardy spaces of Dirichlet series*, Corollary 6.4, arXiv:1602.04729.

Partial. Evidence: Sketch.

Brevig–Perfekt–Seip prove that boundedness on $\mathscr{H}^1$ implies membership in this dual. The converse reduction is now exact for finite-prime symbols, but it leaves a dimension-free dual estimate unresolved; the general inclusion remains open.

25.5. **Off-diagonal and endpoint rigidity on disc Hardy spaces**

Question., arXiv:1607.00113 Laitila, Nieminen, Saksman, and Tylli ask: "Are there results corresponding to our main theorems for $C_\varphi \in \mathcal{L}(H^p, H^q)$ in the case $p \neq q$?" They also ask: "Is there an analogue of Theorem 1.5 for $p = 1$?" Problems 4.3(1)–(2), p. 17.

Here H^p and H^q are disc Hardy spaces; the main theorems identify compactness with strict singularity and characterize the copies of ℓ^2 and L^p fixed by C_φ, while Theorem 1.5 says, for $1 < p < \infty$, $p \neq 2$, that C_φ fixes a copy of L^p exactly when $m(E_\varphi) > 0$, where $E_\varphi = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$.

The resolving work is Shi and Li, *Fixed-Copy Exponent Sets and Strict Singularity of Composition Operators Between Hardy Spaces*, Theorems A and C, arXiv:2608.17663; its localized copies of L^r come from stable integrals and analytic lifting for $r < 2$, and adapted lacunary series for $r = 2$, combined with a lower bound for the pullback boundary measure.

Laitila, Nieminen, Saksman, and Tylli, *Rigidity of composition operators on the Hardy space H^p*, arXiv:1607.00113.

Answered: yes. Evidence: arXiv.

Shi–Li, Theorems A and C, state that they completely resolve both parts for $1 \le p, q < \infty$. Their boundary-measure argument constructs fixed copies of the relevant function spaces when the contact set has positive measure.

25.6. **Strictly singular but noncompact composition operators**

Question., arXiv:1607.00113 Laitila, Nieminen, Saksman, and Tylli ask in Problem 4.3(4): "Is there a Banach space E of scalar-valued analytic functions on $\mathbb{D}$ and an analytic map $\varphi : \mathbb{D} \to \mathbb{D}$, for which $C_\varphi \in \mathcal{S}(E) \setminus \mathcal{K}(E)$?"Laitila, Nieminen, Saksman, and Tylli, *Rigidity of composition operators on the Hardy space H^p*, arXiv:1607.00113.

Answered: yes. Evidence: Sketch.

An explicit weighted coefficient space and the symbol $\varphi(z) = z^2$ give such an example. The proof decomposes the coefficient space into dyadic exponent blocks and verifies strict singularity and noncompactness directly.

Answers, "A strictly singular noncompact composition operator", p. 34.

A strictly singular noncompact composition operator

25.7. **A converse to the BMOA contact-set criterion**

Question., arXiv:1607.00113 Laitila, Nieminen, Saksman, and Tylli ask: "Is the converse of Proposition 4.1(i) also true?" Problem 4.3(3), p. 17. The proposition concerns $E_\varphi = \{\zeta \in \mathbb{T} : |\varphi(\zeta)| = 1\}$.

Proposition 4.1(i) says that if C_φ is ℓ^2-singular on BMOA, meaning that it is not bounded below on any subspace isomorphic to ℓ^2, then $m(E_\varphi) = 0$; m is normalized arc length.

Laitila, Nieminen, Saksman, and Tylli, *Rigidity of composition operators on the Hardy space H^p*, arXiv:1607.00113.

Answered: no. Evidence: Sketch.

A Blaschke-product example has a null contact set but its composition operator is bounded below on a copy of ℓ^2 in BMOA. The construction uses thin zeros approaching the boundary, a feature not detected by contact-set measure.

Answers, "The BMOA contact-set converse", p. 34.

The BMOA contact-set converse

25.8. **Canonical symbols with large parameter**

Source passage., arXiv:1903.08886 For $\varphi_\alpha(s) = 1/2 + \alpha(1 - 2^{-s})/(1 + 2^{-s})$, Brevig and Perfekt write: "We do not know whether the first two of these equalities hold also for $\alpha > \alpha_0$." Here α_0 solves $\alpha\zeta(1 + \alpha) = 2$; §4, p. 16.

The first two equalities are

$$\|C_{\varphi_\alpha}\| = S_{\varphi_\alpha} = S_{\varphi_\alpha^*},$$

where S_φ is the supremum of $\|C_\varphi K_w\|/\|K_w\|$ over $\operatorname{Re} w > 1/2$, and S_{φ^*} is the analogous adjoint test on the Hardy space supported on the powers of two; $K_w(s) = \zeta(s + \overline{w})$.

Brevig, Perfekt, and Pushnitski, *The spectrum of some Hardy kernel matrices*, Theorem D, doi:10.5802/aif.3589 (arXiv:2003.11346v3), Here $N(K_\alpha)$ counts the eigenvalues above $2/\alpha$. They identify $C^*_{\varphi_\alpha} C_{\varphi_\alpha}$ with K_α: if its norm exceeds the continuous-spectrum edge $2/\alpha$, equality in the kernel test would force a power-sequence eigenvector, excluded by Lemma 5.1.

Brevig and Perfekt, *Norms of composition operators on the H^2 space of Dirichlet series*, arXiv:1903.08886v2.

Answered: no. Evidence: Published.

Brevig, Perfekt, and Pushnitski prove that the reproducing-kernel thesis holds exactly when $N(K_\alpha) = 0$; for the canonical symbol it fails when $\alpha \ge 2$. This disproves persistence of the first two equalities throughout the proposed range, although it does not locate every transition point. Sharpness of composition-operator estimates

25.9. **The reproducing-kernel thesis for affine symbols**

Question., arXiv:1903.08886 Brevig and Perfekt ask: "If $\varphi(s) = c + r2^{-s}$ is in $\mathcal{G}$ and $r > 0$, does it hold that $\|C_\varphi\| = S_\varphi = S_{\varphi^*}$?" See §4, p. 16.

Here $\mathcal{G}$ is the Gordon–Hedenmalm class of composition symbols, and

$$S_\varphi = \sup_{\operatorname{Re} w>1/2} \|C_\varphi K_w\|/\|K_w\|, \qquad K_w(s) = \zeta(s+\overline{w}),$$

$$S_{\varphi^*} = \sup_{\operatorname{Re} w>0} \|C_\varphi^* K_w^{(2)}\|/\|K_w^{(2)}\|, \qquad K_w^{(2)}(s) = (1-2^{-s-\overline{w}})^{-1}.$$

Brevig and Perfekt, *Norms of composition operators on the H^2 space of Dirichlet series*, arXiv:1903.08886v2.

Answered: no. Evidence: Sketch.

An affine example in the public note separates the operator norm from the reproducing-kernel tests. The counterexample uses a compactness/eigenvector comparison and is recorded with a proof in the linked note. Affine composition-operator norms

25.10. **The exact norm near the boundary**

Source passage., arXiv:1903.08886 Brevig and Perfekt write that it is "tempting to conjecture that $\|C_\varphi\| = S_\varphi = S_{\varphi^*}$ could hold at least for affine symbols $\varphi(s) = 1/2 + \xi(1-2^{-s})$ with sufficiently small $\xi > 0$." They add that this would imply $\|C_\varphi\| = \xi^{-1/2}$ for all sufficiently small $\xi > 0$. The inferred question is whether that consequence is true; §4, p. 16.

Here S_φ is the normalized reproducing-kernel supremum for C_φ on $\mathscr{H}^2$, and S_{φ^*} is the adjoint supremum on its powers-of-two range space, whose kernel is $(1-2^{-s-\overline{w}})^{-1}$.

Brevig and Perfekt, *Norms of composition operators on the H^2 space of Dirichlet series*, arXiv:1903.08886v2.

Answered: no. Evidence: Sketch.

The public affine-norm analysis gives a strict counterexample to the proposed equality along the small-parameter family. It also determines the relevant norm profile, so the asymptotic statement is not inferred from a leading-order estimate. Affine composition-operator norms

25.11. **Subordination between affine symbols**

Question., arXiv:1903.08886 Brevig and Perfekt ask: "If φ_b and φ_c are two affine symbols with the same mapping properties, is it true that either C_{φ_b} is subordinate to C_{φ_c}, or C_{φ_c} is subordinate to C_{φ_b}?" See §7.2, p. 24.

Here subordination means $\|C_{\varphi_b} f\|_{\mathscr{H}^2} \le \|C_{\varphi_c} f\|_{\mathscr{H}^2}$ for every $f \in \mathscr{H}^2$; the two symbols have the same image disc.

Brevig and Perfekt, *Norms of composition operators on the H^2 space of Dirichlet series*, arXiv:1903.08886v2.

Answered: no. Evidence: Sketch.

The public note gives an explicit pair of affine symbols for which neither operator is subordinate to the other. The paper's moment criterion for subordination makes the two failed inequalities transparent. Affine composition-operator norms

25.12. **Schatten-class criteria for composition operators**

Question., arXiv:2005.05094 "Describe the symbols $\varphi \in \mathcal{G}_0$ such that $C_\varphi \in S_p$." Brevig and Perfekt, Problem 2, §8.2 of *A mean counting function for Dirichlet series and compact composition operators*, p. 35.

Here $\mathcal{G}_0$ consists of Gordon–Hedenmalm symbols of characteristic zero, and S_p means that the singular values of C_φ on $\mathscr{H}^2$ have summable pth powers.

The external criteria are Bayart and Kouroupis, *Schatten class composition operators on the Hardy space of Dirichlet series and a comparison-type principle*, Theorems 1.1 and 1.2, doi:10.4171/RMI/1474 (arXiv:2306.05733).

Partial. Evidence: Sketch.

Bayart–Kouroupis give an S_{2m} criterion for every symbol and a one-fold power criterion for bounded-imaginary-part symbols. For arbitrary symbols, our argument gives a scalar moment-series test for $0 < p < 2$; a local one-fold criterion and the unrestricted non-even range above 2 remain open.

25.13. **Compactness of characteristic-zero operators across p**

Source result., arXiv:2005.05094 Brevig and Perfekt state: "$\mathscr{C}_\varphi : \mathscr{H}^2 \to \mathscr{H}^2$ is compact if and only if $\lim_{\operatorname{Re} w \to 1/2^+} \mathscr{M}_\varphi(w)/(\operatorname{Re} w - 1/2) = 0$." The inferred question is whether this criterion is independent of finite $p \geq 2$; the non-even sufficiency uses arXiv:2609.11560v2.

Here $\mathscr{M}_\varphi(w) = \lim_{\sigma \downarrow 0} \lim_{T\to\infty} (\pi/T) \sum_{\varphi(s)=w,\, |\operatorname{Im} s|<T,\, \operatorname{Re} s>\sigma} \operatorname{Re} s$ is the mean counting function, with zeros counted with multiplicity, and φ has characteristic zero.

The Hilbert-space source is Brevig and Perfekt, *A mean counting function for Dirichlet series and compact composition operators*, Theorem 1.4, arXiv:2005.05094; the conditional local embedding is Chen, Fang, Guo, Hou, Shao, and Zhou, *The local embedding problem for Hardy spaces of Dirichlet series*, Theorem 1.1 and Proposition 10.1, arXiv:2609.11560v2.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2609.11560.

The compactness criterion is necessary for every finite $p \geq 2$, and sufficient at every exponent where the local embedding holds; in particular the converse is unconditional at even exponents. The non-even converse is conditional on the cited preprint.

Answers, "Compactness across Hardy exponents", p. 35.

Compactness across Hardy exponents

25.14. **Operators bounded exactly at even exponents**

Question., arXiv:2103.16186 Brevig, Ortega-Cerdà, and Seip ask in Problem 4.1: "Is there a linear operator, densely defined on $H^p(\mathbb{T}^\infty)$ for $1 \leq p \leq \infty$, which extends boundedly exactly when p is an even integer or $p = \infty$?"§4.1, p. 28.

Andersson, *Exotic operators on the Hardy spaces of the infinite-dimensional torus*, Theorem 1.1, arXiv:2609.12859, constructs rank-one blocks large only on prescribed exponent intervals and tensorizes them on disjoint variables; summing these blocks realizes the relatively closed set of even integers, with infinity included.

The original question is Brevig, Ortega-Cerdà, and Seip, *Idempotent Fourier multipliers acting contractively on H^p spaces*, arXiv:2103.16186.

Answered: yes. Evidence: arXiv.

Andersson's realization theorem constructs every set of exponents whose finite part is relatively closed in $[1, \infty)$. The even integers form such a set, so it gives the requested operator.

25.15. **Cyclicity of an identity plus a weighted shift**

Question., arXiv:2104.06783 Fricain and Mau define T on $H^2 = \mathbb{C}e_0 \oplus H^2_0$ by the matrix with first diagonal entry 1 and forward-shift weights $\alpha_1, \alpha_2, \ldots$, where (α_n) is bounded, then ask: "Is T cyclic?" Question 5.9, p. 21 of their paper on weighted holomorphic Dirichlet series and composition operators with polynomial symbols.

The monomials are $e_n(z) = z^n$, and the operator is $Te_0 = e_0$, $Te_n = \alpha_n e_{n+1}$ for $n \geq 1$; cyclic means that the linear span of $\{T^k x : k \geq 0\}$ is dense for some x.

The published nonzero-weight criterion is Fricain and Mau, *Weighted holomorphic Dirichlet series and composition operators with polynomial symbols*, Theorem 5.11 and Lemma 5.10, doi:10.7146/math.scand.a-129686; the zero-weight exclusion is retained here.

Answered: cyclic exactly when all $\alpha_j \neq 0$ and $\sum_{j\geq 0} |\alpha_1 \cdots \alpha_j|^{-2} = \infty$. Evidence: Sketch.

The criterion agrees with the published nonzero-weight case and adds the necessary zero-weight exclusion. When it holds, $e_0 + e_1$ is cyclic; if a weight vanishes or the reciprocal-product series converges, the relevant adjoint eigenspace has dimension at least two.

25.16. **Positive-characteristic compactness and counting functions**

Question., arXiv:2112.05508 Bayart asks: "On $\mathcal{H}^2$, can we get a necessary condition using some counting function without any extra assumption on $\varphi \in \mathcal{G}_{\geq 1}$? Or at least, if $\varphi \in \mathcal{G}_{\geq 1}$ is supported on a finite set of prime numbers, do we have C_φ is compact on $\mathcal{H}^2$ if and only if $\mathcal{N}_\varphi(w) = o(\operatorname{Re} w)$ as $\operatorname{Re} w \to 0$?" Question 3.6, p. 7.

Here $\mathcal{G}_{\geq 1}$ consists of symbols $\varphi(s) = c_0 s + \psi(s)$ with integer $c_0 \geq 1$ and ψ a Dirichlet series mapping $\mathbb{C}_0$ into $\mathbb{C}_0$ or a purely imaginary constant; $\mathcal{N}_\varphi(w) = \sum_{\varphi(s)=w,\, \operatorname{Re} s>0} \operatorname{Re} s$ counts preimages with multiplicity.

Bayart, *Counting functions for Dirichlet series and compactness of composition operators*, arXiv:2112.05508.

Partial. Evidence: Sketch.

The unrestricted necessary condition is known in an averaged vertical-limit form, and the ordinary criterion is established for one-prime symbols. New work extends it to symbols generated by a single integer frequency, including composite bases, and gives mixed-cusp compact examples; the general finite-prime converse remains open.

25.17. **Spectrum of a vertical translation on $\mathscr{H}^\infty_+$**

Question., arXiv:2406.19737 "What is the spectrum of $\mathcal{C}_\varphi$ provided $\varphi(s) = s + i\tau$?" Bayart and Yao, Question 3.8, p. 15 of *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series.*

Here $\mathscr{H}^\infty_+ = \bigcap_{\varepsilon>0}\{f : f(\varepsilon + \cdot) \in \mathscr{H}^\infty\}$ carries the seminorms $\|f(\varepsilon+\cdot)\|_\infty$, and $\mathcal{C}_\varphi f = f \circ \varphi$.

Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

Answered: $\{1\}$ if $\tau = 0$, and $\mathbb{T}$ otherwise. Evidence: Sketch.

The full argument identifies the dense point spectrum and separately proves that each remaining point of the unit circle is outside the range of $C_\varphi - \lambda I$. This range argument is needed because the spectrum of a continuous operator on a Fréchet space need not be closed.

Answers, "Spectrum of a vertical translation", p. 37.

Bayart–Yao questions and answers

25.18. **Cyclicity forces univalence on the half-plane**

Question., arXiv:2406.19737 "The above proof shows that provided C_φ is cyclic, φ is univalent on $\mathbb{C}_{1/2}$. Is it also univalent on $\mathbb{C}_0$?" Bayart and Yao, Question 4.2, p. 18. Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

Answered: yes. Evidence: Sketch.

For every Gordon–Hedenmalm symbol of positive characteristic that is not univalent on $\mathbb{C}_0$, the range of C_φ has infinite codimension. The proof uses bounded evaluations of finite-prime truncations, extending the cyclicity implication beyond the region where point evaluations on $\mathscr{H}^2$ are bounded.

Answers, "Cyclicity and univalence", p. 37.

Bayart–Yao questions and answers

25.19. **Minimal commutants of diagonal operators**

Question., arXiv:2406.19737 "Does there exist a diagonal operator associated to an injective sequence (λ_n) which fails to have a minimal commutant? If so, can we give a characterization of diagonal operators with a minimal commutant?" Bayart and Yao, Question 5.6, p. 22.

For $De_n = \lambda_n e_n$ on ℓ^2, a minimal commutant means that every bounded operator commuting with D lies in the weak-operator closure of the polynomials in D.

Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

Answered: yes, with a criterion. Evidence: Sketch.

The criterion is that no nonzero $a \in \ell^1$ annihilates every moment $\sum_n a_n \lambda_n^k$, $k \geq 0$. A Vitali-disc packing gives an explicit annihilating measure and hence an injective diagonal example without a minimal commutant; the proof also recovers the classical thin-set cases.
Answers, "Minimal commutants of diagonal operators", p. 38.

Bayart–Yao questions and answers

25.20. **Removing a zero-free hypothesis in the disc Hardy space**

Question., arXiv:2406.19737 Under Corollary 8.5's hypotheses, Bayart and Yao ask: "Can we dispense with the assumption that ψ does not vanish on $\mathbb{D}$ in Corollary 8.5?" Here $\varphi(z) = \lambda z^p \psi(z)$, $p \geq 2$, $\lambda \neq 0$, $\psi(0) = 1$, and $\|\varphi^{[n]}\|_\infty < 1$ for some $n \geq 1$; Question 8.7, p. 33.

Corollary 8.5 concludes that C_φ does not have a minimal commutant, meaning that its commuting operators are not all weak-operator limits of polynomials in C_φ; $\varphi^{[n]}$ is the n-fold iterate.

Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

The local coordinate theorem used in the proof is Milnor, *Dynamics in One Complex Variable*, third edition, Theorem 9.1, doi:10.1515/9781400835539.

Answered: yes. Evidence: Sketch.

A local univalent Böttcher coordinate, composed with a sufficiently deep iterate, gives a small global solution of $v \circ \varphi = v^p$. It produces a commuting composition operator outside the weak-operator closure of the polynomials in C_φ, proving failure of a minimal commutant without requiring ψ to be zero-free.
Answers, "Removing the zero-free hypothesis", p. 38.

Bayart–Yao questions and answers

25.21. **Lens maps and minimal commutants**

Question., arXiv:2406.19737 For $\mathcal{C}(z) = (1+z)/(1-z)$ and $\varphi_\alpha(z) = (\mathcal{C}(z)^\alpha - 1)/(\mathcal{C}(z)^\alpha + 1)$, Bayart and Yao ask: "Does C_{φ_α} have a minimal commutant?" The source writes $0 < \alpha < \pi/2$; the self-map range is $0 < \alpha < 1$. See Question 8.11 and Remark 8.12, p. 36.

A minimal commutant is the weak-operator closure of the polynomials in C_{φ_α}.

Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

Answered: no. Evidence: Sketch.

For every $0 < \alpha < 1$, reflection $Rf(z) = f(-z)$ commutes with the lens composition operator but is not in the weak-operator closure of its polynomial algebra. A bounded functional built from a Cauchy transform separates reflection from that closure.
Answers, "Lens maps and minimal commutants", p. 39.

Bayart–Yao questions and answers

25.22. **Dense range for univalent Dirichlet-series symbols**

Question., arXiv:2406.19737 Bayart and Yao ask: "Let $\varphi \in \mathcal{G}_{>0}$ be univalent. Under which conditions does C_φ have dense range?" Question 4.10, p. 20. Bayart and Yao, *Königs maps and commutants of composition operators on the Hardy–Hilbert space of Dirichlet series*, arXiv:2406.19737.

Partial. Evidence: Sketch.

Univalence on $\mathbb{C}_0$ is necessary, and a one-prime affine family has an exact criterion: density holds precisely when $|a| \log 2 \leq 1$ for $\varphi(s) = s + c + a2^{-s}$. A sufficient condition is also known from univalent extension and image containment; a general criterion for several prime frequencies remains open.

25.23. **Weight-independence on weighted polydisc Bergman spaces**

Question., arXiv:2601.19308 "Let $d \geq 4$ and let $\varphi \in \mathcal{O}(\mathbb{D}^d, \mathbb{D}^d) \cap C^1(\overline{\mathbb{D}^d})$. Is it true that if C_φ is continuous on $A^2_\beta(\mathbb{D}^d)$ for some $\beta \geq d-3$, then it is continuous on $A^2_\beta(\mathbb{D}^d)$ for all $\beta \geq d-3$?" Bayart–Dorval, Question 4.2, p. 21.

Dorval, *Compactness of composition operator on weighted Bergman spaces of the polydisc*, Theorem 1.1/3.6, arXiv:2606.29969, proves necessary boundary-stratum and Jacobian conditions for every weight, then obtains sufficiency at $\beta = d-3$ by local Carleson-window estimates and propagates it upwards using Bayart–Dorval, *Composition operators on weighted Bergman spaces of the polydisc*, Theorem 2.13, arXiv:2601.19308.

Answered: yes. Evidence: arXiv.

Conditional on: arXiv:2606.29969v1 and arXiv:2601.19308.

Dorval's Theorem 1.1 (proved as Theorem 3.6) gives a necessary-and-sufficient boundary-jet criterion independent of β, for all $d \geq 2$. Its hypotheses include the stated C^1 regularity.

25.24. **Boundary jets and weighted Bergman boundedness**

Question. Bayart and Dorval ask in Question 3.2, p. 19: "For which $\beta \geq -1$ are there C^n self-maps φ, ψ with the same contact set and the same boundary jets through order n, for which C_φ is bounded and C_ψ is unbounded on $A^2_\beta(\mathbb{D}^d)$?" Here $d \geq 3$ and $n \in \mathbb{N}$.

The constructions are Bayart and Dorval, *Composition operators on weighted Bergman spaces of the polydisc*, Theorems 1.1 and 3.1, arXiv:2601.19308: their product symbols have matching finite jets but boundary deficits of orders $2n$ and $4n$, so the pullback Carleson thresholds differ. The obstruction above $d-3$ is Dorval's Theorem 3.6, arXiv:2606.29969.

Partial. Evidence: arXiv.

Conditional on: arXiv:2601.19308.

Their construction gives intervals of weights below $d-3$, while Dorval's weight-independent criterion rules out such pairs for $\beta \geq d-3$ and $n \geq 1$. Whether the construction intervals exhaust the remaining range is open, conditional on the cited preprint.

25.25. **Second-order norm asymptotics for a boundary symbol**

Source passage. For fixed $\sigma > 1/2$, Fang, Guo, Mishra, and Muthukumar state: "We also obtain a second-order expansion of the squared norm." See the abstract and Theorem 3.4 of *Critical Norm Profiles for Finite-Prime Composition Operators on the Hardy Space of Dirichlet Series* (arXiv:2608.26041). The inferred follow-up asks for the second-order limit along $\varphi_\xi(s) = 1/2 + \xi(1-2^{-s})$.

Answered: 0**.** Evidence: Sketch.

The written argument establishes the limit by a second-order expansion of the extremal Hankel norm. The cited theorem gives only the leading $1/\xi + O(1)$ term; this limit is an inferred follow-up, not a question stated by its authors.

25.26. **Numerical range of a weighted composition operator**

Question., arXiv:2608.29294 For a non-constant c_0-symbol φ and $\psi \in \mathscr{H}^2$ with $\psi_1 = 0$ such that $C_{\psi,\varphi}$ is bounded on $\mathscr{H}^2$, Lahiri, Ojha, Halder, and Birbonshi ask: "Does 0 lie in the interior of the numerical range $W(C_{\psi,\varphi})$?" Question 2.13, p. 8; Question 3.9 asks a remaining affine-symbol case, p. 11. Lahiri, Ojha, Halder, and Birbonshi, *Numerical range and invariant subspaces of weighted composition operators on the Hardy space of Dirichlet series*, arXiv:2608.29294.

Here $C_{\psi,\varphi} f = \psi(f \circ \varphi)$, $\psi_1 = \psi(+\infty)$, and $W(T) = \{\langle Tx, x\rangle : \|x\| = 1\}$.

Answered: no as stated; yes when $\psi \neq 0$. Evidence: Sketch.

The zero weight gives the counterexample $W(C_{0,\varphi}) = \{0\}$. For every nonzero such weight, a two-dimensional compression has an elliptical numerical range containing a nondegenerate disc about zero.

Answers, "Numerical range of a weighted composition operator", p. 39.

Numerical range of a weighted composition operator

25.27. **Davis–Wielandt shells of affine composition operators**

Question., arXiv:2608.29294 Lahiri, Ojha, Halder, and Birbonshi ask in Question 5.4: "Determine the Davis–Wielandt shell $\mathrm{DW}(C_\varphi)$." Here $\varphi(s) = cs + c_1$, c is a nonnegative integer, and boundedness requires $\operatorname{Re} c_1 \geq 0$ for $c \geq 1$ and $\operatorname{Re} c_1 > 1/2$ for $c = 0$; p. 17. Lahiri, Ojha, Halder, and Birbonshi, *Numerical range and invariant subspaces of weighted composition operators on the Hardy space of Dirichlet series*, arXiv:2608.29294.

Here $\mathrm{DW}(T) = \{(\langle Tx, x\rangle, \|Tx\|^2) : \|x\| = 1\} \subseteq \mathbb{C} \times \mathbb{R}$.

Answered: exact scalar support-function description. Evidence: Sketch.

For positive characteristic and positive real part, the shell is described by an explicit convergent scalar series and an arithmetic maximum over integers that are not c_0-th powers; the only unattained point of its closure is the origin. The remaining characteristic cases reduce to an ellipsoid or a convex hull, with the nonclosed boundary treated separately.

Answers, "Davis–Wielandt shells of affine composition operators", p. 40.

Davis–Wielandt shells of affine composition operators

25.28. **Which boundedness sets can occur?**

Source passage., arXiv:2609.12859 After Theorem 1.1 Andersson writes: "We are unaware of whether this completely characterizes all possible sets A, and find it an interesting question to determine precisely what sets A can occur." The inferred question asks for the exact boundedness sets among the common-domain operators in that theorem; §1 of *Exotic operators on the Hardy spaces of the infinite-dimensional torus*.

Here $A = \{p \in [1, \infty] : T \text{ extends boundedly on } H^p(\mathbb{T}^\infty)\}$, and Theorem 1.1 constructs every such set whose finite part is relatively closed, using one domain dense at every finite exponent and weak-$*$ dense at infinity.

The external block construction used in the proof is Andersson, *Exotic operators on the Hardy spaces of the infinite-dimensional torus*, Lemmas 2.7–2.8 and the proof of Theorem 1.1, arXiv:2609.12859.

Answered: exactly those with $A \setminus \{\infty\}$ an F_σ set in $[1, \infty)$. Evidence: Sketch.

The proof shows the necessary and sufficient condition is F_σ, strengthening the relatively closed sufficient class in the source theorem. The argument uses the common dense-domain convention stated with the theorem.

Answers, "Which boundedness sets can occur?", p. 42.

Which boundedness sets can occur?

25.29. **Are even exponents the only universal boundedness exponents?**

Source passage., doi:10.1007/978-981-15-9351-2 Queffélec and Queffélec formulate the conjecture that even integers are the only p for which every characteristic-zero Gordon–Hedenmalm symbol gives a bounded operator on $\mathscr{H}^p$; see the Remark after Theorem 8.10.1, §8.10.2, pp. 273–274. The question considered here is whether that conjecture holds.

The conditional ingredients are Chen, Fang, Guo, Hou, Shao, and Zhou, *The local embedding problem for Hardy spaces of Dirichlet series*, Theorem 1.1, arXiv:2609.11560v2, and Bayart–Brevig, *Composition operators and embedding theorems for some function spaces of Dirichlet series*, Theorem 3, arXiv:1602.03446v2. The latter transfers a local Hardy embedding to all characteristic-zero composition symbols by comparison with a half-plane Hardy norm.

The source is Queffélec and Queffélec, *Diophantine Approximation and Dirichlet Series*, doi:10.1007/978-981-15-9351-2.

Answered: no, conditional for $p > 2$. Evidence: arXiv.

Conditional on: arXiv:2609.11560.

The cited local-embedding theorem, together with Bayart–Brevig's composition theorem, gives boundedness for every characteristic-zero symbol at every $p \geq 2$. For $p < 2$, local embedding fails and there are unbounded examples; thus the claimed even-integer threshold is false above 2.

25.30. **Approximation numbers of compact composition operators**

Source passage., doi:10.1007/978-981-15-9351-2 Queffélec and Queffélec conclude: "The result of 1. is optimal, and that of 2. as well, up to the logarithmic factor." This follows Theorem 8.9.3, p. 272, in *Diophantine Approximation and Dirichlet Series*. Theorem 8.8.1, p. 268, gives the positive-characteristic lower bound, and Theorem 8.9.3 gives the critical finite-dimensional estimate. The inferred questions are whether the logarithmic loss in the first lower bound can be removed and what the exact approximation-number order is in the second case. The finite-dimensional item in Theorem 8.9.3 is $n^{-(d-1)/2} \ll a_n(C_\varphi) \ll n^{-(d-1)/2}(\log n)^{(d-1)/2}$ for $\varphi(s) = c_1 + \sum_{j=1}^{d} c_j q_j^{-s}$ with independent q_j and $\operatorname{Re} c_1 - \sum_j |c_j| = 1/2$; its infinite-dimensional item asserts membership in every S_p. The separate estimate in Theorem 8.8.1 is $a_n(C_\varphi) \gg (n \log n)^{-\operatorname{Re} c_1}$ when $c_0 \geq 1$; $a_n(T)$ is the distance in operator norm from T to operators of rank less than n.

Answered: yes in (a), and $a_n \asymp n^{-(d-1)/2}$ in (b). Evidence: Sketch.

For (a), an antichain of input monomials gives the sharp polynomial lower bound without the logarithm. For (b), the logarithmic loss is removed by reducing the kernel to a half-plane Laplace embedding and summing operator-norm bounds over dyadic frequency blocks.

Answers, "Approximation numbers of composition operators", p. 43.

Sharpness of composition-operator estimates

25.31. **All-exponent BMOA necessity for Volterra operators**

Source passage. Brevig, Perfekt, and Seip write: "We will use this to prove a rather curious incomplete analogue to part (c) of Theorem 2.3." They explain that their argument yields the BMOA conclusion only for rational p. The inferred question asks whether $T_g : H^p \to H^p$ bounded forces $g \in \mathrm{BMOA}(\mathbb{C}_{1/2})$ for every real $p > 0$; our answer uses the local-embedding claim in arXiv:2609.11560v2, Theorem 1.1.

Theorem 2.3(c) says that boundedness of $\mathbf{T}_g$ on $\mathcal{H}^2$ implies $g \in \mathrm{BMOA}(\mathbb{C}_{1/2})$; $\mathbf{T}_g f(s) = -\int_s^{+\infty} f(w) g'(w)\, \mathrm{d}w$.

The source is Brevig, Perfekt, and Seip, *Volterra operators on Hardy spaces of Dirichlet series*, Theorem 2.3(c) and §5, arXiv:1602.04729.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2609.11560.

The conditional argument lifts from p to $q = np > 2$ by the integer-power inclusions, then applies the local embedding and the BPS Carleson-measure argument. It gives the BMOA norm bound but does not prove $X_p = X_2$.

26. Dirichlet series and frequencies

26.1. **Bohr's theorem and completeness**

Question. Defant and Schoolmann ask: "Is there any relation between the λ's satisfying Bohr's theorem and the λ's for which $\mathscr{D}_\infty(\lambda)$ is complete?" Problem 2.14, p. 10 of arXiv:1902.02073v1.

Bohr's theorem here means that every λ-Dirichlet series with a bounded holomorphic extension to $\mathbb{C}_0$ converges uniformly on each $\mathbb{C}_\varepsilon$, $\varepsilon > 0$; $\mathscr{D}_\infty(\lambda)$ is the bounded convergent-series space with the supremum norm.

Defant and Schoolmann, *Variants of a theorem of Helson on general Dirichlet series*, Theorem 5.1, doi:10.1016/j.jfa.2020.108569 (arXiv:1909.11990), use almost-everywhere convergent twists and Riesz means: equal norms of twists make these means Cauchy in the complete series space, recovering the group Hardy space and hence Bohr's theorem.

Answered: yes. Evidence: Published.

Defant and Schoolmann later prove that, for every frequency, Bohr's theorem is equivalent to completeness of $\mathscr{D}_\infty(\lambda)$ (Theorem 5.1 of *Variants of a theorem of Helson on general Dirichlet series*). This gives the requested relation; completeness of the extended space is a separate issue.

26.2. **Hypercontractivity for general frequencies**

Question., arXiv:1902.02073v1 Defant and Schoolmann ask: "Given $D = \sum a_n e^{\lambda_n s} \in H^p(\lambda)$, $1 \le p < \infty$, under which assumptions on λ is it true that $D_u = \sum a_n e^{\lambda_n u} e^{\lambda_n s} \in H^q(\lambda)$ for all $1 \le q < \infty$ and $u > 0$?" Problem 4.15, p. 21. For the positive translation $P_u D(s) = D(s+u) = \sum a_n e^{-\lambda_n u} e^{-\lambda_n s}$ used here, the displayed plus signs in the source question are misprints.

The result is Carando, Defant, Marceca, Schoolmann, and Sevilla-Peris, *Hypercontractivity and strips of convergence in Hardy spaces of general Dirichlet series*, Theorem 1.1, arXiv:2401.09996; their small-exponent extrapolation in Corollary 3.7 upgrades the even H^2 tests to $H^1 \to H^q$, after which $H^p \subseteq H^1$ covers every finite input exponent.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

Answered: by an all-frequency operator criterion. Evidence: arXiv.

Theorem 1.1 of Carando, Defant, Marceca, Schoolmann, and Sevilla-Peris gives a necessary-and-sufficient criterion: it is enough to test every positive translation from $H^2(\lambda)$ to $H^{2k}(\lambda)$, for each integer $k \ge 1$. Their explicit additive-energy characterization requires an additional growth hypothesis; the later published survey poses a distinct, (LC)-restricted problem.

26.3. **Translated maximal inequalities beyond Bohr's condition**

Question., arXiv:1902.02073v1 Defant and Schoolmann ask: "Does Theorem 4.8 hold for a wider class of frequencies than the class of λ's satisfying (BC)? Is it even true without any condition on λ?" Problem 4.19, p. 22. Here Theorem 4.8 is the estimate $\|\sup_N |\sum_{n \le N} \widehat{f}(h_{\lambda_n}) e^{-u\lambda_n} h_{\lambda_n}|\|_{L^p(G)} \le C(u,\lambda)\|f\|_p$, for $u > 0$ and $1 \le p \le \infty$; (BC) requires $\lambda_{n+1} - \lambda_n \ge C e^{-\ell \lambda_n}$ for some $C, \ell > 0$. Bayart, *Convergence and almost sure properties in Hardy spaces of Dirichlet series*, Theorems 4.1 and 4.9, doi:10.1007/s00208-021-02239-x (arXiv:2101.02990), proves the wider result by linearizing the maximal cutoff and dominating its vertical-convolution kernels by a common integrable kernel; clustered frequencies with rescaled bad Fourier partial sums give the unrestricted counterexample.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

Partial. Evidence: Published.

Bayart proves the strong translated estimate under (NC), a class strictly larger than (LC), and gives an all-frequency counterexample. Thus the unrestricted extension is false and a substantial wider class is covered; whether Bohr's theorem alone suffices remains open.

26.4. **Maximal convergence abscissas for general frequencies**

Question. In arXiv:1902.02073v1, Defant and Schoolmann ask: "Given a frequency λ, give reasonable upper and lower estimates for $\sup_{D \in H_p(\lambda)} \sigma_c(D)$ and $\sup_{D \in H_p(\lambda)} \sigma_c(D)$, for example in terms of $L(\lambda)$." Problem 4.17, p. 21; the printed second σ_c is corrected to σ_a in published Problem 4.21.

Here $L(\lambda) = \limsup_n (\log n)/\lambda_n$, and $\sigma_c(D)$ and $\sigma_a(D)$ are the abscissas of ordinary and absolute convergence; write $C_p = \sup_{D \in H_p(\lambda)} \sigma_c(D)$ and $S_p = \sup_{D \in H_p(\lambda)} \sigma_a(D)$.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

Answered: sharp bounds in terms of $L(\lambda)$ and p. Evidence: Sketch.

The optimal universal bounds are $L/2 \le C_p \le S_p \le L/p$ for $p \le 2$ and $L/p \le C_p \le S_p = L/2$ for $p \ge 2$. Block constructions attain the bounds, including joint profiles for the two maximal abscissas.

Answers, "Maximal convergence abscissas for general frequencies", p. 43.

Maximal convergence abscissas for general frequencies

26.5. **Unconditional bases in $\mathscr{H}^p$**

Question., arXiv:1601.01616 Saksman and Seip ask: "Does H^p have an unconditional basis if $p \in (1,\infty)$ and $p \neq 2$?" Problem 2.5, §§2–3. Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616.

Open. Evidence: Heuristic.

The monomials are not an unconditional basis outside the Hilbert case, but that does not rule out a different basis. Recent results for canonical monomials and finite-dimensional subspaces do not settle the question for arbitrary bases.

26.6. **A Schauder basis for $\mathscr{H}^1$**

Question., arXiv:1601.01616 Saksman and Seip ask in Problem 2.6: "Does H^1 have a basis?" Part (a), §§2–3. Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616.

Open. Evidence: Heuristic.

The monomials form a Markushevich basis, which is weaker than a Schauder basis. This does not decide whether some other Schauder basis exists.

26.7. **An unconditional basis for $\mathscr{H}^1$**

Question., arXiv:1601.01616 Saksman and Seip ask in Problem 2.6: "Does H^1 have an unconditional basis?" Part (b), §§2–3. Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616.

Open. Evidence: Heuristic.

The question concerns arbitrary bases; finite-coordinate constructions and results for the canonical monomials do not provide an infinite-dimensional unconditional basis or an obstruction to one.

26.8. **Completeness of general Dirichlet-series spaces**

Question., arXiv:1902.02073 Defant and Schoolmann ask: "Characterize completeness of $\mathscr{D}_\infty(\lambda)$ and $\mathscr{D}_\infty^{\mathrm{ext}}(\lambda)$." Problem 2.13, p. 10 of the published survey. Its Theorem 2.12 already reduces the unresolved part to whether completeness of the extended space implies completeness of the ordinary space. Here $\mathscr{D}_\infty(\lambda)$ consists of convergent bounded λ-Dirichlet series, while $\mathscr{D}_\infty^{\mathrm{ext}}(\lambda)$ allows bounded holomorphic continuation of a somewhere convergent series; both use the supremum norm on $\mathbb{C}_0$. The published source is Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, Theorem 2.12 and Problem 2.13, doi:10.4064/bc119-6; the later ordinary-space equivalence is their *Variants of a theorem of Helson on general Dirichlet series*, Theorem 5.1 and Remark 5.4, doi:10.1016/j.jfa.2020.108569.

Open. Evidence: Heuristic.

The published survey isolates the residual implication from completeness of $\mathscr{D}_\infty^{\mathrm{ext}}(\lambda)$ to completeness of $\mathscr{D}_\infty(\lambda)$. The cited results characterize the latter through Bohr's theorem but do not resolve the extended-space implication.

26.9. **Banach-valued maximal partial sums**

Question., arXiv:1902.02073 Defant and Schoolmann ask: "Under which assumptions on the Banach space X and the frequency λ" Problem 5.7, p. 26. More precisely, do there exist such assumptions under which, for all $f \in H_p^\lambda(G,X)$ and $u > 0$,

$$\left\| \sup_N \left\| \sum_{n=1}^{N} \widehat{f}(h_{\lambda_n}) e^{-u\lambda_n} h_{\lambda_n} \right\|_X \right\|_{L^p(G)} \leq C \|f\|_{H_p^\lambda(G,X)}?$$

Here (G,β) is a λ-Dirichlet group, h_{λ_n} is the character satisfying $h_{\lambda_n}(\beta(t)) = e^{-i\lambda_n t}$, and $H_p^\lambda(G,X)$ is the X-valued Hardy subspace with Fourier support in these characters; (NC) is Bayart's later-frequency separation condition in Definition 2.2 of arXiv:2101.02990.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

They also ask which of λ, X, p, u determine C.

Partial. Evidence: Sketch.

The new estimate proves the source's suggested case under (NC), for every Banach space and $1 \le p \le \infty$, with a constant independent of X and p. A classification of all pairs (X, λ) remains open.

26.10. Cesàro means in $\mathscr{H}^1$

Question., arXiv:1907.12323 Defant and Schoolmann ask: "Do the first-order Cesàro means of every $D \in \mathscr{H}^1$ converge to D in $\mathscr{H}^1$?" This is the final question in §8, p. 19; the authors answer it immediately afterward.

The source is Defant and Schoolmann, *Henry Helson meets other big shots—A brief survey*, arXiv:1907.12323, §8, the paragraph following (23). Restriction to the powers of two reduces uniform boundedness of these Cesàro operators to uniformly bounded ordinary Taylor partial sums on disc H^1, which fails.

Answered: no. Evidence: arXiv.

The authors answer no immediately after the question: a positive answer would imply norm convergence for every analytic Fourier series in H^1, which is false.

26.11. Dropping the bounded-argument hypothesis for an omitted curve

Question., arXiv:1908.03113 Dan and Guo ask: "In Theorem 7.1, can the assumption that γ has a bounded argument function be dropped?" Question 7.6, p. 42 of *The Periodic Dilation Completeness Problem*.

Theorem 7.1 asserts cyclicity of a zero-free $F \in H^2(\mathbb{D}_2^\infty)$ whose range avoids a simple curve γ joining zero to infinity, provided γ has a bounded continuous argument; $\mathbb{D}_2^\infty = \mathbb{D}^\infty \cap \ell^2$, and cyclicity means that polynomial multiples of F are dense.

Dan and Guo, *The Periodic Dilation Completeness Problem*, arXiv:1908.03113v2.

Answered: yes. Evidence: Sketch.

For a zero-free $F \in H^2(\mathbb{D}_2^\infty)$ whose range on $\mathbb{D}_2^\infty$ avoids the simple curve γ in Theorem 7.1, the answer is affirmative. The proof uses a conformal map onto the component of the complement containing the range and bounded approximate reciprocals.

Answers, "Cyclicity when the omitted curve has unbounded argument", p. 46.

Cyclicity when the omitted curve has unbounded argument

26.12. Kozlov's dilation-completeness conjecture

Source passage., arXiv:1908.03113 Dan and Guo write: "By Proposition 8.1, we conjecture that the dilation system $\mathcal{D}_\theta$ is complete only for $\theta \in \{1, 1/2, 2/3\}$." Remark 8.2, p. 45. The inferred remaining question is whether any irrational θ gives a complete system.

Here $0 < \theta \le 1$ and $\mathcal{D}_\theta = \{\chi_{(0,\theta)}(kx) : k \ge 1\}$, using the odd 2-periodic extension of the indicator; completeness is in $L^2(0,1)$, and Proposition 8.1 treats rational θ.

Dan and Guo, *The Kozlov completeness problem*, Theorem 1.1, arXiv:2005.09779v5, settle the rational-endpoint case by a classification of cyclic Dirichlet series with periodic coefficients and the resulting arithmetic constraints on the interval endpoints; the rationality hypothesis is essential to this application.

Dan and Guo, *The Periodic Dilation Completeness Problem*, arXiv:1908.03113v2.

Partial. Evidence: arXiv.

Dan–Guo's later theorem classifies intervals with rational endpoints; for $(0, \theta)$ this gives exactly $\theta \in \{1, 1/2, 2/3\}$ among rational values. Irrational θ remains unresolved.

26.13. Bohr's theorem and the Helson maximal estimate

Question., arXiv:2101.02990 Bayart asks: "It seems natural to ask whether (11) always follows from (10) or, equivalently, if any frequency λ satisfying Bohr's theorem also satisfies Helson's theorem." Question 4.8, §4.1. Here (10) is $\|\max_{n\le N} |\sum_{k\le n} \widehat{f}(h_{\lambda_k}) h_{\lambda_k}|\|_\infty \le C e^{\delta\lambda_N} \|f\|_\infty$, and (11) is the same estimate in $L^1(G)$ for $f \in H^1_\lambda(G)$, for every $\delta > 0$. Bayart's condition (NC) requires a later index $m > n$ with $\log((\lambda_m + \lambda_n)/(\lambda_m - \lambda_n)) + m - n \le C e^{\delta\lambda_n}$ for every n; the

known result and counterexample are Theorems 4.1 and 4.9 of *Convergence and almost sure properties in Hardy spaces of Dirichlet series*, doi:10.1007/s00208-021-02239-x (arXiv:2101.02990).

Open. Evidence: Heuristic.

Bayart proves the estimate under his condition (NC), which covers a strict subfamily of the Bohr frequencies, and gives a counterexample without Bohr's theorem. Whether Bohr's theorem alone suffices remains open.

26.14. **Multiplier algebras of weighted Dirichlet-series spaces**

Question., arXiv:2301.06413 Sahu asks: "For which weight sequences $\mathbf{w}$ is $M(H_{\mathbf{w}})$ isometrically isomorphic to $H^\infty(\operatorname{Re} s > \delta_{\mathbf{w}}/2) \cap \mathcal{D}$?" Question 2.2, p. 325.

Sahu, *Multipliers of the Hilbert spaces of Dirichlet series*, Theorem 1.3 in arXiv:2301.06413v2, factors the normalized reproducing kernel through a positive kernel whose coefficients are the Möbius transforms of the weights; the resulting tensor-product dilation gives the reverse multiplier-norm inequality. This proves the sufficient condition, not a classification.

Here $H_{\mathbf{w}}$ has reproducing kernel $\sum_{n\geq 2} w_n n^{-s-\overline{u}}$, $M(H_{\mathbf{w}})$ is its multiplier algebra, $\mathcal{D}$ denotes somewhere convergent Dirichlet series, and $\delta_{\mathbf{w}}$ is the infimum of σ for which $\sum_{\operatorname{gpf}(n)\leq p_j} w_n n^{-\sigma} < \infty$ for every j, where p_j is the jth prime. The published sufficient result is Sahu, *Multipliers of the Hilbert spaces of Dirichlet series*, Theorem 2.3, https://nyjm.albany.edu/j/2023/29-14v.pdf; the later examples are Sahu, *On the abscissas of a Dirichlet series and its subseries supported on prime factorization*, Corollary 3.2, doi:10.1090/proc/17879.

Partial. Evidence: Published.

Sahu's Möbius-positivity condition is sufficient but not necessary as a full classification. Later published examples give equality and strict inclusion in additional cases; a general characterization is still open.

26.15. **Completeness for the square-root logarithm frequency**

Question. Defant and Schoolmann ask: "Is $\mathscr{D}_\infty((\sqrt{\log n}))$ complete?" Problem 2.15, p. 10 of arXiv:1902.02073v1.

The source is Defant and Schoolmann, *Variants of a theorem of Helson on general Dirichlet series*, Theorem 5.1 and Corollary 5.5, doi:10.1016/j.jfa.2020.108569 (arXiv:1909.11990); the gaps of $\sqrt{\log n}$ satisfy Landau's condition, and Riesz-mean approximation through a convergent character twist gives the completeness equivalence.

Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, arXiv:1902.02073v1.

Answered: yes. Evidence: Published.

For $\lambda_n = \sqrt{\log n}$, Landau's condition holds, hence Bohr's theorem does. Defant and Schoolmann's Theorem 5.1 equates Bohr's theorem with completeness of $\mathscr{D}_\infty(\lambda)$; their Corollary 5.5 gives this example explicitly.

26.16. **A condition weaker than Landau's condition**

Question. Defant and Schoolmann ask: "Find a reasonable condition on the frequency λ which is weaker than (LC), but is sufficient for the equality $\sigma_b^{\text{ext}}(D) = \sigma_b(D)$ for all λ-Dirichlet series D." Problem 2.16, p. 11. The source is Defant and Schoolmann, *Hardy spaces of general Dirichlet series—a survey*, Problem 2.16, arXiv:1902.02073v1. Here (LC) requires, for each $\delta > 0$, a constant $C > 0$ with $\lambda_{n+1} - \lambda_n \geq C\exp(-e^{\delta\lambda_n})$; σ_b^{ext} is the abscissa of bounded holomorphic extension and σ_b that of bounded convergence. Bayart's (NC) requires, for every $\delta > 0$, a constant $C > 0$ such that every n admits $m > n$ with $\log((\lambda_m + \lambda_n)/(\lambda_m - \lambda_n)) + m - n \leq Ce^{\delta\lambda_n}$. The proof is Bayart, *Convergence and almost sure properties in Hardy spaces of Dirichlet series*, Theorem 3.2 and Corollary 3.3, doi:10.1007/s00208-021-02239-x (arXiv:2101.02990): a trapezoidal vertical-convolution

cutoff controls partial sums through a later separated frequency; (NC) makes their growth subexponential, so positive translations converge uniformly.

Answered: yes. Evidence: Published.

Bayart's condition (NC) is strictly weaker than (LC) and implies Bohr's theorem. Consequently $\sigma_b^{\mathrm{ext}}(D) = \sigma_b(D)$ for every λ-Dirichlet series D; the proof gives the stronger equality $\sigma_b^{\mathrm{ext}} = \sigma_b = \sigma_u$.

26.17. Bass stable rank for multiplicative subsemigroups

Question., doi:10.1007/s00020-026-02825-1 Sasane conjectures: "Does $R(S) = \infty$ imply that the Bass stable rank of $\mathscr{H}_S^\infty$ is infinite?" Conjecture 6.1, §6, p. 11.

Here $S \subseteq \mathbb{N}$ is a multiplicative semigroup containing 1, $\mathscr{H}_S^\infty$ is the algebra of bounded Dirichlet series supported on S, and $R(S)$ is the supremum of the sizes of subsets in which no member belongs to the semigroup generated by the other members.

The classical theorem used here is Treil, *The stable rank of the algebra H^∞ equals 1*, Theorem 1, doi:10.1016/0022-1236(92)90015-B. Sasane, *On Banach subalgebras of the Dirichlet Hardy algebra $\mathscr{H}^\infty$*, doi:10.1007/s00020-026-02825-1.

Answered: no. Evidence: Sketch.

The prime-power semigroup has infinite rank under the source's weak independence definition, but its algebra is isometric to $H^\infty(\mathbb{D})$, whose Bass stable rank is one. The stronger unique-generator hypothesis of the source's Theorem 6.2 is therefore essential to that theorem's conclusion.

Answers, "Bass stable rank of a Dirichlet-series algebra", p. 46.

Bass stable rank of a Dirichlet-series algebra

26.18. Rational dimension of logarithms in a short interval

Source passage., doi:10.1215/S0012-7094-97-08601-4 Hedenmalm, Lindqvist, and Seip write: "The latter translates into the following basic number-theoretic problem: What is the multiplicatively rational dimension of the interval $N \le n < N + k$?" Here this dimension is the dimension over $\mathbb{Q}$ of the span of the logarithms; the inferred question asks for it as a function of N and k. §6, p. 30.

The cited partial classification is Najnudel, *On the distribution of random multiplicative functions*, Propositions 2.2 and 2.4, arXiv:1702.01470.

The source is Hedenmalm, Lindqvist, and Seip, *A Hilbert space of Dirichlet series and systems of dilated functions in $L^2(0,1)$*, doi:10.1215/S0012-7094-97-08601-4.

Partial. Evidence: Sketch.

For fixed length k, full rank is known for all $N \ge (100k)^{k+1}$, and the exceptional length-five starts are classified. A complete formula for the rank as both N and k vary is still open.

26.19. An ordinary Dirichlet series with exactly one zero

Question., https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf Balazard asks: "Does there exist an ordinary Dirichlet series converging in some open half-plane, and having there one zero, and only one?" Problem-session item 8, p. 390 of the Oberwolfach Report 06/2014.

Open. Evidence: Heuristic.

A general Dirichlet series example does not answer the ordinary-series question. The purported ordinary-series construction of Hilberdink and Saias was withdrawn for an irreparable error; no unconditional ordinary example is established in the checked sources.

26.20. Uniform translated partial sums in the Oberwolfach problem session

Question., https://publications.mfo.de/bitstream/handle/mfo/3615/OWR_2017_51.pdf Bayart asks: "Is it the case that $\sigma_N \to 0$ as $N \to \infty$?" Oberwolfach Report 51/2017, problem 4, p. 33. Here $\sigma_N = \sup_{f \in H^\infty(\mathbb{C}_0)} \inf\{\sigma > 0 : \|S_N f\|_{H^\infty(\mathbb{C}_\sigma)} \le \|f\|_{H^\infty(\mathbb{C}_\sigma)}\}$. Bayart's problem appears in Oberwolfach Report 51/2017, https://publications.mfo.de/bitstream/handle/mfo/3615/OWR_2017_51.pdf.

Answered: no. Evidence: Sketch.

For every $N \geq 2$ there is a three-term polynomial for which the defining inequality fails for every $\sigma > 0$; the explicit construction is given in the answers entry.

Answers, "The translated partial-sum inequality", p. 47.

The translated partial-sum inequality

26.21. **A Dirichlet polynomial on the line** $\operatorname{Re} s = 1$

Question., https://publications.mfo.de/bitstream/handle/mfo/3615/OWR_2017_51.pdf Erdős and Ingham ask: "Is it true that $D(s) \neq 0$ for all $\operatorname{Re}(s) = 1$?" Here $D(s) = 1 + 2^{-s} + 3^{-s} + 5^{-s}$; Oberwolfach Report 51/2017, Problem 2, p. 31.

Yip's counterexample is Theorem 1.3 of *On a problem of Erdős and Ingham*, arXiv:2512.16528; its concluding remarks leave the finite set $\{2, 3, 5\}$ open. The finite case is also problem (2), p. 2757, in Oberwolfach Report 51/2025, doi:10.4171/OWR/2025/51.

Yip builds disjoint integer blocks whose sums approximate the remaining complex target with quadratic error and reciprocal mass at most the target size; iterating makes the errors geometric and the total reciprocal mass finite. This construction uses an infinite set and does not settle the four-term polynomial.

Partial. Evidence: arXiv.

The general assertion with an arbitrary summable set of terms is false, but the specific four-term polynomial above remains unresolved. A 2025 Oberwolfach report restates this finite case.

27. Vertical limits

27.1. **An H^1 endpoint for vertical limits**

Question., arXiv:2606.20293 Andersson asks: "Can one take $E = \mathscr{C}_p(f)$ in Theorem 5.1 also in the case $p = 1$?" Question 1, §5, pp. 26–30.

Here $\mathscr{C}_p(f)$ consists of the characters χ for which f_χ continues to $\mathbb{C}_0$ and

$$\sup_{\sigma>0, T\geq 1} (2T)^{-1} \int_{-T}^{T} |f_\chi(\sigma + it)|^p \, \mathrm{d}t < \infty.$$

Theorem 5.1 gives the continuously differentiable vertical p-mean and its Hardy–Stein identity on a full-measure set.

Andersson, *The Littlewood–Paley formula and mean counting function for vertical limits of Dirichlet series*, arXiv:2606.20293v2.

Answered: yes. Evidence: Sketch.

Conditional on: The source's standing hypotheses, including the finite-mean assumption used for $p = 1$.

The endpoint identity holds on the full Carlson set by the linked proof. Its conditionality is inherited from the source's standing lemma; the result does not assert a new external theorem. The Hardy–Stein endpoint

27.2. **One exceptional set for every target value**

Question., arXiv:2606.20293 Andersson asks: "Can one choose the set of full measure in Theorem 1.2 independently of $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 2, pp. 26–30.

Theorem 1.2 identifies the mean counting function $M(f, \zeta)$ with

$$\lim_{T\to\infty} (\pi/T) \sum_{\substack{f_\chi(s)=\zeta,\ |\operatorname{Im} s|<T \\ \operatorname{Re} s>0}} \operatorname{Re} s$$

for almost every χ; $f_\chi(s) = \sum_n a_n \chi(n) n^{-s}$ and $f(+\infty) = a_1$.

Andersson, *The Littlewood–Paley formula and mean counting function for vertical limits of Dirichlet series*, arXiv:2606.20293v2.

Open. Evidence: Heuristic.

The source proves the limit for each fixed target with a target-dependent full-measure set. Its ordered cutoff theorem and the related completed-class result take limits in a different order, so they do not imply the common-set boundary-first statement.

27.3. **Counting functions in the completed Nevanlinna class**

Question. Andersson asks: "Let $f \in \mathcal{N}$. Does there exist a set $E \subseteq \mathbb{T}^\infty$ of full measure such that if $\chi \in E$, then f_χ has an analytic continuation to $\mathbb{C}_0$ for which

$$\lim_{\sigma_0 \downarrow 0} \lim_{T \to \infty} \frac{\pi}{T} \sum_{\substack{s \in \mathscr{Z}(f_\chi - \zeta) \\ |\operatorname{Im} s| < T \\ \operatorname{Re} s > \sigma_0}} (\operatorname{Re} s - \sigma_0)$$

exists and is finite for all $\zeta \in \mathbb{C} \setminus \{f(+\infty)\}$?" Question 3, pp. 26–30; zeros are

Here $\mathcal{N}$ is the completion of uniformly convergent Nevanlinna Dirichlet series for the logarithmic metric $\|f - g\|_0 = \limsup_{\sigma \downarrow 0} \lim_{T \to \infty} (2T)^{-1} \int_{-T}^{T} \log(1 + |f - g|(\sigma + it))\, \mathrm{d}t$, and f_χ denotes the character twist, with $f(+\infty)$ its constant coefficient.

The source is Andersson, *The Littlewood–Paley formula and mean counting function for vertical limits of Dirichlet series*, Question 3 and Lemma 3.1, arXiv:2606.20293v2; that lemma is the extended nonnegative ergodic identity for the prime Kronecker flow.

repeated according to multiplicity.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2606.20293v2, Lemma 3.1.

The proof constructs one conull set controlling all targets through a countable dense set and analytic dependence, then passes to arbitrary targets by the completed-class counting estimate. It uses Lemma 3.1 of the source and preserves the order $T \to \infty$ followed by the boundary cutoff.

Answers, "Counting functions in the completed Nevanlinna class", p. 47.

Counting functions in the completed Nevanlinna class

27.4. **The optimal H^p order-to-mean abscissa for $1 \le p \le 2$**

Source passage. Brevig and Kouroupis state: "Fix $\beta > 0$. If an element f of $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ such that $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(f) \le \beta/(1 + 2\beta)$." Their Theorem 4 proves sharpness in $\mathscr{H}^2$. The inferred question is whether the same optimal bound holds in $\mathscr{H}^p$ for $1 \le p \le 2$.

Here $\mu_f(\sigma) = \limsup_{|t| \to \infty} \log|f(\sigma + it)| / \log|t|$ is the Lindelöf order, and $\sigma_{\mathrm{m}}(f)$ is the infimum of κ such that the continuation has finite order on $\mathbb{C}_\kappa$ and its vertical mean equals the Hardy norm of $f(\sigma + \cdot)$ for every $\sigma > \kappa$.

The source is Brevig and Kouroupis, *Mean values, order, and convergence in the Hardy space of Dirichlet series*, Theorems 1 and 4, arXiv:2609.35496.

Answered: $\beta/(1 + 2\beta)$, sharply. Evidence: Sketch.

Conditional on: arXiv:2609.35496.

The bound equals the H^2 value and is attained by the Brevig–Kouroupis extremal function. The proof uses Helson's inequality to transfer to H^2 and the source's sharp example; it relies on the source hypotheses and theorem.

Answers, "The optimal H^p order-to-mean abscissa for $1 \le p \le 2$", p. 49.

27.5. **The order-to-mean abscissa for $p > 2$**

Source passage., arXiv:2609.35496 Brevig and Kouroupis state: "Fix $\beta > 0$. If an element f of $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ such that $\mu_f(\sigma) \le \beta$ for every $\sigma > 0$, then $\sigma_{\mathrm{m}}(f) \le \beta/(1 + 2\beta)$." The inferred question asks for the optimal analogue when $p > 2$.

Here $\mu_f(\sigma) = \limsup_{|t| \to \infty} \log|f(\sigma + it)| / \log|t|$ is the Lindelöf order, and $\sigma_{\mathrm{m}}(f)$ is the infimum of κ such that the continuation has finite order on $\mathbb{C}_\kappa$ and its vertical mean equals the Hardy norm of $f(\sigma + \cdot)$ for every $\sigma > \kappa$.

The source is Brevig and Kouroupis, *Mean values, order, and convergence in the Hardy space of Dirichlet series*, Theorems 1 and 4, arXiv:2609.35496.

Open. Evidence: Heuristic.

The source proves the sharp H^2 estimate $\beta/(1+2\beta)$. Power reduction yields bounds at even exponents, but does not identify the optimal general real-$p>2$ value or its extremizers.

27.6. The order-zero mean-value formula in H^p

Source passage. Brevig and Kouroupis state: "If f in $\mathscr{H}^2$ has an analytic continuation to $\mathbb{C}_0$ of order 0, then

$$\|f\|_{\mathscr{H}^2}^2 = \lim_{\sigma\to 0^+}\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|f(\sigma+it)|^2\,\mathrm{d}t.$$

" Corollary 2, §2. The inferred question asks whether this ordered formula extends to every $1\le p<\infty$.

Order zero means $\mu_f(\sigma)\le 0$ for every $\sigma>0$, where $\mu_f(\sigma)=\limsup_{|t|\to\infty}\log|f(\sigma+it)|/\log|t|$; the proposed H^p formula replaces the squared norms and integrands by their pth powers.

The source is Brevig and Kouroupis, *Mean values, order, and convergence in the Hardy space of Dirichlet series*, Corollary 2, arXiv:2609.35496; the shift theorem is Carando, Defant, Marceca, Schoolmann, and Sevilla-Peris, *Hypercontractivity and strips of convergence in Hardy spaces of general Dirichlet series*, Corollary 1.2, arXiv:2401.09996.

Answered: yes. Evidence: Sketch.

Conditional on: arXiv:2609.35496 and arXiv:2401.09996.

For $p=2k$, apply the H^2 formula to f^k. For general p, the proof combines hypercontractive shifts and the mean-value theorem for an even exponent above p, then lets the shift tend to zero. Both cited results are standing hypotheses.

Answers, "The order-zero mean-value formula in H^p", p. 49.

The order-zero mean-value formula in H^p

28. Optimal polynomial approximants and cyclicity

28.1. A common zero-free compact set

> Let $\{p_n^*\}$ be optimal approximants to $f\in D_\alpha\setminus\{0\}$. Is there a compact set $K\subset\mathbb{D}^2$ such that $p_n^*(z)\neq 0$ for $z\in K$ and all n? Similarly, if $\{p_n^*\}$ are optimal approximants to $f\in H_d^2$, is there a compact set $K\subset B_d$ such that $p_n^*(z)\neq 0$ for $z\in K$ and all n?

Sargent–Sola, *Optimal approximants and orthogonal polynomials in several variables*, arXiv:2002.08790v2, Question 24, p. 21. Here D_α has norm $\sum_{j,k\ge 0}(j+1)^\alpha(k+1)^\alpha|a_{jk}|^2$, and H_d^2 is the Drury–Arveson space with kernel $(1-\langle z,w\rangle)^{-1}$ on the unit ball B_d; an optimal approximant minimizes $\|pf-1\|$ in the prescribed polynomial subspace.

Answered: iff $f(0)\neq 0$ for nonempty K. Evidence: Sketch.

Our full written projection argument proves a common zero-free neighbourhood of 0 when $f(0)\neq 0$. If $f(0)=0$, every approximant is identically zero, so no nonempty K exists. The source does not require K to be nonempty; allowing $K=\varnothing$ makes its literal question vacuous.

28.2. When approximants are Taylor polynomials

> Given $g\in H$ and a power series $\varphi(z)=\sum a_kz^k$, can we characterize $f\in H$ such that the nth optimal polynomial approximants to g/f are given by $T_n(\varphi)$ for all n greater than some $M>0$?

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Question 1, p. 5. Here H is a reproducing-kernel Hilbert space of analytic functions on $\mathbb{D}$ with bounded shift, $[f]=\overline{\operatorname{span}}\{z^kf:k\ge 0\}$, $\Pi_{[f]}$ is orthogonal projection, and $T_n(\varphi)=\sum_{k=0}^n a_kz^k$; the approximant minimizes $\|pf-g\|_H$ among polynomials of degree at most n.

Partial. Evidence: Sketch.

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Theorem 3.8 classifies the normalized kernel-at-zero target. Our full written partial argument supplies Hardy-space and selected weighted polynomial or sparse-level examples. The classification for general g remains open; all-degree banded-recurrence rigidity theorems do not apply to sparse orthogonal levels.

28.3. When the cyclic projection is a polynomial multiple

Given $g \in H$ and supposing p is a polynomial, can we characterize f such that $\Pi_{[f]}(g) = pf$?

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Question 2, p. 5. Here H is a reproducing-kernel Hilbert space of analytic functions on $\mathbb{D}$ with bounded shift, $[f] = \overline{\operatorname{span}}\{z^k f : k \geq 0\}$, $\Pi_{[f]}$ is orthogonal projection, and $T_n(\varphi) = \sum_{k=0}^{n} a_k z^k$; the approximant minimizes $\|pf - g\|_H$ among polynomials of degree at most n.

Partial. Evidence: Sketch.

Felder, *Optimal polynomial approximants and orthogonal polynomials*, arXiv:2003.10015v2, Theorem 3.8 classifies the normalized kernel-at-zero target. Our full written partial argument supplies Hardy-space and selected weighted polynomial or sparse-level examples. The classification for general g remains open; all-degree banded-recurrence rigidity theorems do not apply to sparse orthogonal levels.

28.4. The extremal Jacobi norm

What is M for the spaces D_α when $\alpha < 0$, $\alpha \neq -1$? What is the corresponding extremal function?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 1, p. 18. Here D_α has $\|f\|_\alpha^2 = \sum_{k\geq 0}(k+1)^\alpha |a_k|^2$, M is the infimum of the moduli of all approximant zeros, and the symmetric Jacobi matrix has $J_{j,j+1} = J_{j+1,j} = ((j+1)/(j+2))^{\alpha/2}$ and other entries zero, for $j \geq 0$.

Open. Evidence: Heuristic.

The source defines M as the infimum of $|z|$ over zeros of all optimal polynomial approximants in the space. For $\alpha < 0$, $M = 2/\|J\|$. Bénéteau–Khavinson–Liaw–Seco–Simanek, *Zeros of optimal polynomial approximants: Jacobi matrices and Jentzsch-type theorems*, doi:10.4171/rmi/1064, Theorem 4.2 and Corollary 4.5, identify the extremal constant with the norm of a symmetric Jacobi matrix and prove existence of an extremizer. The case $\alpha = -1$ is explicit. These results do not evaluate the norm for the other negative parameters; no proposed value is supported here.

28.5. Zero-free sectors at the circle

Given a cyclic function $f \in H^2_\omega$, and given $\zeta \in \mathbb{T}$, is there a sector with vertex at ζ that is devoid of zeros of opas? More generally, are there regions or rays that the zeros avoid?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 2, p. 19. Here H^2_ω has $\|f\|_\omega^2 = \sum_{k\geq 0} |a_k|^2 \omega_k$, with $\omega_0 = 1$, $\omega_k > 0$ and $\omega_k/\omega_{k+1} \to 1$; q_n minimizes $\|qf - 1\|_\omega$ over polynomials of degree at most n, and cyclic means that polynomial multiples of f are dense.

Partial. Evidence: Sketch.

Our written argument gives an admissible block weight and a cyclic $f = (1-z)^2$ for which every sector at every $\zeta \neq 1$ contains zeros of approximants. The weight is not quasi-integral. Thus the quasi-integral setting motivating the source question, including D_α with $\alpha < 0$, remains open; the example also leaves $\zeta = 1$ untreated.

Full written argument.

28.6. **Rates for functions without analytic extension**

Given a function $f \in D_\alpha$ such that f does not have an analytic extension to the closed disk, find the rate of decay of $\|q_n f - 1\|_\alpha$.

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 5, p. 22. Here D_α has $\|f\|_\alpha^2 = \sum_{k\geq 0}(k+1)^\alpha |a_k|^2$, and q_n minimizes $\|qf-1\|_\alpha$ among polynomials of degree at most n.

Partial. Evidence: Sketch.

Our full written partial argument constructs cyclic functions with arbitrarily slow decay, for every real α, and functions realizing prescribed subgeometric upper bounds. Consequently nonextension alone imposes no universal rate. A characterization of the rate for an individual f, as asked in the source, remains open.

28.7. **Boundary convergence for polynomial reciprocals**

Suppose f is a polynomial with zeros $Z(f)$ in the complement of $\mathbb{D}$ and let q_n be the n-th opa of $1/f$ in H^2_ω. For which weights $\{\omega_k\}_{k\in\mathbb{N}}$ is it true that $1 - q_n f$ converges to 0 uniformly on compact subsets of $\overline{\mathbb{D}} \setminus Z(f)$?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 6, p. 24. Here H^2_ω has $\|f\|_\omega^2 = \sum_{k\geq 0} |a_k|^2 \omega_k$, with $\omega_0 = 1$, $\omega_k > 0$ and $\omega_k/\omega_{k+1} \to 1$; q_n minimizes $\|qf-1\|_\omega$ over polynomials of degree at most n, and cyclic means that polynomial multiples of f are dense.

Partial. Evidence: Sketch.

Bénéteau–Manolaki–Seco, *Boundary behavior of optimal polynomial approximants*, doi:10.1007/s00365-020-09508-z, Theorems 1.8 and 6.1, treat simple zeros in Hardy and Bergman spaces and all-boundary simple zeros under monotone doubling weights. Our written argument gives the threshold $\beta < 1/m$ for $f = (1-z)^m$ and $\omega_k = e^{-ck^\beta}$, $c > 0$, $0 < \beta < 1$. Arbitrary admissible weights and mixed multiple zeros remain open.

28.8. **Boundary convergence beyond polynomials**

Does Theorem 5.5 hold for functions other than polynomials, and if so, for which ones?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 7, first question, p. 24. Here Theorem 5.5 says that, for a polynomial f with simple zeros outside $\mathbb{D}$, its optimal approximants q_n in H^2 or A^2 satisfy $1 - q_n f \to 0$ uniformly on compact subsets of $\overline{\mathbb{D}} \setminus Z(f)$.

Partial. Evidence: Published.

Bénéteau–Manolaki–Seco, *Boundary behavior of optimal polynomial approximants*, doi:10.1007/s00365-020-09508-z, Theorem 1.8, proves the quoted convergence for polynomials with simple zeros. Bénéteau–Ivrii–Manolaki–Seco, *Simultaneous zero-free approximation and universal optimal polynomial approximants*, Theorems 1.2 and 4.1 (arXiv:1811.04308v2), published in *J. Approx. Theory* 256 (2020), 105389, gives a dense G_δ universality class. Baxter, *A convergence equivalence related to polynomials orthogonal on the unit circle*, doi:10.1090/S0002-9947-1961-0126126-8, supplies nonpolynomial sufficient conditions in H^2. The universality proof combines simultaneous zero-free approximation with continuity of the finite-degree normal equations and Baire category. The polynomial convergence proof represents $1 - q_n f$ by an inverse kernel Gram matrix and estimates its entries. These sufficient and generic results do not characterize either requested class.

28.9. **Characterizing approximant universality**

On the other hand, can we characterize the functions for which the universality behavior as in Theorem 5.4 occurs?

Bénéteau–Centner, *A survey of optimal polynomial approximants, applications to digital filter design, and related open problems*, arXiv:2102.01725, Open Question 7, second question,

p. 24. Here Theorem 5.4 says that, for each $\zeta \in \mathbb{T}$, a dense G_δ set of f in H^2 or the Dirichlet space has optimal approximants q_n with $\{q_n(\zeta) : n \geq 0\}$ dense in $\mathbb{C}$.
Partial. Evidence: Published.

Bénéteau–Manolaki–Seco, *Boundary behavior of optimal polynomial approximants*, doi:10.1007/s00365-020-09508-z, Theorem 1.8, proves the quoted convergence for polynomials with simple zeros. Bénéteau–Ivrii–Manolaki–Seco, *Simultaneous zero-free approximation and universal optimal polynomial approximants*, Theorems 1.2 and 4.1 (arXiv:1811.04308v2), published in *J. Approx. Theory* 256 (2020), 105389, gives a dense G_δ universality class. Baxter, *A convergence equivalence related to polynomials orthogonal on the unit circle*, doi:10.1090/S0002-9947-1961-0126126-8, supplies nonpolynomial sufficient conditions in H^2. The universality proof combines simultaneous zero-free approximation with continuity of the finite-degree normal equations and Baire category. The polynomial convergence proof represents $1 - q_n f$ by an inverse kernel Gram matrix and estimates its entries. These sufficient and generic results do not characterize either requested class.

28.10. **Zero exclusion for Hardy approximants**

Let $1 < p < \infty$, $f \in H^p$, and $n \in \mathbb{N}$. If $f(0) \neq 0$, then $q_{n,p}[f, 1]$ is zero-free in $\overline{\mathbb{D}}$.

Centner, *Optimal polynomial approximants in L^p*, arXiv:2112.14002v3, Conjecture 7.1, p. 23. Here $q_{n,p}[f, 1]$ minimizes $\|qf - 1\|_{H^p}$ over polynomials q of degree at most n.
Answered: no. Evidence: Sketch.

Our full written counterexample has a degree-one approximant with a zero inside $\mathbb{D}$, at an exponent $p \neq 2$. This refutes even open-disc zero exclusion. The classical Hilbert-space conclusion at $p = 2$ is unaffected.

Full written argument.

28.11. **Pairing spurious zeros with electrostatic partners**

Let $\{\xi_{n,k}\}$ be the set of spurious roots of φ_n and $\{s_{n,\ell}\}$ the set of the roots of the electrostatic partner S_n which are inside the disk and do not accumulate on $\mathbb{T}$. Then both sets has the same amount of elements and we can reorganize all of them in pairs $(\xi_{n,k}, s_{n,k})$ in such a way that

$$\xi_{n,k} = s_{n,k} - \frac{s_{n,k}}{n} + o(1/n),$$

for any of these pairs.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, Conjecture 1, p. 20. Here φ_n is the monic orthogonal polynomial for $|f|^2\,dm$ on $\mathbb{T}$, with f a polynomial whose zeros lie on $\mathbb{T}$; spurious roots do not accumulate on $\mathbb{T}$. The electrostatic partner is the monic multiple of $\det\left[\begin{smallmatrix} \varphi_n & \widehat{\varphi}_n \\ A\varphi_n' & A\widehat{\varphi}_n' - B_n\widehat{\varphi}_n \end{smallmatrix}\right]$, where $\widehat{\varphi}_n$ is the second-kind Cauchy integral for the source's contour weight w_n and $B_n/A = w_n'/w_n$.
Answered: no. Evidence: Sketch.

Our full written counterexample uses a fixed cubic with three simple zeros on $\mathbb{T}$. Along a subsequence, two spurious zeros coalesce on the $1/n$ scale, whereas the two partner zeros have different limiting displacements. The example refutes the pairing clause; it does not refute the separate equal-cardinality assertion.
Answers, "Pairing spurious zeros with electrostatic partners", p. 50.

28.12. **Weak limits of approximant zeros**

It is conjectured that the same should hold for the OPA $\{p_n\}$ themselves.

Orive–Sánchez-Lara–Seco, *Electrostatic interpretation of the zeros of optimal polynomial approximants*, arXiv:2507.15488, unnumbered conjecture, p. 3. Here "the same means weak convergence to normalized arc length on $\mathbb{T}$ of $\deg(p_n)^{-1} \sum_{p_n(z)=0} \delta_z$, with zeros counted with

multiplicity; p_n minimizes $ormpf - 1_{H^2}$ and the source considers polynomials f with all zeros on $\mathbb{T}$.

Answered: Haar measure. Evidence: Sketch.

"The same" refers to convergence of normalized zero measures to uniform arc length on $\mathbb{T}$. Our full written deduction proves this for every critical polynomial and positive power of one, using the published asymptotics of Martínez-Finkelshtein–McLaughlin–Saff, *Asymptotics of orthogonal polynomials with respect to an analytic weight with algebraic singularities on the circle*, doi:10.1155/IMRN/2006/91426, Theorem 1(iii) and Corollary 1. The argument includes the bounded number of zeros that may escape after reversal.

Answers, "Weak limits of approximant zeros", p. 51.

28.13. **Removing an inner factor by a scalar**

Let $1 < p < \infty$ and $f \in H^p$ with $f(0) \neq 0$. If J is any non-constant inner function, then there exists a constant $c = c_f \in \mathbb{T}$ such that $\|1 - Jf\|_p > \|1 - cf\|_p$.

Bénéteau–Cheng–Felder–Khavinson–Manolaki–Maronikolakis, *Metric projections, zeros of optimal polynomial approximants, and some extremal problems in Hardy spaces*, arXiv:2511.08000v2, Conjecture 4.3, p. 12.

Answered: no. Evidence: Sketch.

Our full written counterexample uses a scale parameter and defeats every unimodular scalar. It refutes the literal existential comparison, independently of whether the notation c_f is intended to prohibit dependence on J.

Full written argument.

28.14. **Bidisc approximant zeros near the origin**

It has been suggested by A. Dayan [7] that, without imposing symmetry, the zeros may approach the origin arbitrarily closely.

Navío–Seco, *Polynomial counterexamples to Shanks' conjecture in Dirichlet-type spaces over the bidisc*, arXiv:2609.37180, Introduction after Theorem 1.1, p. 3. Here [7] is A. Dayan's private communication, and the approximants are linear polynomials p minimizing $\|pf - 1\|_{H^2(\mathbb{D}^2)}$ for a stable polynomial f, meaning f has no zeros on $\overline{\mathbb{D}}^2$.

Answered: yes. Evidence: Sketch.

The motivating passage concerns linear optimal approximants to reciprocals of stable bidisc polynomials. Our full written construction uses high powers of $(1 + z^2/2)(1 + w^2/2)(1 + zw/2)(1 + it(z - w))$. The normalized boundary mass concentrates at two symmetric torus maxima, and the resulting linear approximant has a zero tending to the origin as $t \downarrow 0$.

28.15. **Cyclic polynomials on the bidisc**

Characterize cyclic polynomials in the Dirichlet spaces D_α.

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, problem session, Bénéteau, item 1, pp. 383–384.

Answered: classification. Evidence: Published.

Here $\|\sum a_{kl} z_1^k z_2^l\|_\alpha^2 = \sum (k+1)^\alpha (l+1)^\alpha |a_{kl}|^2$. Bénéteau–Knese–Kosiński–Liaw–Seco–Sola, *Cyclic polynomials in two variables*, doi:10.1090/tran6689, Main Theorem, gives the full classification. The thresholds depend on the irreducible factors and their boundary zero sets; in particular, $1 - z_1$ is cyclic exactly for $\alpha \leq 1$, whereas $1 - z_1 z_2$ is cyclic exactly for $\alpha \leq 1/2$.

The proof reduces to irreducible factors, constructs cyclic approximations using stable determinantal representations, and obtains the converse from positive-capacity boundary curves and their Fourier decay.

28.16. **A real-variable proof of the Nyman equivalence**

Is there a real variable proof of the equivalence of (2) and (3)?

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, Balazard's talk, Question 1, p. 350.

Open. Evidence: Heuristic.

In the source, (2) is $\pi(x) = \mathrm{li}(x) + O(x^{1/2+\varepsilon})$ for every $\varepsilon > 0$, and (3) says $\chi \in \overline{\mathrm{span}}\{\{t/\alpha\} : \alpha \geq 1\}$ in $L^2(0,\infty; t^{-2}\,\mathrm{d}t)$. Nyman's thesis *On the one-dimensional translation group and semi-group in certain function spaces* (1950) proves the equivalence with the Riemann hypothesis, and Báez-Duarte, *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*, arXiv:math/0202141v2, main theorem, restricts α to integers. These theorems do not provide the requested real-variable proof; the methodological restriction remains unspecified.

28.17. **Irrational local maxima of the autocorrelation**

Does there exist a positive irrational where A has a strict local maximum?

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, Balazard's talk, Question 2, p. 350. Here $A(\lambda) = \int_0^\infty \{t\}\{\lambda t\} t^{-2}\,\mathrm{d}t$, where $\{t\}$ denotes the fractional part.

Answered: no. Evidence: Sketch.

Our full written proof gives a positive symmetric second difference at every positive irrational along a subsequence of continued-fraction scales. Thus no positive irrational is a local maximum, strict or otherwise. Báez-Duarte–Balazard–Landreau–Saias, *Étude de l'autocorrélation multiplicative de la fonction "partie fractionnaire"*, doi:10.1007/s11139-005-0834-4, with the detailed rational expansion in arXiv:math/0306251, Proposition 98, then identifies all positive local maxima.

Answers, "Irrational local maxima of the autocorrelation", p. 51.

28.18. **The discrete Nyman invariant subspace**

Is it true that $V = \{D \in E^2,\ D(s)/\zeta(s) \text{ is holomorphic for } \Re s > 1/2\}$?

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, Balazard's talk, Question 3, p. 352.

Open. Evidence: Heuristic.

Here E^2 consists of Dirichlet series with $D(s)/s \in H^2(\Re s > 1/2)$ and V is the closed span of $\zeta(s)\sum_{k\leq n} c_k k^{-s}$ with $\sum c_k/k = 0$. Báez-Duarte, *A strengthening of the Nyman–Beurling criterion for the Riemann hypothesis*, arXiv:math/0202141v2, main theorem, proves the equality under the Riemann hypothesis. Bercovici–Foias, *A real variable restatement of Riemann's hypothesis*, doi:10.1007/BF02760524, and Balazard–Saias, *The Nyman–Beurling equivalent form for the Riemann hypothesis*, https://www.esi.ac.at/preprints/esi623.pdf, Theorem 0, treat the continuous Nyman analogue; their invariant-subspace methods do not supply a theorem for the integer-dilation semigroup. The unconditional equality is open.

28.19. **Continuity of Nyman distances**

Is $D(\lambda)$ a continuous function of λ?

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, problem session, Balazard, item 8, second question, p. 390.

Partial. Evidence: Sketch.

Here $D(\lambda) = \mathrm{dist}_L(\chi, \overline{\mathrm{span}}\{e_\alpha : 1 \leq \alpha \leq \lambda\})$, $L = L^2(0,\infty; t^{-2}\,\mathrm{d}t)$, $\chi(t) = [t \geq 1]$, and $e_\alpha(t) = \{t/\alpha\}$. Our full written partial argument proves left continuity, continuity at 1, and strict improvement over $D(1)$ for sufficiently large cutoffs. Right continuity at arbitrary $\lambda > 1$ and strict decrease on every interval remain open.

28.20. **Strict decrease of Nyman distances**

Is $D(\lambda)$ a strictly decreasing function of λ?

Dirichlet Series and Function Theory in Polydiscs, https://publications.mfo.de/bitstream/handle/mfo/3396/OWR_2014_06.pdf, Oberwolfach Report 06/2014, problem session, Balazard, item 8, third question, p. 390.

Partial. Evidence: Sketch.

Here $D(\lambda) = \operatorname{dist}_L(\chi, \overline{\operatorname{span}}\{e_\alpha : 1 \le \alpha \le \lambda\})$, $L = L^2(0,\infty; t^{-2}\,\mathrm{d}t)$, $\chi(t) = [t \ge 1]$, and $e_\alpha(t) = \{t/\alpha\}$. Our full written partial argument proves left continuity, continuity at 1, and strict improvement over $D(1)$ for sufficiently large cutoffs. Right continuity at arbitrary $\lambda > 1$ and strict decrease on every interval remain open.